

Part VI

OSPF

Chapter 21: Understanding OSPF Concepts

Chapter 22: Implementing Basic OSPF Features

Chapter 23: Implementing Optional OSPF Features

Chapter 24: OSPF Neighbors and Route Selection

Part VI Review

[Part IV](#) began the story in this book about IP Version 4 (IPv4) addressing. [Part V](#) continued that story with how to implement addressing in Cisco routers, along with a variety of methods to route packets between local interfaces. But those topics delayed the discussion of one of the most important topics in TCP/IP, namely IP routing protocols, as discussed in [Part VI](#).

Routers use IP routing protocols to learn about the subnets in an internetwork, choose the current-best routes to reach each subnet, and add those routes to each router's IP routing table. Cisco chose to include one and only one IP routing protocol in the CCNA 200-301 Version 1.1 blueprint:

the Open Shortest Path First (OSPF) routing protocol. This entire part focuses on OSPF as an end to itself and to show the principles of routing protocols.

Chapter 21

Understanding OSPF Concepts

This chapter covers the following exam topics:

3.0 IP Connectivity

3.2 Determine how a router makes a forwarding decision by default

3.2.b Administrative distance

3.2.c Routing protocol metric

3.4 Configure and verify single area OSPFv2

3.4.a Neighbor adjacencies

3.4.b Point-to-point

3.4.c Broadcast (DR/BR selection)

3.4.d Router ID

Every enterprise uses some dynamic routing protocol inside their network so that the routers cooperatively learn routes to all subnets. But in the decades that led to TCP/IP becoming the common networking model used on all computers, several routing protocols emerged as candidates

to be used by those enterprises. As a result, even today, enterprises choose from a small set of alternative routing protocols. Of those, Cisco includes Open Shortest Path First (OSPF) in the CCNA 200-301 V1.1 blueprint.

To establish some context, this chapter begins by examining the different routing protocols, their similar goals, and their differences in implementation. With that context in mind, the rest of the chapter then examines the basic concepts of how OSPF operates. The second major section of the chapter gets into the foundations of OSPF: running OSPF on each router, becoming neighbors, exchanging data about routes, and calculating IP routes to be used by the IP routing table. The final major section then looks more closely at how OSPF internally represents network topologies as a database of network links and their states—the OSPF link-state database (LSDB).

“Do I Know This Already?” Quiz

Take the quiz (either here or use the PTP software) if you want to use the score to help you decide how much time to spend on this chapter. The letter answers are listed at the bottom of the page following the quiz. [Appendix C](#), found both at the end of the book as well as on the companion website, includes both the answers and explanations. You can also find both answers and explanations in the PTP testing software.

Table 21-1 “Do I Know This Already?” Foundation Topics Section-to-Question Mapping

Foundation Topics Section	Questions
Comparing Dynamic Routing Protocol Features	1-3
OSPF Concepts and Operation	4, 5

Foundation Topics Section	Questions
OSPF Areas and LSAs	6

- 1.** Which of the following routing protocols uses link-state logic?
 - a.** RIPv1
 - b.** RIPv2
 - c.** EIGRP
 - d.** OSPF

- 2.** Which of the following routing protocols use a metric that is, by default, at least partially affected by link bandwidth? (Choose two answers.)
 - a.** RIPv1
 - b.** RIPv2
 - c.** EIGRP
 - d.** OSPF

- 3.** Which of the following interior routing protocols support VLSM? (Choose three answers.)
 - a.** RIPv1
 - b.** RIPv2
 - c.** EIGRP
 - d.** OSPF

- 4.** Two routers using OSPFv2 have become neighbors and exchanged all LSAs. As a result, Router R1 now lists some OSPF-learned routes in its routing table. Which of the following best describes how R1 uses those

recently learned LSAs to choose which IP routes to add to its IP routing table?

- a.** R1 copies a route from every LSA into its routing table.
 - b.** R1 copies a route from certain types of LSAs into its routing table.
 - c.** R1 runs SPF against the LSAs to calculate the routes.
 - d.** R1 does not use the LSAs when choosing what routes to add.
- 5.** Which of the following OSPF neighbor states is expected when the exchange of topology information is complete between two OSPF neighbors?
- a.** 2-way
 - b.** Full
 - c.** Up/up
 - d.** Final
- 6.** A company has a small/medium-sized network with 15 routers and 40 subnets and uses OSPFv2. Which of the following is considered an advantage of using a single-area design as opposed to a multiarea design?
- a.** It reduces the CPU processing overhead on most routers.
 - b.** It reduces the frequency of running the SPF algorithm due to interface status changes.
 - c.** It allows for simpler planning and operations.
 - d.** It reduces memory consumption.

Answers to the “Do I Know This Already?” quiz:

- 1 D
- 2 C, D
- 3 B, C, D
- 4 C
- 5 B
- 6 C

Foundation Topics

Comparing Dynamic Routing Protocol Features

Routers add IP routes to their routing tables using three methods: connected routes, static routes, and routes learned by using dynamic routing protocols. Before we get too far into the discussion, however, it is important to define a few related terms and clear up any misconceptions about the terms *routing protocol*, *routed protocol*, and *routable protocol*. These terms are generally defined as follows:

- **Routing protocol:** A set of messages, rules, and algorithms used by routers for the overall purpose of learning routes. This process includes the exchange and analysis of routing information. Each router chooses the best route to each subnet (path selection) and finally places those best routes in its IP routing table. Examples include RIP, EIGRP, OSPF, and BGP.
- **Routed protocol and routable protocol:** Synonyms, both terms refer to a protocol that defines packets that can be routed (forwarded) by a router. Routers forward packets defined by routed protocols. Examples include IP Version 4 (IPv4) and IP Version 6 (IPv6).

Note

The term *path selection* sometimes refers to part of the job of a routing protocol, in which the routing protocol chooses the best route.

Even though routing protocols (such as OSPF) are different from routed protocols (such as IP), they do work together very closely. The routing process forwards IP packets, but if a router does not have any routes in its IP routing table that match a packet's destination address, the router discards the packet. Routers need routing protocols so that the routers can learn all the possible routes and add them to the routing table so that the routing process can forward (route) routable protocols such as IP.

Routing Protocol Functions

Cisco IOS software supports several IP routing protocols, performing the same general functions:

1. Learn routing information about IP subnets from neighboring routers.



2. Advertise routing information about IP subnets to neighboring routers.
3. If more than one possible route exists to reach one subnet, pick the best route based on a **metric**.
4. If the network topology changes—for example, a link fails—react by advertising that some routes have failed

and pick a new currently best route. (This process is called **convergence**.)

Note

A neighboring router connects to the same link as another router, such as the same WAN link or the same Ethernet LAN.

Figure 21-1 shows an example of three of the four functions in the list. Router R1, in the lower left of the figure, must choose the best route to reach the subnet connected off Router R2, on the bottom right of the figure. Following the steps in the figure:

- Step 1.** R2 advertises a route to the lower right subnet—172.16.3.0/24—to both Router R1 and R3.
- Step 2.** After R3 learns about the route to 172.16.3.0/24 from R2, R3 advertises that route to R1.
- Step 3.** R1 must choose between the two routes it learned about for reaching subnet 172.16.3.0/24—one with metric 1 from R2 and one with metric 2 from R3. R1 chooses the lower metric route through R2 (function 3).

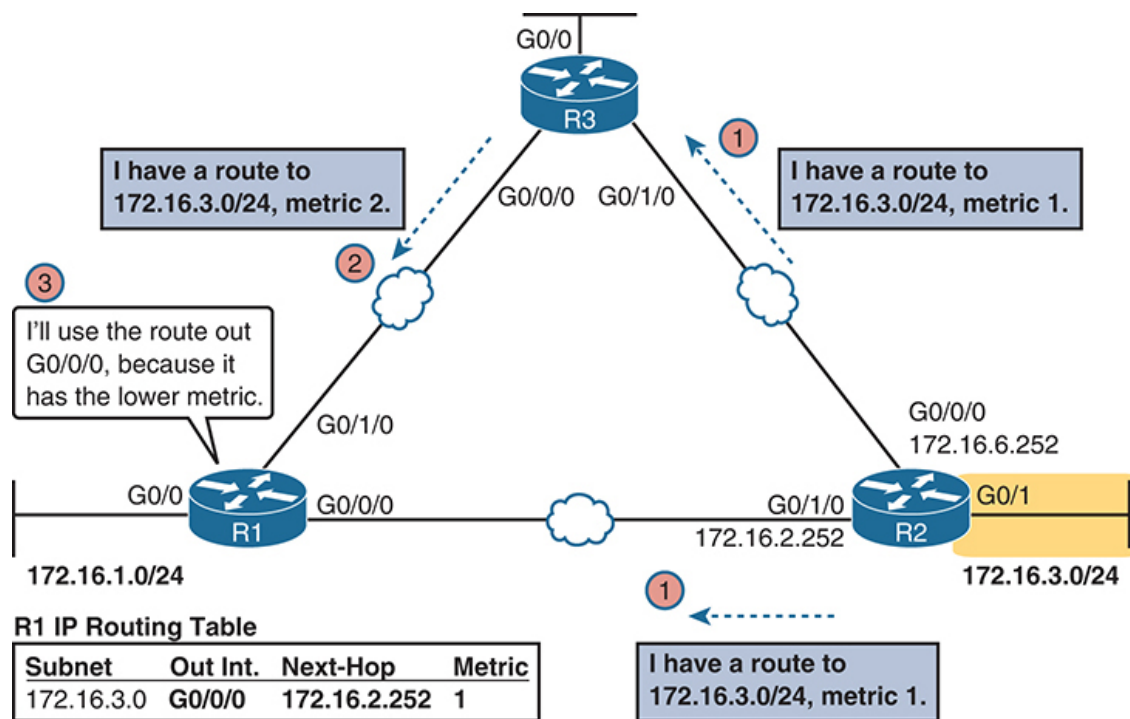


Figure 21-1 Three of the Four Basic Functions of Routing Protocols

The other routing protocol function, *convergence*, occurs when the topology changes—that is, when either a router or link fails or comes back up again. When something changes, the best routes available in the network can change. Convergence simply refers to the process by which all the routers collectively realize something has changed, advertise the information about the changes to all the other routers, and all the routers then choose the currently best routes for each subnet. The ability to converge quickly, without causing loops, is one of the most important considerations when choosing which IP routing protocol to use.

In [Figure 21-1](#), convergence might occur if the link between R1 and R2 failed. In that case, R1 should stop using its old route for subnet 172.16.3.0/24 (directly through R2) and begin sending packets to R3.

Interior and Exterior Routing Protocols

IP routing protocols fall into one of two major categories: **interior gateway protocols (IGP)** or *exterior gateway protocols* (EGP). The definitions of each are as follows:

- **IGP:** A routing protocol that was designed and intended for use inside a single autonomous system (AS)



- **EGP:** A routing protocol that was designed and intended for use between different autonomous systems

Note

The terms *IGP* and *EGP* include the word *gateway* because routers used to be called gateways.

These definitions use another new term: *autonomous system* (AS). An AS is a network under the administrative control of a single organization. For example, a network created and paid for by a single company is probably a single AS, and a network created by a single school system is probably a single AS. Other examples include large divisions of a state or national government, where different government agencies might be able to build their own networks. Each ISP is also typically a single different AS.

Some routing protocols work best inside a single AS by design, so these routing protocols are called IGP. Conversely, routing protocols designed to exchange routes

between routers in different autonomous systems are called EGPs. Today, Border Gateway Protocol (BGP) is the only EGP.

Each AS can be assigned a number called (unsurprisingly) an *AS number* (ASN). Like public IP addresses, the Internet Assigned Numbers Authority (IANA, www.iana.org) controls the worldwide rights to assigning ASNs. It delegates that authority to other organizations around the world, typically to the same organizations that assign public IP addresses. For example, in North America, the American Registry for Internet Numbers (ARIN, www.arin.net) assigns public IP address ranges and ASNs.

Figure 21-2 shows a small view of the worldwide Internet. The figure shows two enterprises and three ISPs using IGPs (OSPF and EIGRP) inside their own networks and with BGP being used between the ASNs.

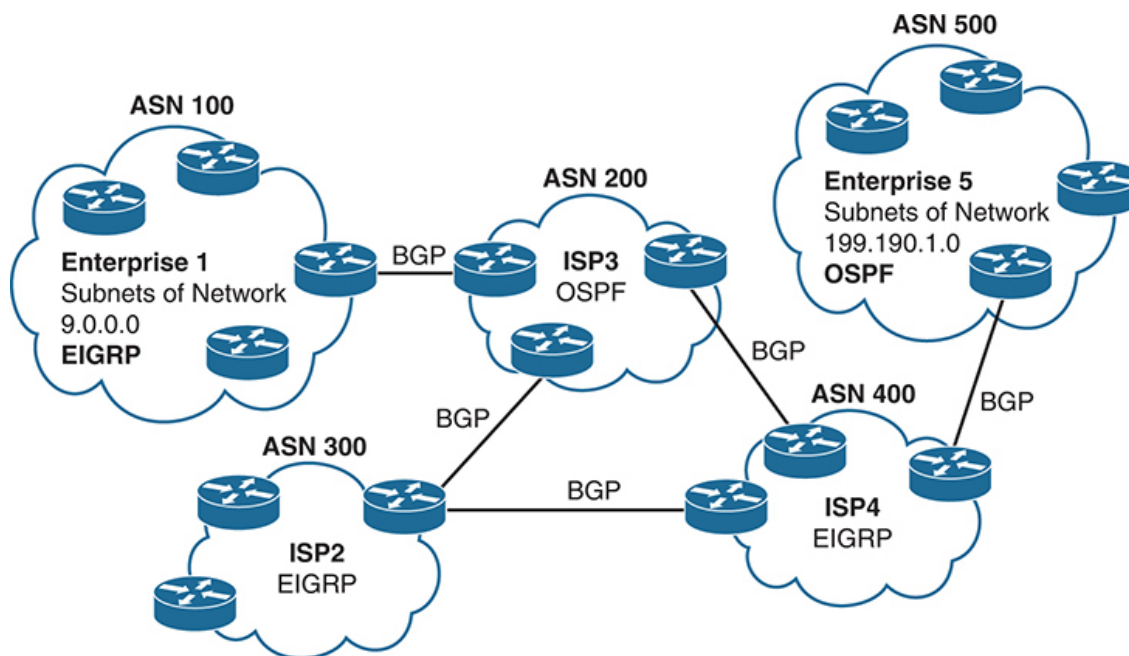


Figure 21-2 Comparing Locations for Using IGPs and EGPs

Comparing IGPs

Before getting into the details of OSPF, it helps to consider some comparisons between OSPF and the other IGP options. This section takes a brief look at all three routing protocols.

IGP Routing Protocol Algorithms

A routing protocol's underlying algorithm determines how the routing protocol does its job. The term *routing protocol algorithm* simply refers to the logic and processes used by different routing protocols to solve the problem of learning all routes, choosing the best route to each subnet, and converging in reaction to changes in the internetwork. Three main branches of routing protocol algorithms exist for IGP routing protocols:

- **Distance vector** (sometimes called Bellman-Ford after its creators)



- Advanced distance vector (sometimes called balanced hybrid)
- **Link-state**

Historically speaking, distance vector protocols were invented first, mainly in the early 1980s. Routing Information Protocol (RIP) was the first popularly used IP distance vector protocol, with the Cisco-proprietary Interior Gateway Routing Protocol (IGRP) being introduced a little later.

By the early 1990s, distance vector protocols' somewhat slow convergence and potential for routing loops drove the development of new alternative routing protocols that used new algorithms. Link-state protocols—in particular, Open Shortest Path First (OSPF) and Integrated Intermediate

System to Intermediate System (IS-IS)—solved the main issues. They also came with a price: they required extra CPU and memory on routers, with more planning required from the network engineers.

Note

All references to OSPF in this chapter refer to OSPF Version 2 (OSPFv2) unless otherwise stated.

Around the same time as the introduction of OSPF, Cisco created a proprietary routing protocol called Enhanced Interior Gateway Routing Protocol (EIGRP), which used some features of the earlier IGRP protocol. EIGRP solved the same problems as did link-state routing protocols, but EIGRP required less planning and less CPU/RAM overhead. As time went on, EIGRP was classified as a unique type of routing protocol. However, it used more distance vector features than link-state, so the industry refers to its algorithm as either an advanced distance vector protocol or as a balanced hybrid protocol.

Metrics

Routing protocols choose the best route to reach a subnet by choosing the route with the lowest metric. For example, RIP uses a counter of the number of routers (hops) between a router and the destination subnet, as shown in the example of [Figure 21-1](#). OSPF totals the cost associated with each interface in the end-to-end route, with the cost based on link bandwidth. [Table 21-2](#) lists the most common IP routing protocols and some details about the metric in each case.



Table 21-2 IP IGP Metrics

IGP	Metric	Description
RIPv2	Hop count	The number of routers (hops) between a router and the destination subnet
OSPF	Cost	The sum of all interface cost settings for all links in a route, with the cost default based on interface bandwidth
EIGRP	Calculation based on bandwidth and delay	Calculated based on the route's slowest link and the cumulative delay associated with each interface in the route

A brief comparison of the metric used by the older RIP versus the metric used by OSPF shows some insight into why OSPF and EIGRP surpassed RIP. [Figure 21-3](#) shows an example in which Router B has two possible routes to subnet 10.1.1.0 on the left side of the network: a shorter route over a very slow serial link at 1544 Kbps, or a longer route over two Gigabit Ethernet WAN links.

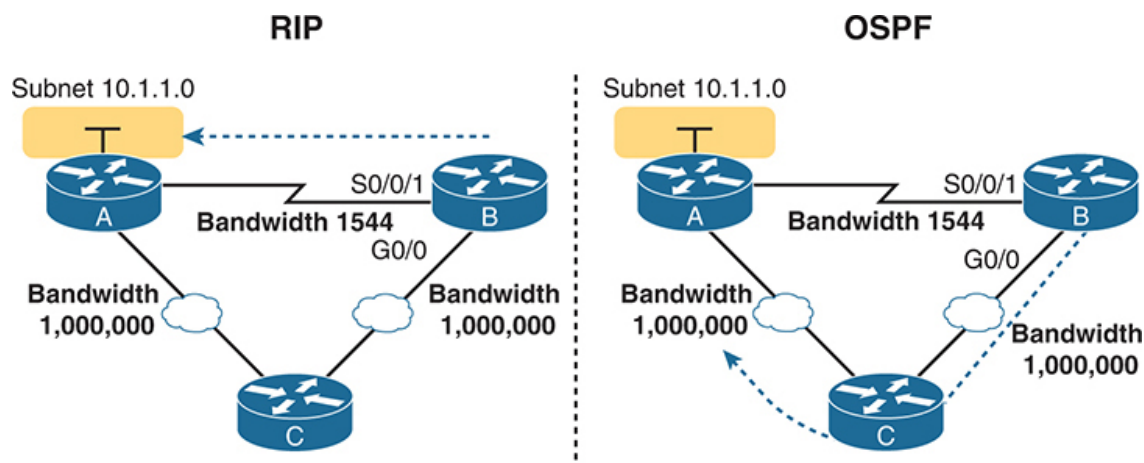


Figure 21-3 *RIP and OSPF Metrics Compared*

The left side of the figure shows the results of RIP in this network. Using hop count, Router B learns of a one-hop route directly to Router A through B's S0/0/1 interface. B also learns of a two-hop route through Router C, through B's G0/0 interface. Router B chooses the lower hop count route, which happens to go over the slow-speed serial link.

The right side of the figure shows the better choice made by OSPF based on its better metric. To cause OSPF to make the right choice, the engineer could use default settings based on the correct interface bandwidth to match the actual link speeds, thereby allowing OSPF to choose the faster route. (The **bandwidth** interface subcommand does not change the actual physical speed of the interface. It just tells IOS what speed to assume the interface is using.)

Other IGP Comparisons

Routing protocols can be compared based on many features, some of which matter to the current CCNA exam, whereas some do not. [Table 21-3](#) introduces a few more points and lists the comparison points mentioned in this book for easier study, with a few supporting comments following the table.

Table 21-3 Interior IP Routing Protocols Compared

Feature	RIPv2	EIGRP	OSPF
Classless/sends mask in updates/supports VLSM	Yes	Yes	Yes
Algorithm (DV, advanced DV, LS)	DV	Advanced DV	LS
Supports manual summarization	Yes	Yes	Yes

Feature	RIPv2	EIGRP	OSPF
Cisco-proprietary	No	Yes*	No
Routing updates are sent to a multicast IP address	Yes	Yes	Yes
Convergence	Slow	Fast	Fast
Multicast addresses used	224.0.0.9	224.0.0.10	224.0.0.5, 224.0.0.6

* Although Cisco created EIGRP and has kept it as a proprietary protocol for many years, Cisco chose to publish EIGRP as an informational RFC in 2013. This allows other vendors to implement EIGRP, while Cisco retains the rights to the protocol.

Regarding the top row of the table, classless routing protocols support variable-length subnet masks (VLSM) as well as manual route summarization by sending routing protocol messages that include the subnet masks in the message. The older RIPv1 and IGRP routing protocols—both classful routing protocols—do not.

Also, note that the older routing protocols (RIPv1, IGRP) sent routing protocol messages to IP broadcast addresses, while the newer routing protocols in the table all use IP multicast destination addresses. The use of multicasts makes the protocol more efficient and causes less overhead and fewer issues with the devices in the subnet that are not running the routing protocol.

OSPF Concepts and Operation

Routing protocols basically exchange information so routers can learn routes. The routers learn information about subnets, routes to those subnets, and metric information about how good each route is compared to others. The

routing protocol can then choose the currently best route to each subnet, building the IP routing table.

Link-state protocols like OSPF take a little different approach to the particulars of what information they exchange and what the routers do with that information once learned. This next (second) major section narrows the focus to only link-state protocols, specifically OSPFv2.

This section begins with an overview of what OSPF does by exchanging data about the network in data structures called **link-state advertisements (LSAs)**. Then the discussion backs up a bit to provide more details about each of three fundamental parts of how OSPF operates: how OSPF routers use *neighbor* relationships, how routers exchange LSAs with neighbors, and then how routers calculate the best routes once they learn all the LSAs.

OSPF Overview

Link-state protocols build IP routes with a couple of major steps. First, the routers together build a lot of information about the network: routers, links, IP addresses, status information, and so on. Then the routers flood the information, so all routers know the same information. At that point, each router can calculate routes to all subnets, but from each router's own perspective.

Topology Information and LSAs

Routers using link-state routing protocols need to collectively advertise practically every detail about the internetwork to all the other routers. At the end of the process of *flooding* the information to all routers, every router in the internetwork has the exact same information about the internetwork. Flooding a lot of detailed information to every router sounds like a lot of work, and relative to distance vector routing protocols, it is.

Open Shortest Path First (OSPF), the most popular link-state IP routing protocol, organizes topology information using LSAs and the **link-state database (LSDB)**. Figure 21-4 represents the ideas. Each LSA is a data structure with some specific information about the network topology; the LSDB is simply the collection of all the LSAs known to a router.

Link-State Database (LSDB)

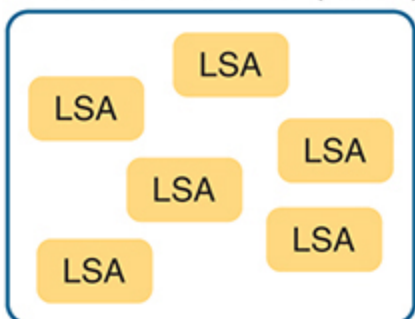


Figure 21-4 LSA and LSDB Relationship

Figure 21-5 shows the general idea of the flooding process, with R8 creating and flooding its *router LSA*. The router LSA for Router R8 describes the router itself, including the existence of subnet 172.16.3.0/24, as seen on the right side of the figure. (Note that Figure 21-5 shows only a subset of the information in R8's router LSA.)

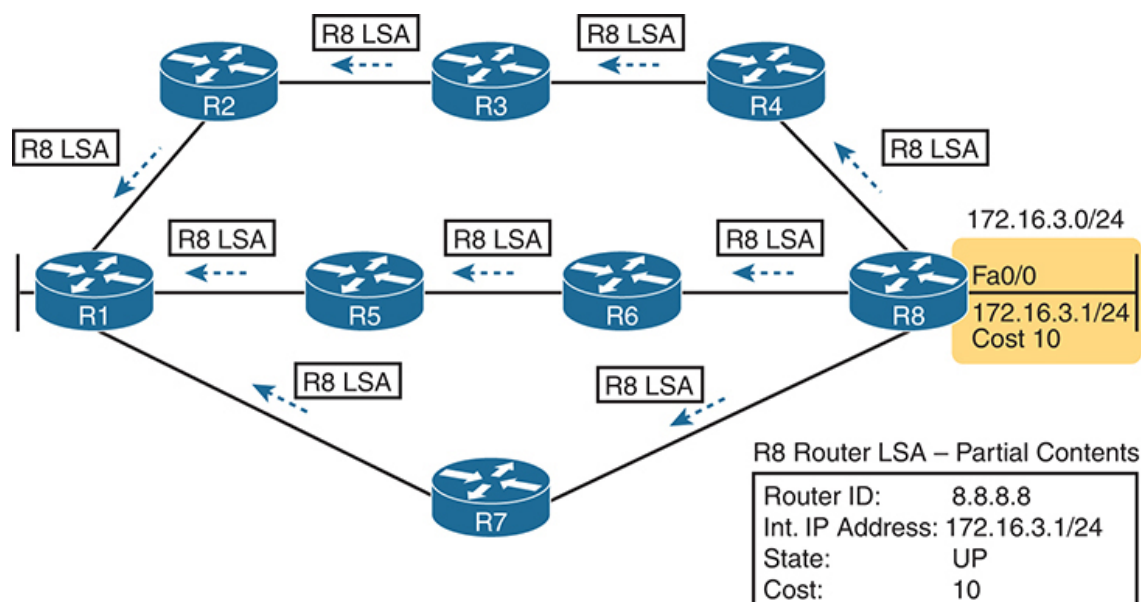


Figure 21-5 *Flooding LSAs Using a Link-State Routing Protocol*

Figure 21-5 shows the rather basic flooding process, with R8 sending the original LSA for itself, and the other routers flooding the LSA by forwarding it until every router has a copy. The flooding process causes every router to learn the contents of the LSA while preventing the LSA from being flooded around in circles. Basically, before sending an LSA to yet another neighbor, routers communicate, asking “Do you already have this LSA?” and then sending the LSA to the next neighbor only if the neighbor has not yet learned about the LSA.

Once flooded, routers do occasionally reflood each LSA. Routers reflood an LSA when some information changes (for example, when a link goes up or comes down). They also reflood each LSA based on each LSA’s separate aging timer (default 30 minutes).

Applying Dijkstra SPF Math to Find the Best Routes

The link-state flooding process results in every router having an identical copy of the LSDB in memory, but the flooding process alone does not cause a router to learn what routes to add to the IP routing table. Although incredibly detailed and useful, the information in the LSDB does not explicitly state each router’s best route to reach a destination.

To build routes, link-state routers have to do some math. Thankfully, you and I do not have to know the math! However, all link-state protocols use a type of math algorithm, called the Dijkstra **Shortest Path First (SPF) algorithm**, to process the LSDB. That algorithm analyzes (with math) the LSDB and builds the routes that the local router should add to the IP routing table—routes that list a

subnet number and mask, an outgoing interface, and a next-hop router IP address.

Now that you have the big ideas down, the next several topics walk through the three main phases of how OSPF routers accomplish the work of exchanging LSAs and calculating routes. Those three phases are

Becoming neighbors: A relationship between two routers that connect to the same data link, created so that the neighboring routers have a means to exchange their LSDBs.

Exchanging databases: The process of sending LSAs to neighbors so that all routers learn the same LSAs.

Adding the best routes: The process of each router independently running SPF, on their local copy of the LSDB, calculating the best routes, and adding those to the IPv4 routing table.

Becoming OSPF Neighbors

Of everything you learn about OSPF in this chapter, OSPF neighbor concepts have the most to do with how you will configure and troubleshoot OSPF in Cisco routers. You configure OSPF to cause routers to run OSPF and become neighbors with other routers. Once that happens, OSPF does the rest of the work to exchange LSAs and calculate routes in the background, with no additional configuration required. This section discusses the fundamental concepts of OSPF neighbors.

The Basics of OSPF Neighbors

Two routers must meet some compatibility requirements to become neighbors. First, they must both use OSPF and both connect to the same data link. Two routers can become

OSPF neighbors if connected to the same VLAN, or same serial link, or same Ethernet WAN link.

Additionally, the two routers must send OSPF messages that declare some OSPF settings, and those settings must be compatible. To do so, the routers send OSPF Hello messages, introducing themselves to the potential neighbor. Assuming the two potential neighbors have compatible OSPF parameters, the two form an OSPF neighbor relationship, and would be displayed in the output of the **show ip ospf neighbor** command.

The OSPF neighbor relationship also lets OSPF know when a neighbor might not be a good option for routing packets right now. Imagine R1 and R2 form a neighbor relationship, learn LSAs, and calculate routes that send packets through the other router. Months later, R1 notices that the neighbor relationship with R2 fails. That failed neighbor connection to R2 makes R1 react: R1 refloods LSAs impacted by the failed link, and R1 runs SPF to recalculate its own routes.

Finally, the OSPF neighbor model allows new routers to be dynamically discovered. That means new routers can be added to a network without requiring every router to be reconfigured. Instead, OSPF routers listen for OSPF Hello messages from new routers and react to those messages, attempting to become neighbors and exchange LSDBs.

Meeting Neighbors and Learning Their Router ID

The OSPF neighbor relationship begins by exchanging OSPF *Hello* messages, which list each router's **router ID (RID)**. OSPF RIDs are 32-bit numbers, so most command output lists these as dotted-decimal numbers (DDN). By default, IOS chooses one of the router interface's IPv4 addresses to use as its OSPF RID. However, the OSPF RID can be directly configured, as covered in the section "[Configuring the OSPF](#)

Router ID” in [Chapter 22, “Implementing Basic OSPF Features.”](#)

As soon as a router has chosen its OSPF RID and some interfaces come up, the router is ready to meet its OSPF neighbors. OSPF routers can become neighbors if they are connected to the same subnet. To discover other OSPF-speaking routers, a router sends multicast OSPF Hello packets to each interface and hopes to receive OSPF Hello packets from other routers connected to those interfaces. [Figure 21-6](#) outlines the basic concept.

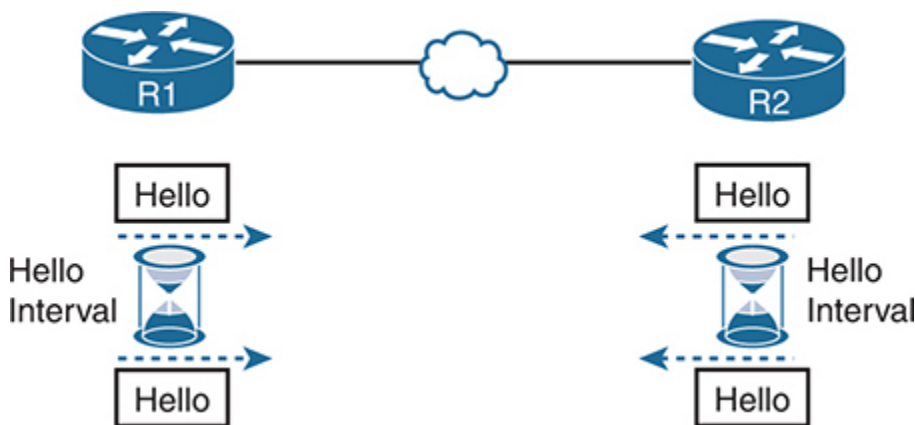


Figure 21-6 *OSPF Hello Packets*

Routers R1 and R2 both send Hello messages onto the link. They continue to send Hellos at a regular interval based on their Hello timer settings. The Hello messages themselves have the following features:

- The Hello message follows the IP packet header, with IP protocol type 89.
- Hello packets are sent to multicast IP address 224.0.0.5, a multicast IP address intended for all OSPF-speaking routers.
- OSPF routers listen for packets sent to IP multicast address 224.0.0.5, in part hoping to receive Hello packets and learn about new neighbors.

Taking a closer look, [Figure 21-7](#) shows several of the neighbor states used by the early formation of an OSPF neighbor relationship. The figure shows the Hello messages in the center and the resulting neighbor states on the left and right edges of the figure. Each router keeps an OSPF state variable for how it views the neighbor.

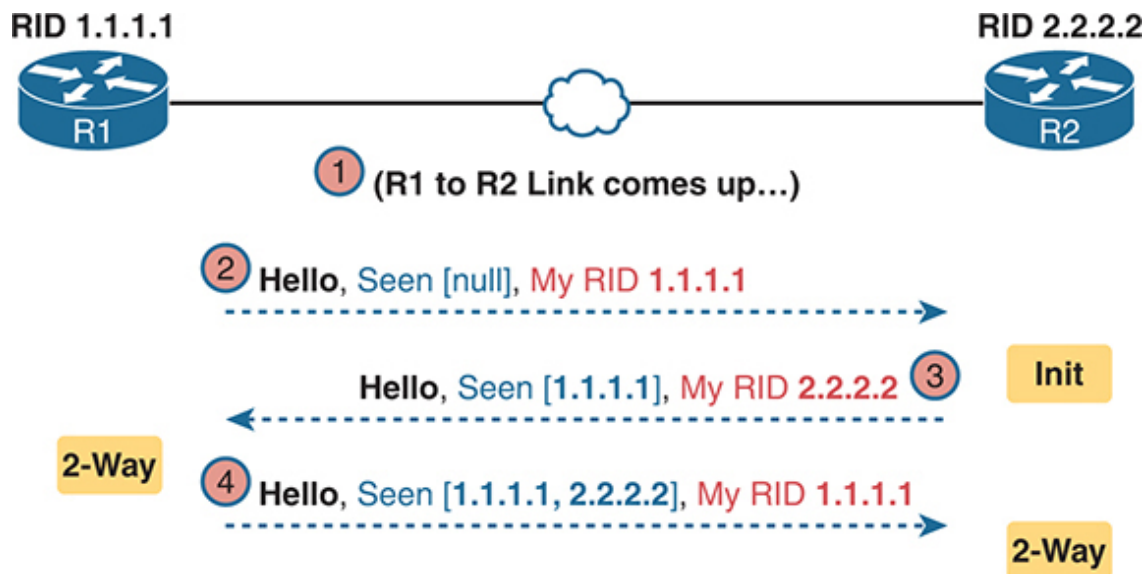


Figure 21-7 *Early Neighbor States*

Following the steps in the figure, the scenario begins with the link down, so the routers have no knowledge of each other as OSPF neighbors. As a result, they have no state (status) information about each other as neighbors, and they would not list each other in the output of the **show ip ospf neighbor** command. At Step 2, R1 sends the first Hello, so R2 learns of the existence of R1 as an OSPF router. At that point, R2 lists R1 as a neighbor, with an interim beginning state of init.

The process continues at Step 3, with R2 sending back a Hello. This message tells R1 that R2 exists, and it allows R1 to move through the init state and quickly to a 2-way state. At Step 4, R2 receives the next Hello from R1, and R2 can also move to a 2-way state.

The **2-way state** is a particularly important OSPF state. At that point, the following major facts are true:

- The router received a Hello from the neighbor, with that router's own RID listed as being seen by the neighbor.



- The router has performed all checks of settings in the Hello and considers the potential neighbor to have passed all checks so they can become neighbors.
- If both routers reach a 2-way state with each other, they are neighbors and ready to exchange their LSDB with each other.

Exchanging the LSDB Between Neighbors

One purpose of forming OSPF neighbor relationships is to allow the two neighbors to exchange their databases. This next topic works through some of the details of OSPF database exchange.

Fully Exchanging LSAs with Neighbors

Once two routers on a link reach the 2-way state, they can immediately move on to the process of database exchange. The database exchange process can be quite involved, with several OSPF messages and several interim neighbor states. This chapter is more concerned with a few of the messages and the final state when database exchange has completed: the **full state**.

After two routers decide to exchange databases, they do not simply send the contents of the entire database. First, they tell each other a list of LSAs in their respective databases—not all the details of the LSAs, just a list. (Think of these lists

as checklists.) Next, each router can check which LSAs it already has and then ask the other router for only the LSAs that are not known yet.

For instance, R1 might send R2 a checklist that lists ten LSAs (using an OSPF Database Description, or DD, packet). R2 then checks its LSDB and finds six of those ten LSAs. So, R2 asks R1 (using a Link-State Request packet) to send the four additional LSAs.

Thankfully, most OSPFv2 work does not require detailed knowledge of these specific protocol steps. However, a few of the terms are used quite a bit and should be remembered. In particular, the OSPF messages that actually send the LSAs between neighbors are called **link-state update** (LSU) packets. That is, the LSU packet holds data structures called *link-state advertisements (LSAs)*. The LSAs are not packets, but rather data structures that sit inside the LSDB and describe the topology.

[Figure 21-8](#) pulls some of these terms and processes together, with a general example. The story picks up the example shown in [Figure 21-7](#), with [Figure 21-8](#) showing an example of the database exchange process between Routers R1 and R2. The center shows the protocol messages, and the outer items show the neighbor states at different points in the process. Focus on two items in particular:

- The routers exchange the LSAs inside LSU packets.
- When finished, the routers reach a full state, meaning they have fully exchanged the contents of their LSDBs.

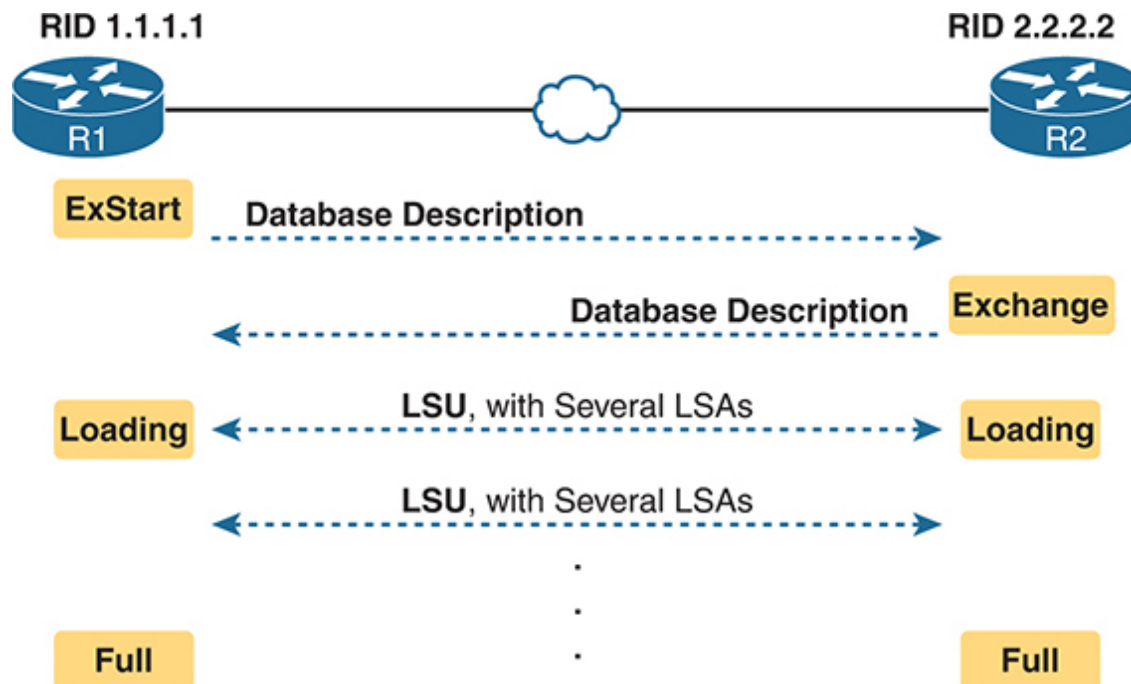


Figure 21-8 Database Exchange Example, Ending in a Full State

Maintaining Neighbors and the LSDB

Once two neighbors reach a full state, they have done all the initial work to exchange OSPF information between them. However, neighbors still have to do some small ongoing tasks to maintain the neighbor relationship.

First, routers monitor each neighbor relationship using Hello messages and two related timers: the **Hello interval** and the **Dead interval**. Routers send Hellos every Hello interval to each neighbor. Each router expects to receive a Hello from each neighbor based on the Hello interval, so if a neighbor is silent for the length of the Dead interval (by default, four times as long as the Hello interval), the loss of Hellos means that the neighbor has failed.

Next, routers must react when the topology changes as well, and neighbors play a key role in that process. When something changes, one or more routers change one or

more LSAs. Then the routers must flood the changed LSAs to each neighbor so that the neighbor can change its LSDB.

For example, imagine a LAN switch loses power, so a router's G0/0 interface fails from up/up to down/down. That router updates an LSA that shows the router's G0/0 as being down. That router then sends the LSA to its neighbors, and that neighbor in turn sends it to its neighbors, until all routers again have an identical copy of the LSDB. Each router's LSDB now reflects the fact that the original router's G0/0 interface failed, so each router will then use SPF to recalculate any routes affected by the failed interface.

A third maintenance task done by neighbors is to reflood each LSA occasionally, even when the network is completely stable. By default, each router that creates an LSA also has the responsibility to reflood the LSA every 30 minutes (the default), even if no changes occur. (Note that each LSA has a separate timer, based on when the LSA was created, so there is no single big event where the network is overloaded with flooding LSAs.)

The following list summarizes these three maintenance tasks for easier review:

- Maintain neighbor state by sending Hello messages based on the Hello interval and listening for Hellos before the Dead interval expires
- Flood any changed LSAs to each neighbor
- Reflood unchanged LSAs as their lifetime expires (default 30 minutes)

Using Designated Routers on Ethernet Links

OSPF behaves differently on some types of interfaces based on a per-interface setting called the OSPF *network type*. On Ethernet links, OSPF defaults to use a network type of

broadcast, which causes OSPF to elect one of the routers on the same subnet to act as the **designated router (DR)**. The DR plays a key role in how the database exchange process works, with different rules than with point-to-point links.

To see how, consider the example that begins with [Figure 21-9](#). The figure shows five OSPFv2 routers on the same Ethernet VLAN. These five OSPF routers elect one router to act as the DR and one router to be a **backup designated router (BDR)**. The figure shows A and B as DR and BDR, for no other reason than the subnet with OSPF network type broadcast will have one of each.

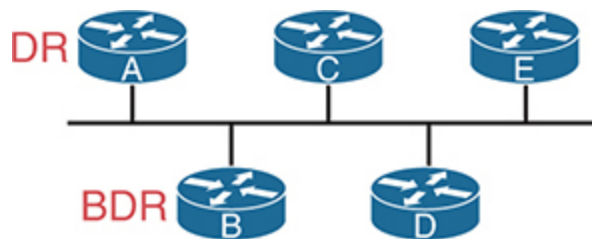


Figure 21-9 *Routers A and B Elected as DR and BDR*

The database exchange process on an Ethernet link does not happen between every pair of routers on the same VLAN/subnet. Instead, it happens between the DR and each of the other routers, with the DR making sure that all the other routers get a copy of each LSA. In other words, the database exchange happens over the flows shown in [Figure 21-10](#).

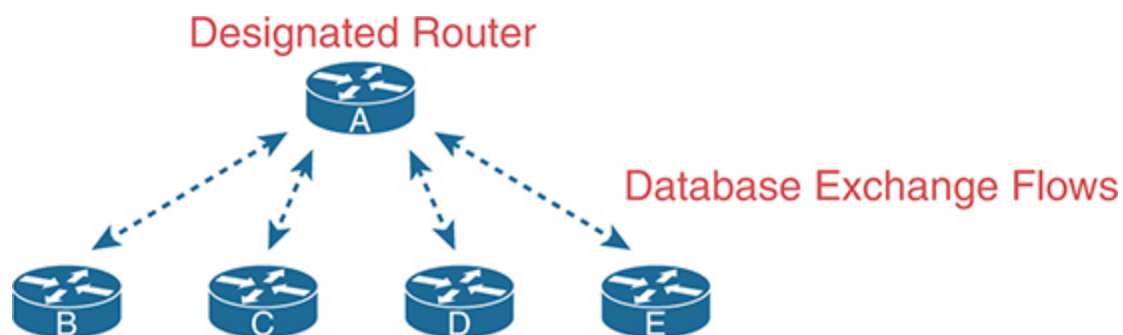


Figure 21-10 *Database Exchange to and from the DR on an Ethernet*

OSPF defines the backup designated router (BDR) role so that some router can take over for the DR should it fail. When the DR fails, the BDR takes over as DR, and some other router is elected as the new BDR.

The use of a DR/BDR, along with the use of multicast IP addresses, makes the exchange of OSPF LSDBs more efficient on networks that allow more than two routers on the same link. The DR can send a packet to all OSPF routers in the subnet by using multicast IP address 224.0.0.5. IANA reserves this address as the “All SPF Routers” multicast address just for this purpose. For instance, in [Figure 21-10](#), the DR can send one set of messages to all the OSPF routers rather than sending one message to each router.

Similarly, any OSPF router needing to send a message to the DR and also to the BDR (so it remains ready to take over for the DR) can send those messages to the “All SPF DRs” multicast address 224.0.0.6. So, instead of having to send one set of messages to the DR and another set to the BDR, an OSPF router can send one set of messages, making the exchange more efficient.

You will see quite a bit of the DR and BDR theory in **show** commands on a router. Because the DR and BDR both do full database exchange with all the other OSPF routers in the LAN, they reach a full state with all neighbors. However, routers that are neither a DR nor a BDR—called *DROthers* by OSPF—never reach a full state because they do not exchange LSDBs directly with each other. As a result, the **show ip ospf neighbor** command on these DROther routers lists some neighbors in a 2-way state, remaining in that state under normal operation.

For instance, with OSPF working normally on the Ethernet LAN in [Figure 21-10](#), a **show ip ospf neighbor** command on Router C (which is a DROther router) would show the following:

- Two neighbors (A and B, the DR and BDR, respectively) with a full state (called *fully adjacent neighbors*)
- Two neighbors (D and E, which are DROthers) with a 2-way state (called **neighbors**)

OSPF requires some terms to describe all neighbors versus the subset of all neighbors that reach the full state. First, all OSPF routers on the same link that reach the 2-way state—that is, they send Hello messages and the parameters match—are called *neighbors*. The subset of neighbors for which the neighbor relationship reaches the full state are called *adjacent neighbors*. Additionally, OSPFv2 RFC 2328 emphasizes this point by defining two synonyms to the term *adjacent neighbor*: **fully adjacent** and *fully adjacent neighbor*. Finally, while the terms so far refer to the neighbor, two other terms refer to the relationship: *neighbor relationship* refers to any OSPF neighbor relationship, while the term *adjacency* refers to neighbor relationships that reach a full state. [Table 21-4](#) details the terms.



Table 21-4 Stable OSPF Neighbor States and Their Meanings

Neighbor State	Term for Neighbor	Term for Relationship
2-way	Neighbor	Neighbor Relationship

Neighbor State	Term for Neighbor	Term for Relationship
Full	Adjacent Neighbor Fully Adjacent Neighbor	Adjacency

Calculating the Best Routes with SPF

OSPF LSAs contain useful information, but they do not contain the specific information that a router needs to add to its IPv4 routing table. In other words, a router cannot just copy information from the LSDB into a route in the IPv4 routing table. The LSAs individually are more like pieces of a jigsaw puzzle, with the picture shown by the completed puzzle showing a topology map of the entire network. So, to know what routes to add to the routing table, each router must do some SPF math to choose the best routes from that router's perspective. The router then adds each route to its routing table: a route with a subnet number and mask, an outgoing interface, and a next-hop router IP address.

Although engineers do not need to know the details of how SPF does the math, they do need to know how to predict which routes SPF will choose as the best route. The SPF algorithm calculates all the routes for a subnet—that is, all possible routes from the router to the destination subnet. If more than one route exists, the router compares the metrics, picking the best (lowest) metric route to add to the routing table. Although the SPF math can be complex, engineers with a network diagram, router status information, and simple addition can calculate the metric for each route, predicting what SPF will choose.

Once SPF has identified a route, OSPF calculates the metric for a route as follows:

The sum of the OSPF interface costs for all outgoing interfaces in the route.

**Key
Topic**

Figure 21-11 shows an example with three possible routes from R1 to Subnet X (172.16.3.0/24) at the bottom of the figure.

**Key
Topic**

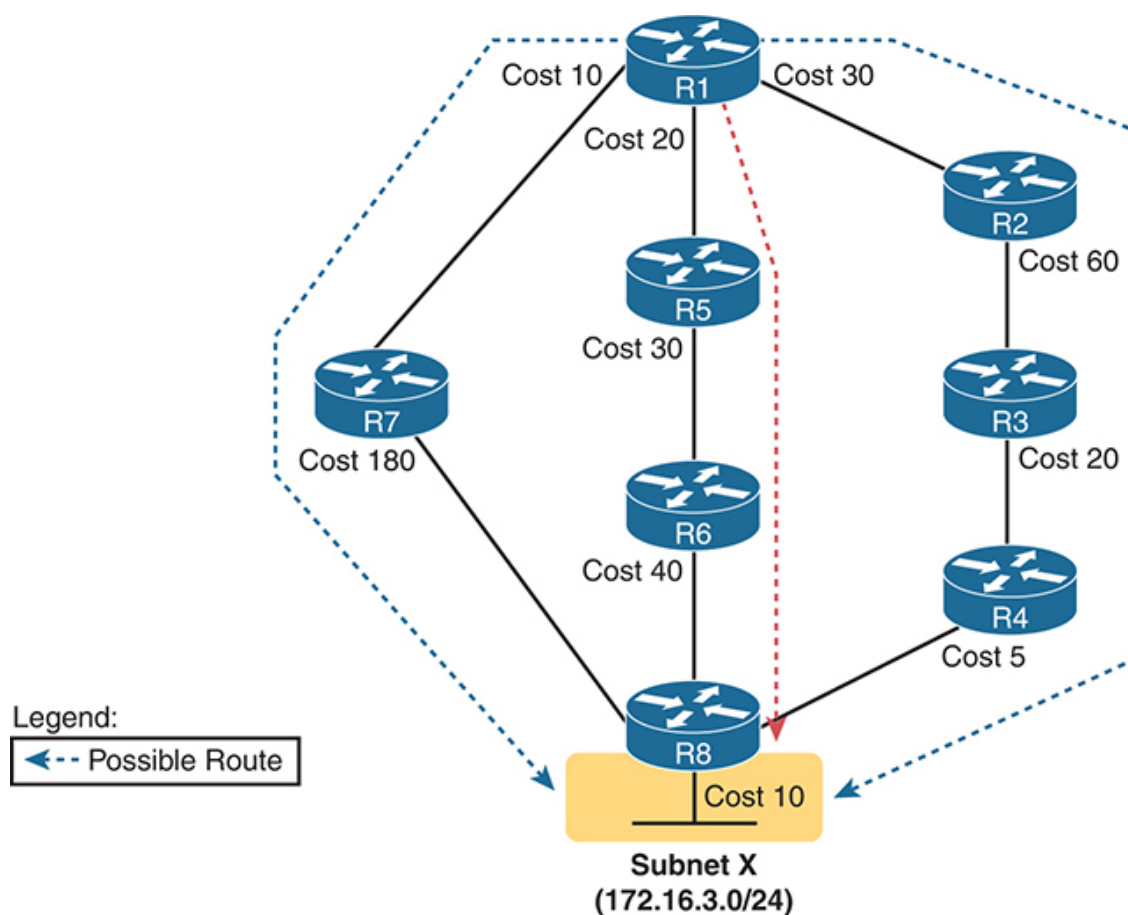


Figure 21-11 SPF Tree to Find R1's Route to 172.16.3.0/24

Note

OSPF considers the costs of the outgoing interfaces (only) in each route. It does not add the cost for incoming interfaces in the route.

Table 21-5 lists the three routes shown in Figure 21-11, with their cumulative costs, showing that R1's best route to 172.16.3.0/24 starts by going through R5.

Table 21-5 Comparing R1's Three Alternatives for the Route to 172.16.3.0/24

Route	Location in Figure 21-11	Cumulative Cost
R1-R7-R8	Left	$10 + 180 + 10 = 200$
R1-R5-R6-R8	Middle	$20 + 30 + 40 + 10 = 100$
R1-R2-R3-R4-R8	Right	$30 + 60 + 20 + 5 + 10 = 125$

As a result of the SPF algorithm's analysis of the LSDB, R1 adds a route to subnet 172.16.3.0/24 to its routing table, with the next-hop router of R5.

In real OSPF networks, an engineer can do the same process by knowing the OSPF cost for each interface. Armed with a network diagram, the engineer can examine all routes, add the costs, and predict the metric for each route.

OSPF Areas and LSAs

OSPF can be used in some networks with very little thought about design issues. You just turn on OSPF in all the routers,

put all interfaces into the same area (usually area 0), and it works! [Figure 21-12](#) shows one such network example, with 11 routers and all interfaces in area 0.

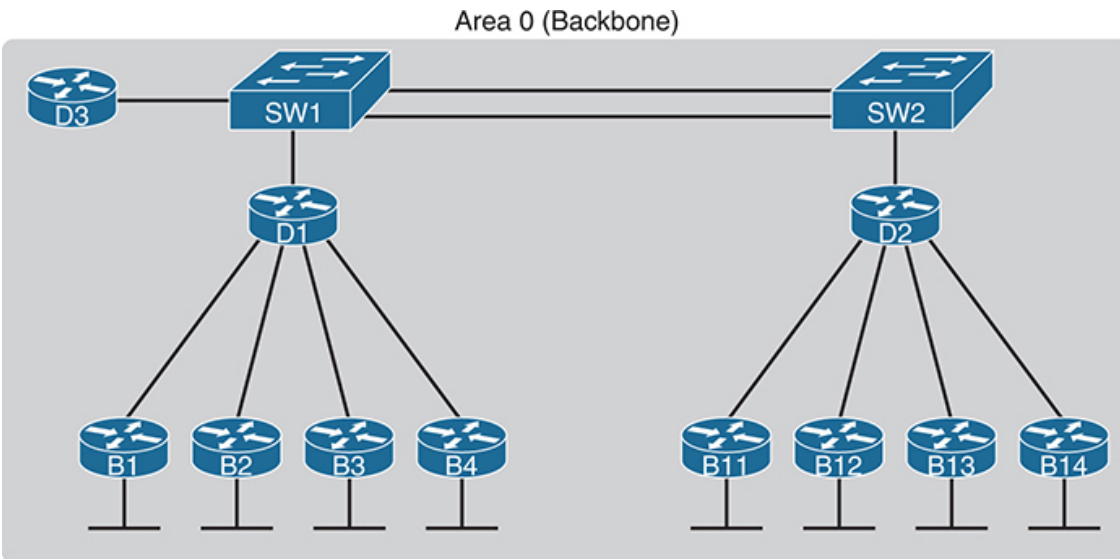


Figure 21-12 *Single-Area OSPF*

Using a single-area design works well in small- to medium-sized networks. In fact, the CCNA 200-301 V1.1 blueprint specifically mentions **single-area OSPF**, omitting **multiarea OSPF**. However, it helps to think through the concept of areas even for CCNA preparation. The next few pages look at how OSPF area design works, with more reasons as to why areas help make larger OSPF networks work better.

OSPF Areas

OSPF area design follows a couple of basic rules. To apply the rules, start with a clean drawing of the internetwork, with routers, and all interfaces. Then choose the area for each router interface, as follows:

- Put all interfaces connected to the same subnet inside the same area.



- An area should be contiguous.
- Some routers may be internal to an area, with all interfaces assigned to that single area.
- Some routers may be Area Border Routers (ABRs) because some interfaces connect to the backbone area, and some connect to nonbackbone areas.
- All nonbackbone areas must have a path to reach the backbone area (area 0) by having at least one ABR connected to both the backbone area and the nonbackbone area.

Figure 21-13 shows one example. An engineer started with a network diagram that showed all 11 routers and their links. On the left, the engineer put four WAN links and the LANs connected to branch routers B1 through B4 into area 1. Similarly, he placed the links to branches B11 through B14 and their LANs in area 2. Both areas need a connection to the backbone area, area 0, so he put the LAN interfaces of D1 and D2 into area 0, along with D3, creating the backbone area.



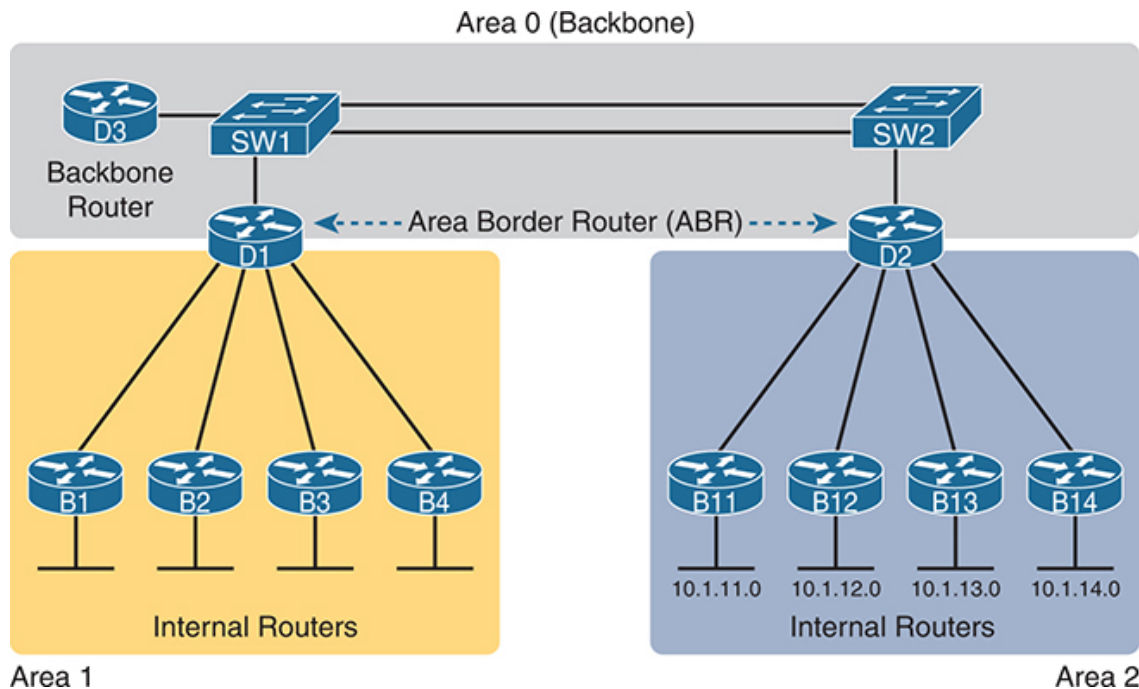


Figure 21-13 *Three-Area OSPF with D1 and D2 as ABRs*

The figure also shows a few important OSPF area design terms. [Table 21-6](#) summarizes the meaning of these terms, plus some other related terms, but pay closest attention to the terms from the figure.

**Key
Topic**

Table 21-6 OSPF Design Terminology

Term	Description
Area Border Router (ABR)	An OSPF router with interfaces connected to the backbone area and to at least one other area
<i>Backbone router</i>	A router connected to the backbone area (includes ABRs)

Term	Description
Internal router	A router in one area (not the backbone area)
Area	A set of routers and links that shares the same detailed LSDB information, but not with routers in other areas, for better efficiency
Backbone area	A special OSPF area to which all other areas must connect—area 0
Intra-area route	A route to a subnet inside the same area as the router
Interarea route	A route to a subnet in an area of which the router is not a part

How Areas Reduce SPF Calculation Time

[Figure 21-13](#) shows a sample area design and some terminology related to areas, but it does not show the power and benefit of the areas. To understand how areas reduce the work SPF has to do, you need to understand what changes about the LSDB inside an area, as a result of the area design.

SPF spends most of its processing time working through all the topology details, namely routers and the links that connect routers. Areas reduce SPF's workload because, for a given area, the LSDB lists only routers and links inside that area, as shown on the left side of [Figure 21-14](#).

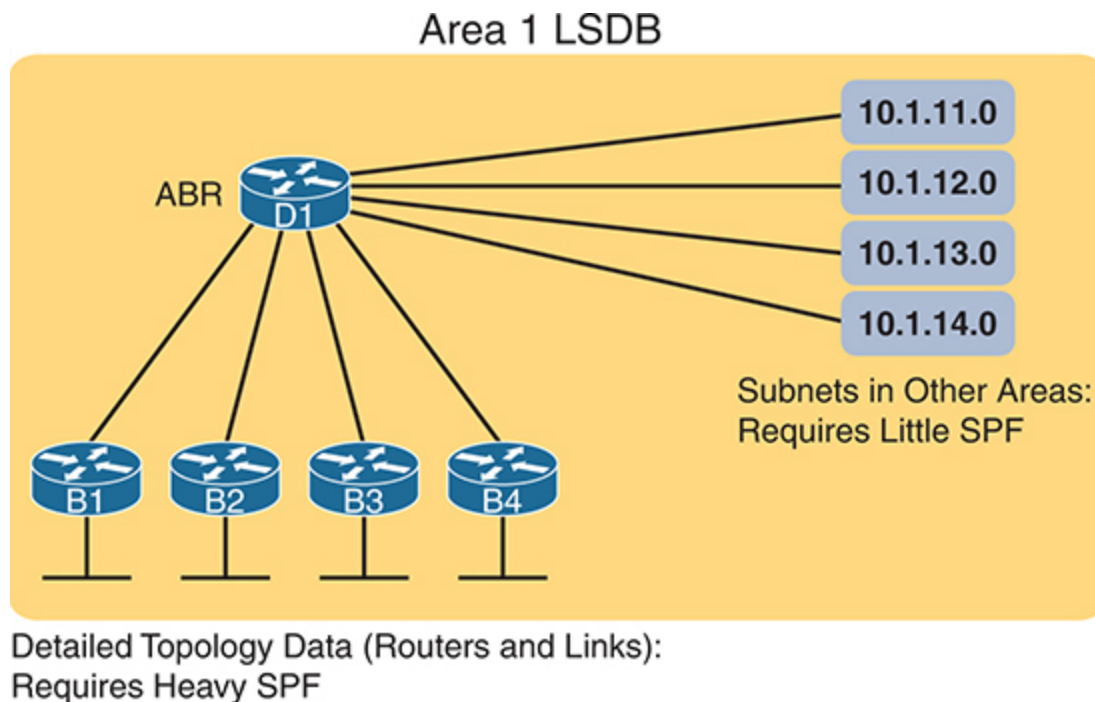


Figure 21-14 *Smaller Area 1 LSDB Concept*

While the LSDB has less topology information, it still needs information about all subnets in all areas so that each router can create IPv4 routes for all subnets. So, with an area design, OSPFv2 uses brief summary information about the subnets in other areas. These summary LSAs do not include topology information about the other areas; however, each summary LSA *does* list a subnet ID and mask of a subnet in some other area. Summary LSAs do not require SPF processing at all. Instead, these subnets all appear like subnets connected to the ABR (in [Figure 21-14](#), ABR D1).

Using multiple areas improves OSPF operations in many ways for larger networks. The following list summarizes some of the key points arguing for the use of multiple areas in larger OSPF networks:

- Routers require fewer CPU cycles to process the smaller per-area LSDB with the SPF algorithm, reducing CPU overhead and improving convergence time.
- The smaller per-area LSDB requires less memory.

- Changes in the network (for example, links failing and recovering) require SPF calculations only on routers in the area where the link changed state, reducing the number of routers that must rerun SPF.
- Less information must be advertised between areas, reducing the bandwidth required to send LSAs.

(OSPFv2) Link-State Advertisements

Many people tend to get a little intimidated by OSPF LSAs when first learning about them. Commands like **show ip ospf database** in its many variations list a lot of information about the LSDB. Those details appear to be in some kind of code, using lots of numbers. It can seem like a bit of a mess.

However, if you examine LSAs while thinking about OSPF areas and area design, some of the most common LSA types will make a lot more sense. For instance, think about the LSDB in one area. The topology in one area includes routers and the links between the routers. As it turns out, OSPF defines the first two types of LSAs to define those exact details, as follows:

- One *router LSA* for each router in the area
- One *network LSA* for each network that has a DR plus one neighbor of the DR

Next, think about the subnets in the other areas. The ABR creates summary information about each subnet in one area to advertise into other areas—basically just the subnet IDs and masks—as a third type of LSA:

- One *summary LSA* for each subnet ID that exists in a different area

The next few pages discuss these three LSA types in a little more detail; [Table 21-7](#) lists some information about all three for easier reference and study.

Table 21-7 The Three OSPFv2 LSA Types Seen with a Multiarea OSPF Design

LSA Name	LSA Type	Primary Purpose	Contents of LSA
Router	1	Describe a router	RID, interfaces, IP address/mask, current interface state (status)
Network	2	Describe a network that has a DR and BDR	DR and BDR IP addresses, subnet ID, mask
Summary	3	Describe a subnet in another area	Subnet ID, mask, RID of ABR that advertises the LSA

Router LSAs Build Most of the Intra-Area Topology

OSPF needs very detailed topology information inside each area. The routers inside area X need to know all the details about the topology inside area X. And the mechanism to give routers all these details is for the routers to create and flood router (Type 1) and network (Type 2) LSAs about the routers and links in the area.

Router LSAs, also known as Type 1 LSAs, describe the router in detail. Each lists a router's RID, its interfaces, its IPv4 addresses and masks, its interface state, and notes about what neighbors the router knows about via each of its interfaces.

To see a specific instance, first review [Figure 21-15](#). It lists internetwork topology, with subnets listed. Because it's a small internetwork, the engineer chose a single-area design, with all interfaces in backbone area 0.

With the single-area design planned for this small internetwork, the LSDB will contain four router LSAs. Each router creates a router LSA for itself, with its own RID as the LSA identifier. The LSA lists that router's own interfaces, IP address/mask, with pointers to neighbors.

Once all four routers have copies of all four router LSAs, SPF can mathematically analyze the LSAs to create a model. The model looks a lot like the concept drawing in [Figure 21-16](#). Note that the drawing shows each router with an obvious RID value. Each router has pointers that represent each of its interfaces, and because the LSAs identify neighbors, SPF can figure out which interfaces connect to which other routers.

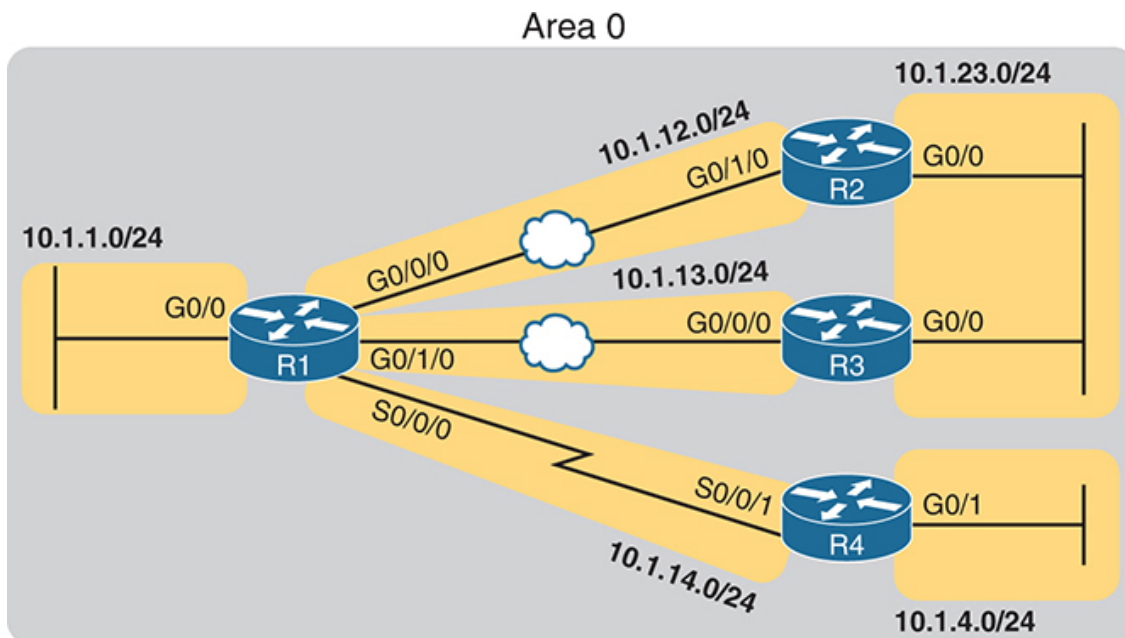


Figure 21-15 Enterprise Network with Six IPv4 Subnets

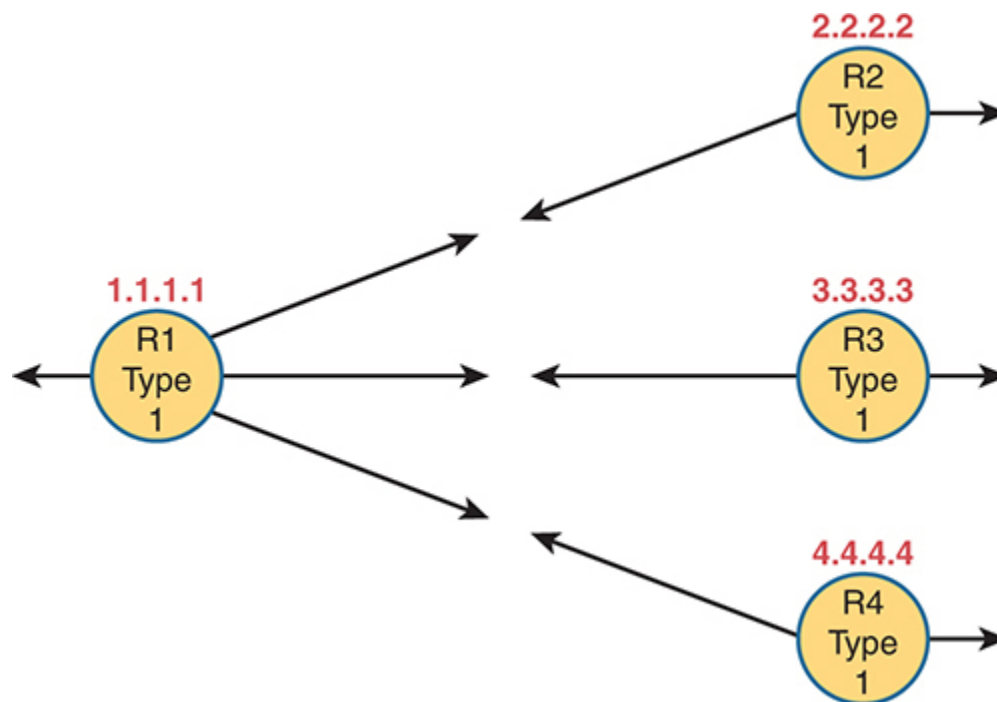


Figure 21-16 *Type 1 LSAs, Assuming a Single-Area Design*

Network LSAs Complete the Intra-Area Topology

Whereas router LSAs define most of the intra-area topology, network LSAs define the rest. As it turns out, when OSPF elects a DR on some subnet *and* that DR has at least one neighbor, OSPF treats that subnet as another node in its mathematical model of the network. To represent that network, the DR creates and floods a network (Type 2) LSA for that network (subnet).

For instance, back in [Figure 21-15](#), one Ethernet LAN and two Ethernet WANs exist. The Ethernet LAN between R2 and R3 will elect a DR, and the two routers will become neighbors; so, whichever router is the DR will create a network LSA. Similarly, R1 and R2 connect with an Ethernet WAN, so the DR on that link will create a network LSA.

Likewise, the DR on the Ethernet WAN link between R1 and R3 will also create a network LSA.

Figure 21-17 shows the completed version of the intra-area LSAs in area 0 with this design. Note that the router LSAs reference the network LSAs when they exist, which lets the SPF processes connect the pieces together.

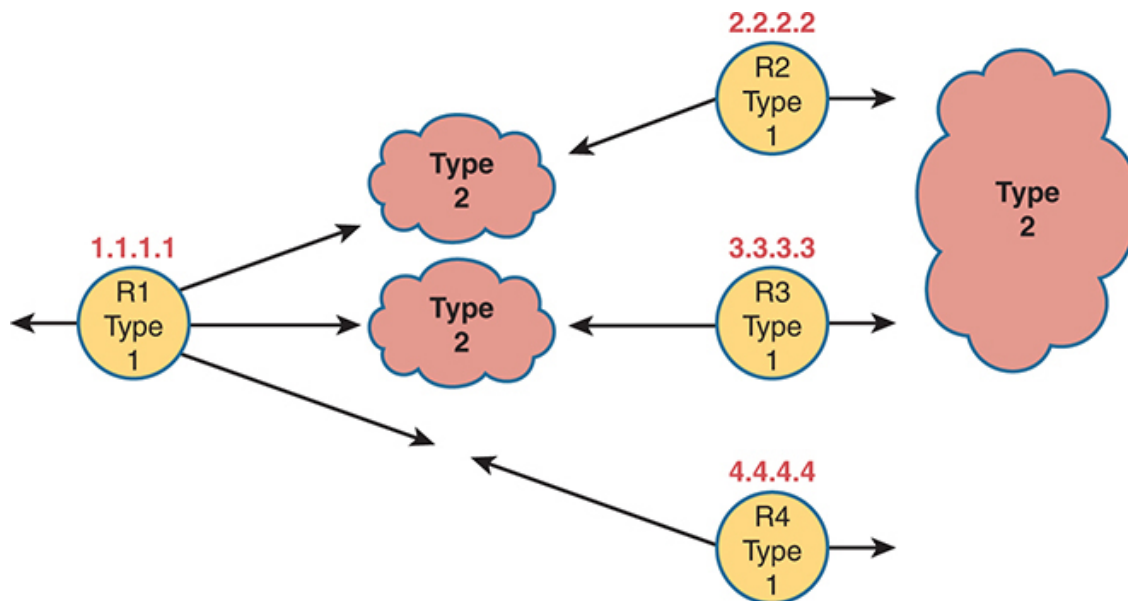


Figure 21-17 Type 1 and Type 2 LSAs in Area 0, Assuming a Single-Area Design

Finally, note that in this single-area design example, no summary (Type 3) LSAs exist at all. These LSAs represent subnets in other areas, and there are no other areas. Given that the CCNA 200-301 V1.1 exam blueprint refers specifically to single-area OSPF designs, this section stops at showing the details of the intra-area LSAs (Types 1 and 2).

Chapter Review

One key to doing well on the exams is to perform repetitive spaced review sessions. Review this chapter's material using either the tools in the book or interactive tools for the

same material found on the book’s companion website. Refer to the “[Your Study Plan](#)” element for more details. [Table 21-8](#) outlines the key review elements and where you can find them. To better track your study progress, record when you completed these activities in the second column.

Table 21-8 Chapter Review Tracking

Review Element	Review Date(s)	Resource Used:
Review key topics		Book, website
Review key terms		Book, website
Answer DIKTA questions		Book, PTP
Review memory tables		Website

Review All the Key Topics



Table 21-9 Key Topics for [Chapter 21](#)

Key Topic Element	Description	Page Number
List	Functions of IP routing protocols	541
List	Definitions of IGP and EGP	542
List	Types of IGP routing protocols	543
Table 21-2	IGP metrics	544

Key Topic Element	Description	Page Number
List	Key facts about the OSPF 2-way state	550
Table 21-4	Key OSPF neighbor states	553
Item	Definition of how OSPF calculates the cost for a route	554
Figure 21-11	Example of calculating the cost for multiple competing routes	554
List	OSPF area design rules	555
Figure 21-13	Sample OSPF multiarea design with terminology	556
Table 21-6	OSPF design terms and definitions	556

Key Terms You Should Know

2-way state
 Area Border Router (ABR)
 backbone area
 backup designated router (BDR)
 convergence
 Dead interval
 designated router (DR)
 distance vector
 full state
 fully adjacent
 Hello interval
 interior gateway protocol (IGP)
 internal router
 link-state

link-state advertisement (LSA)
link-state database (LSDB)
link-state update
metric
multiarea OSPF
neighbor
router ID (RID)
Shortest Path First (SPF) algorithm
single-area OSPF

Chapter 22

Implementing Basic OSPF Features

This chapter covers the following exam topics:

3.0 IP Connectivity

3.2 Determine how a router makes a forwarding decision by default

3.2.b Administrative distance

3.2.c Routing protocol metric

3.4 Configure and verify single area OSPFv2

3.4.a Neighbor adjacencies

3.4.b Point-to-point

3.4.c Broadcast (DR/BR selection)

3.4.d Router ID

OSPFv2 requires only a few configuration commands if you rely on default settings. To use OSPF, all you need to do is enable OSPF on each interface you intend to use in the network, and OSPF uses messages to discover neighbors and learn routes through those neighbors. OSPF performs

many background tasks, and you can discover details about that work using a large number of OSPF **show** commands. However, configuring OSPF, using mostly default settings for all the optional features, requires only a few commands. This chapter sets about to help you learn those minimal settings.

The first major section of this chapter focuses on traditional OSPFv2 configuration using the **network** command, along with the large variety of associated **show** commands. This section teaches you how to make OSPFv2 operate with default settings and convince yourself that it really is working through use of those **show** commands.

The second major section shows an alternative configuration option called OSPF interface mode, in contrast with the traditional OSPF configuration shown in the first section of the chapter. This mode uses the **ip ospf process-id area area-number** configuration command instead of the **network** command.

Along the way, the first major section includes the detail of how to set the OSPF router ID (RID). While optional, configuring a predictable and stable OSPF RID allows easier operation and troubleshooting of OSPF and may be the most important of the optional OSPF settings.

“Do I Know This Already?” Quiz

Take the quiz (either here or use the PTP software) if you want to use the score to help you decide how much time to spend on this chapter. The letter answers are listed at the bottom of the page following the quiz. [Appendix C](#), found both at the end of the book as well as on the companion website, includes both the answers and explanations. You can also find both answers and explanations in the PTP testing software.

Table 22-1 “Do I Know This Already?” Foundation Topics
Section-to-Question Mapping

Foundation Topics Section	Questions
Implementing OSPFv2 Using network Commands	1-4
Implementing OSPFv2 Using Interface Subcommands	5, 6

1. Which of the following **network** commands, following the command **router ospf 1**, enables OSPF on interfaces whose IP addresses are 10.1.1.1, 10.1.100.1, and 10.1.120.1?
 - a. **network 10.0.0.0 0.0.0.0 area 0**
 - b. **network 10.0.0.0 0.255.255.255 area 0**
 - c. **network 10.0.0.0 0.0.0.255 area 0**
 - d. **network 10.0.0.0 0.0.255.255 area 0**
2. Which of the following **network** commands, following the command **router ospf 1**, tells this router to start using OSPF on interfaces whose IP addresses are 10.1.1.1, 10.1.100.1, and 10.1.120.1?
 - a. **network 10.1.0.0 0.0.255.255 area 0**
 - b. **network 10.0.0.0 0.255.255.0 area 0**
 - c. **network 10.1.1.0 0.x.1x.0 area 0**
 - d. **network 10.1.1.0 255.0.0.0 area 0**
3. Which of the following commands list the OSPF neighbors off interface serial 0/0? (Choose two answers.)

- a. **show ip ospf neighbor**
- b. **show ip ospf interface brief**
- c. **show ip neighbor**
- d. **show ip interface**
- e. **show ip ospf neighbor serial 0/0**

4. When reloading and choosing a new OSPF router ID (RID), a router had working interfaces loopback 1 with IP address 10.8.8.8, loopback 2 with address 10.7.7.7, and GigabitEthernet0/0/0 with 10.9.9.9. The router did not have a **router-id** command in the OSPF process configuration. What RID did the router choose?

- a. 10.7.7.7
- b. 10.8.8.8
- c. 10.9.9.9
- d. The router would fail to choose an RID.

5. An engineer migrates from a more traditional OSPFv2 configuration that uses **network** commands in OSPF configuration mode to instead use OSPFv2 interface configuration. Which of the following commands configures the area number assigned to an interface in this new configuration?

- a. The **area** command in interface configuration mode
- b. The **ip ospf** command in interface configuration mode
- c. The **router ospf** command in interface configuration mode
- d. The **network** command in interface configuration mode

6. An enterprise avoids using the OSPF **network** command, instead preferring to enable OSPF per-interface with the **ip ospf process-id area area-id** interface subcommand. Which **show** command identifies whether an interface has been configured with the **ip ospf process-id area area-id** interface subcommand? (Choose two answers.)
- a. The **show ip ospf interface** command
 - b. The **show ip ospf interface brief** command
 - c. The **show ip ospf neighbor** command
 - d. The **show ip protocols** command

Answers to the “Do I Know This Already?” quiz:

- 1 B
- 2 A
- 3 A, E
- 4 B
- 5 B
- 6 A, D

Foundation Topics

Implementing OSPFv2 Using network Commands

After an OSPF design has been chosen—a task that can be complex in larger IP internetworks—the configuration can be as simple as enabling OSPF on each router interface and placing that interface in the correct OSPF area. This first

major section of the chapter focuses on the required configuration using the traditional OSPFv2 **network** command along with one optional configuration setting: how to set the **OSPF router-id**. Additionally, this section works through how to show the various lists and tables that confirm how OSPF is working.

For reference and study, the following list outlines the configuration steps covered in this first major section of the chapter:

Step 1. Use the **router ospf** *process-id* global command to enter OSPF configuration mode for a particular OSPF process.

Config
Checklist

Step 2. (Optional) Configure the OSPF router ID by doing the following:

- a. Use the **router-id** *id-value* router subcommand to define the router ID, or
- b. Use the **interface loopback** *number* global command, along with an **ip address** *address mask* command, to configure an IP address on a loopback interface (chooses the highest IP address of all working loopbacks), or
- c. Rely on an interface IP address (chooses the highest IP address of all working nonloopbacks).

Step 3. Use one or more **network** *ip-address wildcard-mask area area-id* router subcommands to enable OSPFv2 on any interfaces matched by the configured address and mask, enabling OSPF on the interface for the listed area.

Figure 22-1 shows the relationship between the OSPF configuration commands, with the idea that the configuration creates a routing process in one part of the configuration, and then indirectly enables OSPF on each interface. The configuration does not name the interfaces on which OSPF is enabled, instead requiring IOS to apply some logic by comparing the OSPF **network** command to the interface **ip address** commands. The upcoming example discusses more about this logic.

Key Topic

Configuration

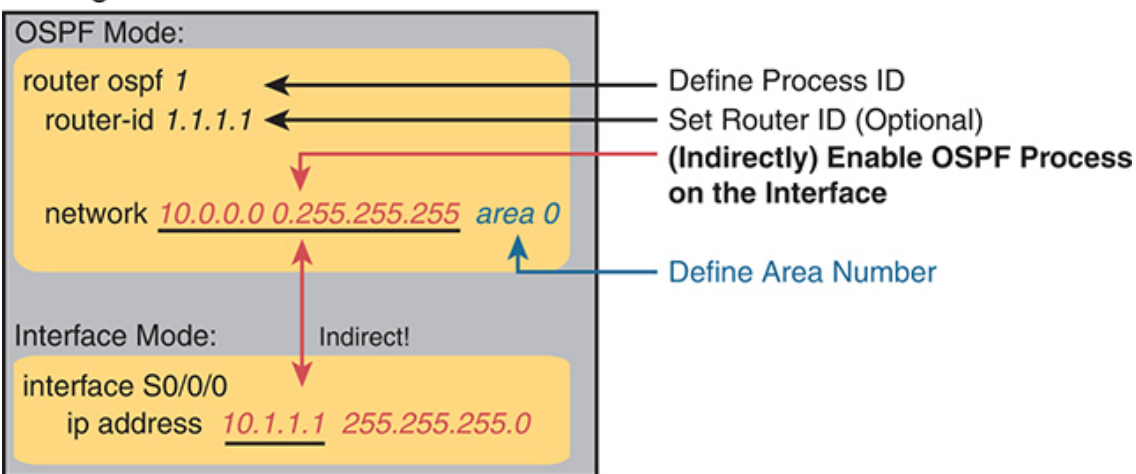


Figure 22-1 Organization of OSPFv2 Configuration with the **network** Command

OSPF Single-Area Configuration

Figure 22-2 shows a sample network that will be used for most examples throughout this chapter. All links reside in area 0, making the area design a single-area design, with four routers. You can think of Router R1 as a router at a central site, with WAN links to each remote site. Routers R2 and R3 might be at one large remote site that needs two

WAN links and two routers for WAN redundancy, with both routers connected to the LAN at that remote site. Router R4 might be a typical smaller remote site with a single router needed for that site.

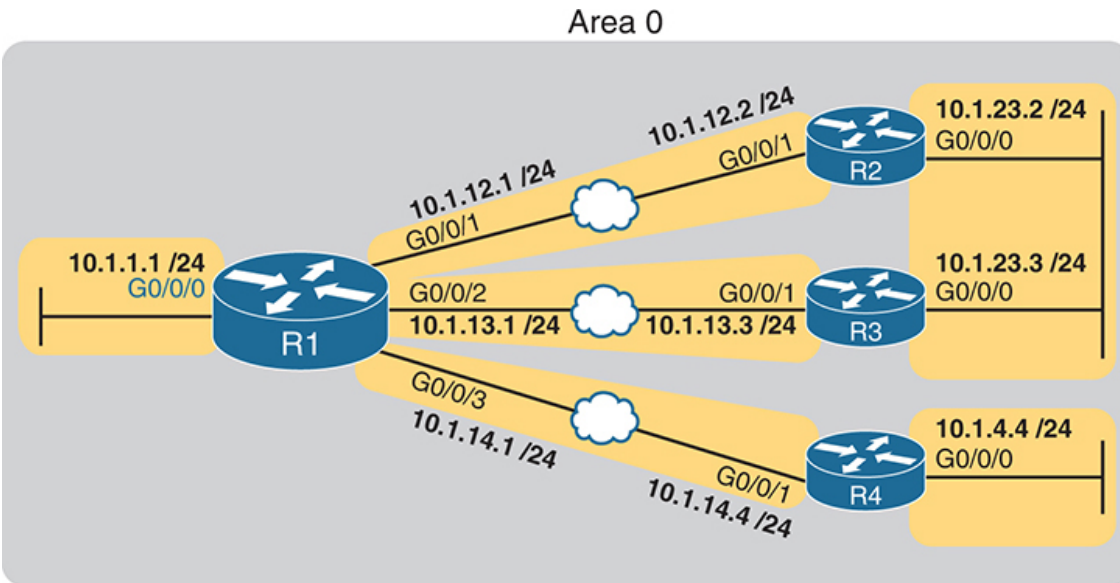


Figure 22-2 Sample Network for OSPF Single-Area Configuration

[Example 22-1](#) shows the IPv4 addressing configuration on Router R1, before getting into the OSPF detail.

Example 22-1 IPv4 Address Configuration on R1

[Click here to view code image](#)

```
interface GigabitEthernet0/0/0
 ip address 10.1.1.1 255.255.255.0
!
interface GigabitEthernet0/0/1
 ip address 10.1.12.1 255.255.255.0
!
interface GigabitEthernet0/0/2
 ip address 10.1.13.1 255.255.255.0
!
```

```
interface GigabitEthernet0/0/3
ip address 10.1.14.1 255.255.255.0
```

The OSPF configuration begins with the **router ospf** *process-id* global command, which puts the user in OSPF configuration mode, and sets the OSPF *process-id* value. The *process-id* number just needs to be unique on the local router, matching between various commands in a router. The *process-id* does not need to match between neighboring routers or other routers in the same area. The value can be any integer between 1 and 65,535.

Second, the configuration needs one or more **network** commands in OSPF mode. These commands tell the router to find its local interfaces that match the first two parameters on the **network** command. Then, for each matched interface, the router enables OSPF on those interfaces, discovers neighbors, creates neighbor relationships, and assigns the interface to the area listed in the **network** command. (Note that the area can be configured as either an integer or a dotted-decimal number, but this book makes a habit of configuring the area number as an integer. The integer area numbers range from 0 through 4,294,967,295.)

[Example 22-2](#) shows an example configuration on Router R1 from [Figure 22-2](#). The **router ospf 1** command enables OSPF process 1, and the single **network** command enables OSPF on all interfaces shown in the figure.

Example 22-2 *OSPF Single-Area Configuration on R1 Using One **network** Command*

[Click here to view code image](#)

```
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
```

For the specific **network** command in [Example 22-2](#), any matched interfaces are assigned to area 0. However, the first two parameters—the *ip_address* and *wildcard_mask* parameter values of 10.0.0.0 and 0.255.255.255—need some explaining. In this case, the command matches all interfaces shown for Router R1; the next topic explains why.

Wildcard Matching with the network Command

The key to understanding the traditional OSPFv2 configuration shown in this first example is to understand the OSPF **network** command. The OSPF **network** command compares the first parameter in the command to each interface IP address on the local router, trying to find a match. However, rather than comparing the entire number in the **network** command to the entire IPv4 address on the interface, the router can compare a subset of the octets, based on the wildcard mask, as follows:

Wildcard 0.0.0.0: Compare all four octets. In other words, the numbers must exactly match.



Wildcard 0.0.0.255: Compare the first three octets only. Ignore the last octet when comparing the numbers.

Wildcard 0.0.255.255: Compare the first two octets only. Ignore the last two octets when comparing the numbers.

Wildcard 0.255.255.255: Compare the first octet only. Ignore the last three octets when comparing the numbers.

Wildcard 255.255.255.255: Compare nothing; this wildcard mask means that all addresses will match the **network** command.

Basically, a wildcard mask value of decimal 0 in an octet tells IOS to compare to see if the numbers match, and a value of 255 tells IOS to ignore that octet when comparing the numbers.

The **network** command provides many flexible options because of the wildcard mask. For example, in Router R1, many **network** commands could be used, with some matching all interfaces, and some matching a subset of interfaces. [Table 22-2](#) shows a sampling of options, with notes.

Table 22-2 Example OSPF **network** Commands on R1, with Expected Results

Command	Logic in Command	Matched Interfaces
network 10.1.0.0 0.0.255.255	Match addresses that begin with 10.1	G0/0/0 G0/0/1 G0/0/1 G0/0/2
network 10.0.0.0 0.255.255.255	Match addresses that begin with 10	G0/0/0 G0/0/1 G0/0/1 G0/0/2

Command	Logic in Command	Matched Interfaces
network 0.0.0.0 255.255.255.255	Match all addresses	G0/0/0 G0/0/1 G0/0/1 G0/0/2
network 10.1.13.0 0.0.0.255	Match addresses that begin with 10.1.13	G0/0/2
network 10.1.13.1 0.0.0.0	Match one address: 10.1.13.1	G0/0/2

The wildcard mask gives the local router its rules for matching its own interfaces. To show examples of the different options, [Example 22-3](#) shows the configuration on routers R2, R3, and R4, each using different wildcard masks. Note that all three routers (R2, R3, and R4) enable OSPF on all the interfaces shown in [Figure 22-2](#).

Example 22-3 *OSPF Configuration on Routers R2, R3, and R4*

[Click here to view code image](#)

```
! R2 configuration next - one network command enables OSPF on both
interface GigabitEthernet0/0/0
 ip address 10.1.23.2 255.255.255.0
!
interface GigabitEthernet0/0/1
 ip address 10.1.12.2 255.255.255.0
!
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
```

```
! R3 configuration next - One network command per interface
interface GigabitEthernet0/0/0
 ip address 10.1.23.3 255.255.255.0
!
interface GigabitEthernet0/0/1
 ip address 10.1.13.3 255.255.255.0
!
router ospf 1
 network 10.1.13.3 0.0.0.0 area 0
network 10.1.23.3 0.0.0.0 area 0
! R4 configuration next - One network command per interface with
interface GigabitEthernet0/0/0
 ip address 10.1.4.4 255.255.255.0
!
interface GigabitEthernet0/0/1
 ip address 10.1.14.4 255.255.255.0
!
router ospf 1
 network 10.1.14.0 0.0.0.255 area 0
 network 10.1.4.0 0.0.0.255 area 0
```

Finally, note that OSPF uses the same wildcard mask logic as defined by Cisco IOS access control lists. The section titled “Finding the Right Wildcard Mask to Match a Subnet” section in [Chapter 6](#) of the *CCNA 200-301 Official Cert Guide, Volume 2*, Second Edition, provides more detail about wildcard masks.

Note

If the wildcard mask octet in a **network** command is 255, the matching address octet should be configured as a 0. Interestingly, IOS will accept a **network** command that breaks this rule, but if you configure a wildcard mask octet as 255, then IOS changes the

corresponding address octet to a 0 before putting it into the running configuration file. For example, IOS will change a typed command that begins with **network 1.2.3.4 0.0.255.255** to **network 1.2.0.0 0.0.255.255**.

Verifying OSPF Operation

As mentioned in [Chapter 21, “Understanding OSPF Concepts,”](#) OSPF routers use a three-step process to eventually add OSPF-learned routes to the IP routing table. First, they create neighbor relationships. Then they build and flood LSAs between those neighbors so each router in the same area has a copy of the same LSDB. Finally, each router independently computes its own IP routes using the SPF algorithm and adds them to its routing table. This next topic works through how to display the results of each of those steps, which lets you confirm whether OSPF has worked correctly or not.

The **show ip ospf neighbor**, **show ip ospf database**, and **show ip route** commands display information to match each of these three steps, respectively. [Figure 22-3](#) summarizes the commands you can use (and others) when verifying OSPF.



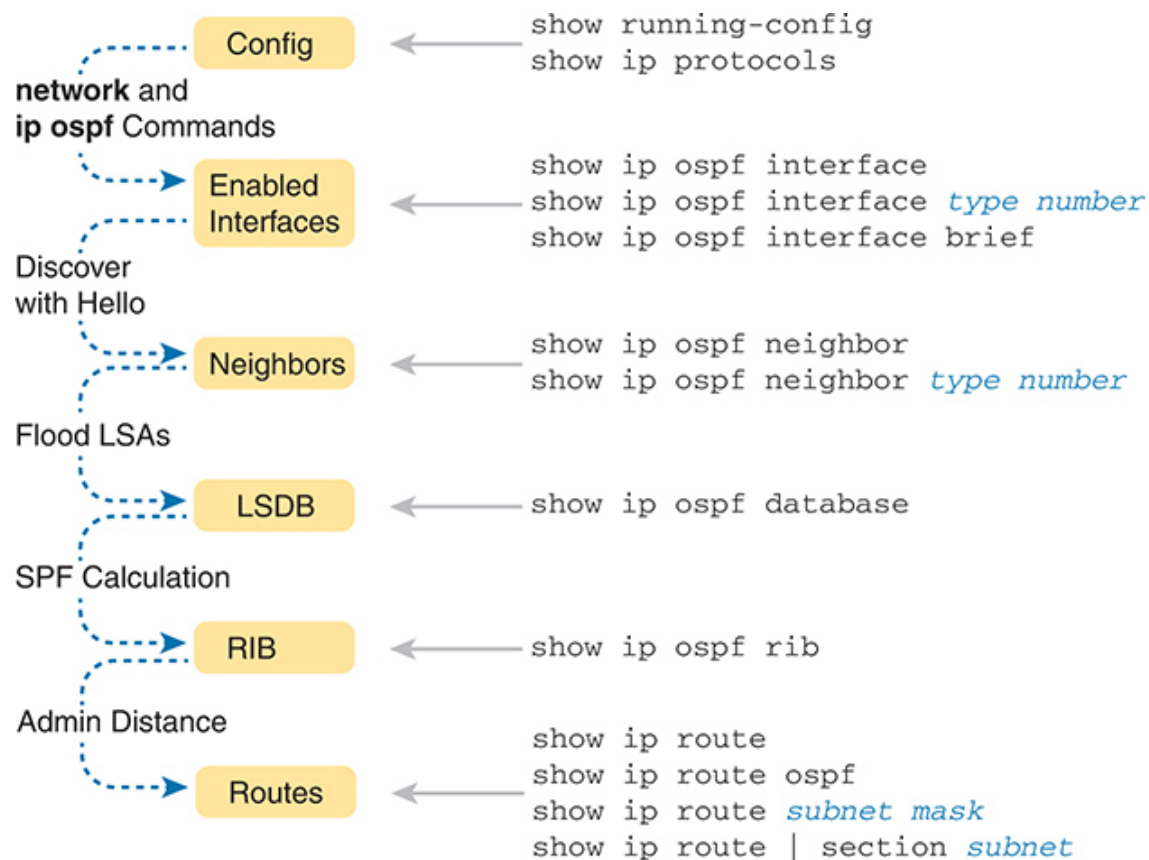


Figure 22-3 *OSPF Verification Commands*

Many engineers begin OSPF verification by looking at the output of the **show ip ospf neighbor** command. For instance, [Example 22-4](#) shows a sample from Router R1, which should have one neighbor relationship each with routers R2, R3, and R4. [Example 22-4](#) shows all three.



Example 22-4 *OSPF Neighbors on Router R1 from [Figure 22-2](#)*

[Click here to view code image](#)

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:37	10.1.12.2	G0/0/1
3.3.3.3	1	FULL/DR	00:00:37	10.1.13.3	G0/0/1
4.4.4.4	1	FULL/BDR	00:00:34	10.1.14.4	G0/0/1

The detail in the output mentions several important facts, and for most people, working right to left works best in this case. For example, look at the headings:

Interface: This is the local router's interface connected to the neighbor. For example, the first neighbor in the list is reachable through R1's G0/0/1 interface.

Address: This is the neighbor's IP address on that link. Again, this first neighbor, the neighbor, which is R2, uses IP address 10.1.12.2.

State: While many possible states exist, for the details discussed in this chapter, FULL is the correct and fully working state in this case.

Neighbor ID: This is the router ID of the neighbor.

Note

Examples 22-4 through 22-8 use configuration not shown here that sets the RID values to easily identify the routers, using 2.2.2.2 for Router R2, 3.3.3.3 for Router R3, and so on. The upcoming section “Configuring the OSPF Router ID” shows how to set the RID.

Once OSPF convergence has completed, a router should list each neighbor. On links that use a designated router (DR), the state will also list the role of the neighboring router after the / (DR, BDR, or DROther). As a result, the normal working states will be:

FULL/ -: The neighbor state is full, with the “-” instead of letters meaning that the link does not use a DR/BDR.

Key Topic

FULL/DR: The neighbor state is full, and the neighbor is the DR.

FULL/BDR: The neighbor state is full, and the neighbor is the backup DR (BDR).

FULL/DROTHER: The neighbor state is full, and the neighbor is neither the DR nor BDR. (It also implies that the local router is a DR or BDR because the state is FULL.)

2WAY/DROTHER: The neighbor state is 2-way, and the neighbor is neither the DR nor BDR—that is, a DROther router. (It also implies that the local router is also a DROther router because otherwise the state would reach a full state.)

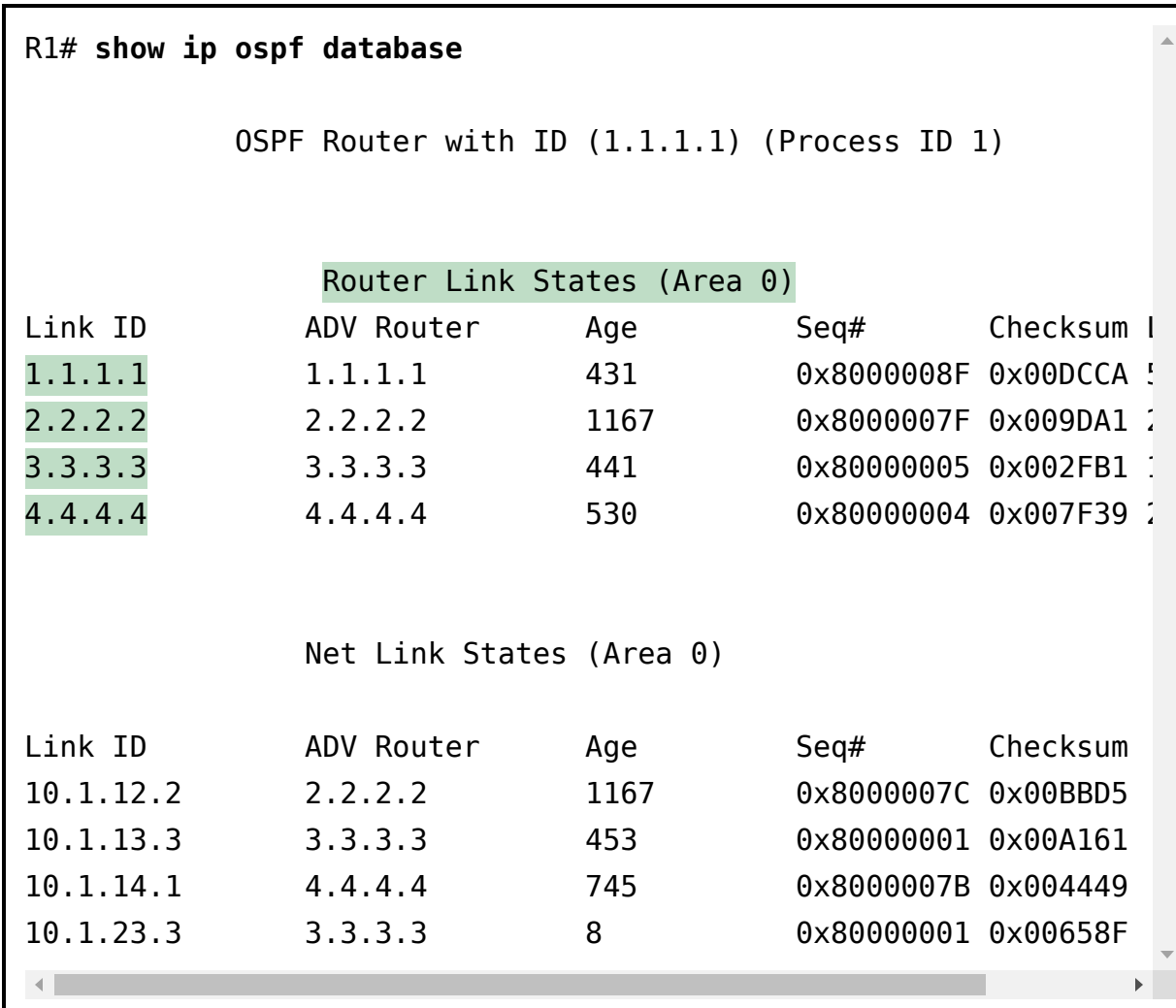
Once a router’s OSPF process forms a working neighbor relationship, the routers exchange the contents of their LSDBs, either directly or through the DR on the subnet.

[Example 22-5](#) shows the contents of the LSDB on Router R1. Interestingly, with a single-area design, all the routers will have the same LSDB contents once all neighbors are up and all LSAs have been exchanged. So, the **show ip ospf database** command in [Example 22-5](#) should list the same

exact information, no matter on which of the four routers it is issued.

Example 22-5 *OSPF Database on Router R1 from [Figure 22-2](#)*

[Click here to view code image](#)



```
R1# show ip ospf database

      OSPF Router with ID (1.1.1.1) (Process ID 1)

      Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum
1.1.1.1      1.1.1.1      431         0x8000008F  0x00DCCA
2.2.2.2      2.2.2.2      1167        0x8000007F  0x009DA1
3.3.3.3      3.3.3.3      441         0x80000005  0x002FB1
4.4.4.4      4.4.4.4      530         0x80000004  0x007F39

      Net Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum
10.1.12.2    2.2.2.2      1167        0x8000007C  0x00BBD5
10.1.13.3    3.3.3.3      453         0x80000001  0x00A161
10.1.14.1    4.4.4.4      745         0x8000007B  0x004449
10.1.23.3    3.3.3.3      8           0x80000001  0x00658F
```

For the purposes of this book, do not be concerned about the specifics in the output of this command. However, for perspective, note that the LSDB should list one “Router Link State” (Type 1 Router LSA) for each of the routers in the same area, so with the design based on [Figure 22-2](#), the

output lists four Type 1 LSAs. Also, with all default settings in this design, the routers will create a total of four Type 2 Network LSAs as shown, one each for the subnets that have a DR and contain at least two routers in that subnet (the three WAN links plus the LAN to which both R2 and R3 connect).

Next, [Example 22-6](#) shows R4's IPv4 routing table with the **show ip route** command. As configured, with all links working, R4 has connected routes to two of those subnets and should learn OSPF routes to the other subnets.

Example 22-6 *IPv4 Routes Added by OSPF on Router R4 from [Figure 22-2](#)*

[Click here to view code image](#)

```
R4# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external ty
       E1 - OSPF external type 1, E2 - OSPF external type 2
! Additional legend lines omitted for brevity

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.1.1.0/24 [110/2] via 10.1.14.1, 00:27:24, GigabitEthe
C       10.1.4.0/24 is directly connected, GigabitEthernet0/0/0
L       10.1.4.4/32 is directly connected, GigabitEthernet0/0/0
O       10.1.12.0/24 [110/2] via 10.1.14.1, 00:27:24, GigabitEtl
O       10.1.13.0/24 [110/2] via 10.1.14.1, 00:25:15, GigabitEtl
C       10.1.14.0/24 is directly connected, GigabitEthernet0/0/0
L       10.1.14.4/32 is directly connected, GigabitEthernet0/0/0
O       10.1.23.0/24 [110/3] via 10.1.14.1, 00:27:24, GigabitEtl
```

Any time you want to check OSPF on a router in a small design like the ones in the book, you can count all the subnets, then count the subnets connected to the local router, and know that OSPF should learn routes to the rest of the subnets. Then just use the **show ip route** command and add up how many connected and OSPF routes exist as a quick check of whether all the routes have been learned or not.

In this case, Router R4 has two connected subnets, but six subnets exist per the figure, so Router R4 should learn four OSPF routes. Next look for the code of “O” on the left, which identifies a route as being learned by OSPF. The output lists four such IP routes: one for the LAN subnet off Router R1, one for the LAN subnet connected to both R2 and R3, and one each for the WAN subnets from R1 to R2 and R1 to R3.

Next, examine the first route (to subnet 10.1.1.0/24). It lists the subnet ID and mask, identifying the subnet. It also lists two numbers in brackets. The first, 110, is the administrative distance of the route. All the OSPF routes in this example use the default of 110 (see [Table 24-4](#) in [Chapter 24, “OSPF Neighbors and Route Selection,”](#) for the list of administrative distance values). The second number, 2, is the OSPF metric for this route. The route also lists the forwarding instructions: the next-hop IP address (10.1.14.1) and R4’s outgoing interface (G0/0/1).

Note

The section “[Floating Static Routes](#)” in [Chapter 17, “Configuring IPv4 Addresses and Static Routes,”](#) introduced the concept of administrative distance; however, the section “[Multiple Routes Learned from Competing Sources](#),” in [Chapter 24](#) discusses the topic in more depth.

Verifying OSPF Configuration

Once you can configure OSPF with confidence, you will likely verify OSPF focusing on **OSPF neighbors** and the IP routing table as just discussed. However, if OSPF does not work immediately, you may need to circle back and check the configuration. To do so, you can use these steps:

- If you have enable mode access, use the **show running-config** command to examine the configuration.
- If you have only user mode access, use the **show ip protocols** command to re-create the OSPF configuration.
- Use the **show ip ospf interface [brief]** command to determine whether the router enabled OSPF on the correct interfaces or not based on the configuration.

The best way to verify the configuration begins with the **show running-config** command, of course. However, the **show ip protocols** command repeats the details of the OSPFv2 configuration and does not require enable mode access. [Example 22-7](#) does just that for Router R3.

Example 22-7 *Router R3 Configuration and the **show ip protocols** Command*

[Click here to view code image](#)

```
R3# show running-config | section router ospf 1
router ospf 1
  network 10.1.13.3 0.0.0.0 area 0
  network 10.1.23.3 0.0.0.0 area 0
  router-id 3.3.3.3

R3# show ip protocols
```

```
*** IP Routing is NSF aware ***
```

```
Routing Protocol is "ospf 1"
```

```
Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
```

```
Router ID 3.3.3.3
```

```
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
Maximum path: 4
```

```
Routing for Networks:
```

```
10.1.13.3 0.0.0.0 area 0
```

```
10.1.23.3 0.0.0.0 area 0
```

```
Routing Information Sources:
```

Gateway	Distance	Last Update
1.1.1.1	110	02:05:26
4.4.4.4	110	02:05:26
2.2.2.2	110	01:51:16

Distance: (default is 110)

The highlighted output emphasizes some of the configuration. The first highlighted line repeats the parameters on the **router ospf 1** global configuration command. (The second highlighted item points out the router's router ID, which will be discussed in the next section.) The third set of highlighted lines begins with a heading of "Routing for Networks:" followed by two lines that closely resemble the parameters on the configured **network** commands. In fact, closely compare those last two highlighted lines with the **network** configuration commands at the top of the example, and you will see that they mirror each other, but the **show** command just leaves out the word *network*. For instance:

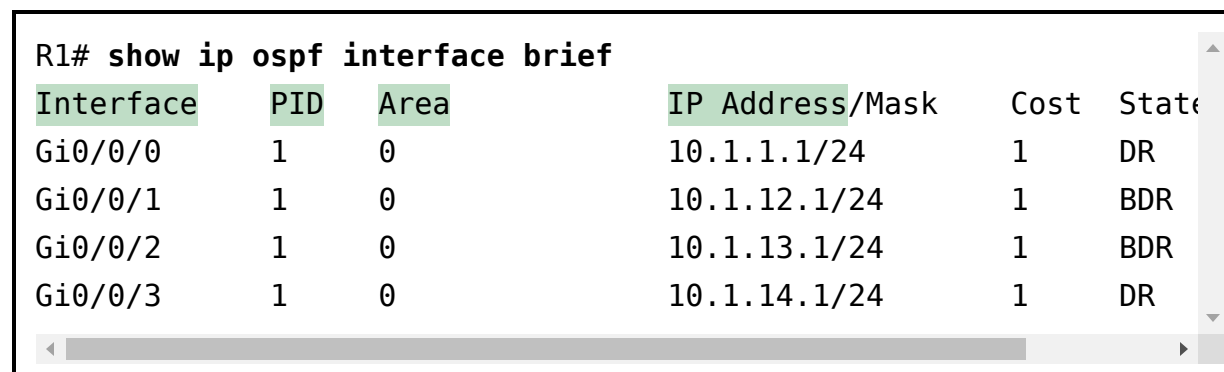
Configuration: **network 10.1.13.3 0.0.0.0 area 0**

show Command: **10.1.13.3 0.0.0.0 area 0**

IOS interprets the **network** commands to choose interfaces on which to run OSPF, so it could be that IOS chooses a different set of interfaces than you predicted. To check the list of interfaces chosen by IOS, use the **show ip ospf interface brief** command, which lists all interfaces that have been enabled for OSPF processing. Verifying the interfaces can be a useful step if you have issues with OSPF neighbors because OSPF must first be enabled on an interface before a router will attempt to discover neighbors on that interface. [Example 22-8](#) shows a sample from Router R1.

Example 22-8 Router R1 **show ip ospf interface brief** Command

[Click here to view code image](#)



R1# show ip ospf interface brief					
Interface	PID	Area	IP Address/Mask	Cost	State
Gi0/0/0	1	0	10.1.1.1/24	1	DR
Gi0/0/1	1	0	10.1.12.1/24	1	BDR
Gi0/0/2	1	0	10.1.13.1/24	1	BDR
Gi0/0/3	1	0	10.1.14.1/24	1	DR

The **show ip ospf interface brief** command lists one line per interface, showing all the interfaces on which OSPF has been enabled. Each line identifies the OSPF process ID (per the **router ospf process-id** command), the area, the interface IP address, and the number of neighbors found via each interface.

You may use the command in [Example 22-8](#) quite often, but the **show ip ospf interface** command (without the **brief** keyword) gives much more detail about OSPF per-interface settings. [Example 23-4](#) in [Chapter 23](#), “Implementing

[Optional OSPF Features](#),” shows an example of the entire output of that command.

Configuring the OSPF Router ID

While OSPF has many other optional features, most enterprise networks that use OSPF choose to configure each router’s OSPF router ID. OSPF-speaking routers must have a router ID (RID) for proper operation. By default, routers will choose an interface IP address to use as the RID. However, many network engineers prefer to choose each router’s router ID, so command output from commands like **show ip ospf neighbor** lists more recognizable router IDs.

To choose its RID, a Cisco router uses the following process when the router reloads and brings up the OSPF process. Note that the router stops looking for a router ID to use once one of the steps identifies a value to use.

1. If the **router-id** *rid* OSPF subcommand is configured, this value is used as the RID.



2. If any loopback interfaces have an IP address configured, and the interface has an interface status of up, the router picks the highest numeric IP address among these loopback interfaces.
3. The router picks the highest numeric IP address from all other interfaces whose interface status code (first status code) is up. (In other words, an interface in up/down state will be included by OSPF when choosing its router ID.)

The first and third criteria should make some sense right away: the RID is either configured or is taken from a working

interface's IP address. However, this book has not yet explained the concept of a *loopback interface*, as mentioned in Step 2.

A loopback interface is a virtual interface that can be configured with the **interface loopback** *interface-number* command, where *interface-number* is an integer. Loopback interfaces are always in an “up and up” state unless administratively placed in a shutdown state. For example, a simple configuration of the command **interface loopback 0**, followed by **ip address 2.2.2.2 255.255.255.0**, would create a loopback interface and assign it an IP address. Because loopback interfaces do not rely on any hardware, these interfaces can be up/up whenever IOS is running, making them good interfaces on which to base an OSPF RID.

[Example 22-9](#) shows the configuration that existed in Routers R1 and R2 before the creation of the **show** command output earlier in this chapter. R1 set its router ID using the direct method, while R2 used a loopback IP address. [Example 22-10](#) that follows shows the output of the **show ip ospf** command on R1, which identifies the OSPF RID used by R1.

Example 22-9 OSPF Router ID Configuration Examples

[Click here to view code image](#)

```
! R1 Configuration first
router ospf 1
  router-id 1.1.1.1
network 10.1.0.0 0.0.255.255 area 0
```

```
! R2 Configuration next
!
interface Loopback2
  ip address 2.2.2.2 255.255.255.255
```

Example 22-10 *Confirming the Current OSPF Router ID*

[Click here to view code image](#)

```
R1# show ip ospf
Routing Process "ospf 1" with ID 1.1.1.1
! lines omitted for brevity
```

Routers need a stable OSPF RID because any change to the OSPF RID causes a router to close existing neighbor relationships and remove all routes learned through those neighbors. To keep the RID stable, a router chooses its RID when the router first initializes (at power-on or per the **reload** command). So the RID might change at the next reload when the router re-evaluates the RID choice rules based on the current conditions.

However, routers do support one scenario to update their RID without a **reload**, which can be useful for testing in lab. To do so, configure the OSPF **router-id** OSPF subcommand followed by the **clear ip ospf process** EXEC command.

Implementing Multiarea OSPF

Even though the current CCNA 200-301 V1.1 exam blueprint mentions single area but not multiarea OSPF, you only need to learn one more idea to know how to configure multiarea OSPF. So, this chapter takes a brief page to show how.

For example, consider a multiarea OSPF design as shown in [Figure 22-4](#). It uses the same routers and IP addresses as shown earlier in [Figure 22-2](#), on which all the examples in this chapter have been based so far. However, the design shows three areas instead of the single-area design shown in [Figure 22-2](#).

Configuring the routers in a multiarea design is almost just like configuring OSPFv2 for a single area. To configure multiarea OSPF, all you need is a valid OSPF area design (for instance, like [Figure 22-4](#)) and a configuration that places each router interface into the correct area per that design. For example, both of R4's interfaces connect to links in area 4, making R4 an internal router, so any **network** commands on Router R4 will list area 4.

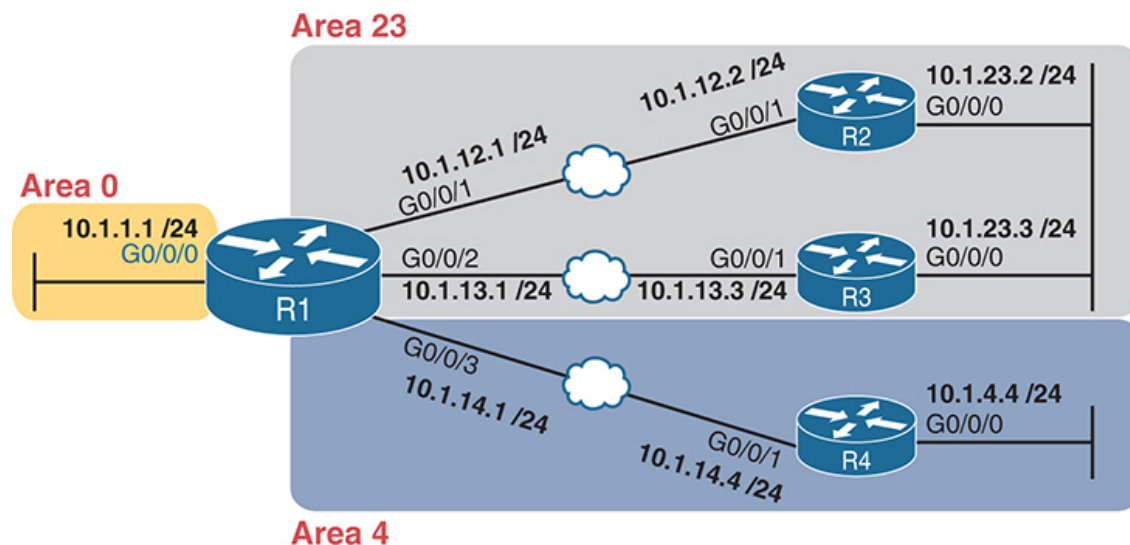


Figure 22-4 Area Design for an Example Multiarea OSPF Configuration

[Example 22-11](#) shows a sample configuration for Router R1. To make the configuration clear, it uses **network** commands with a wildcard mask of 0.0.0.0, meaning each **network** command matches a single interface. Each interface will be placed into either area 0, 23, or 4 to match the figure.

Example 22-11 OSPF Configuration on R1, Placing Interfaces into Different Areas

[Click here to view code image](#)

```
router ospf 1
network 10.1.1.1 0.0.0.0 area 0
```

```
network 10.1.12.1 0.0.0.0 area 23
network 10.1.13.1 0.0.0.0 area 23
network 10.1.14.1 0.0.0.0 area 4
```

Implementing OSPFv2 Using Interface Subcommands

From the earliest days of OSPFv2 support in Cisco routers, the configuration used the OSPF **network** command as discussed in this chapter. However, that configuration style can be confusing, and it does require some interpretation. As a result, Cisco added another option for OSPFv2 configuration called OSPF interface configuration.

The newer interface-style OSPF configuration still enables OSPF on interfaces, but it does so directly with the **ip ospf** interface subcommand. Instead of matching interfaces with indirect logic using **network** commands, you directly enable OSPFv2 on interfaces by configuring an interface subcommand on each interface.

OSPF Interface Configuration Example

To show how OSPF interface configuration works, this example basically repeats the example shown earlier in the chapter using the traditional OSPFv2 configuration with **network** commands. So, before looking at the OSPFv2 interface configuration, take a moment to look back to review traditional OSPFv2 configuration with [Figure 22-2](#) and [Examples 22-2](#) and [22-3](#).

After reviewing the traditional configuration, consider this checklist, which details how to convert from the old-style configuration in [Example 22-2](#) and [Example 22-3](#) to use interface configuration:

Step 1. Use the **no network network-id area area-id** subcommands in OSPF configuration mode to remove the **network** commands.

Config
Checklist

Step 2. Add one **ip ospf process-id area area-id** command in interface configuration mode under each interface on which OSPF should operate, with the correct OSPF process (*process-id*) and the correct OSPF area number.

Figure 22-5 repeats the design for both the original examples in this chapter and for this upcoming interface configuration example.

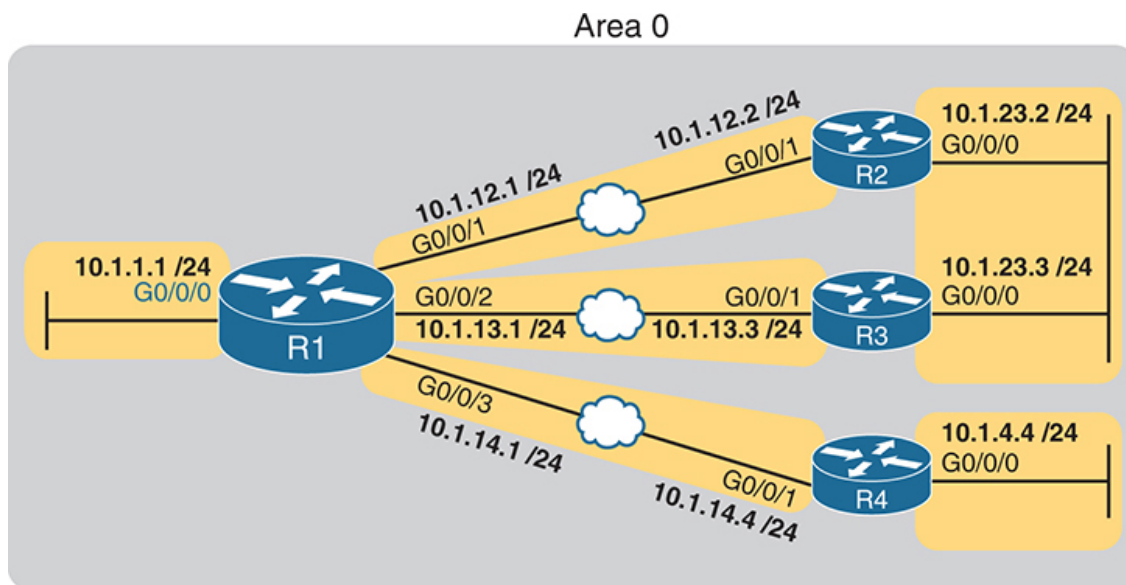


Figure 22-5 Area Design Used in the Upcoming OSPF Interface Config Example

Example 22-2 shows a single **network** command: **network 10.0.0.0 0.255.255.255 area 0**. Example 22-12 follows the steps in the migration checklist, beginning with the removal of the previous configuration using the **no network 10.0.0.0 0.255.255.255 area 0** command. The

example then shows the addition of the **ip ospf 1 area 0** command on each of the interfaces on Router R1, enabling OSPF process 1 on the interface and placing each interface into area 0.

Example 22-12 *Migrating to Use OSPF Interface Subcommand Configuration*

[Click here to view code image](#)

```
R1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# router ospf 1
R1(config-router)# no network 10.0.0.0 0.255.255.255 area 0
R1(config-router)#
*Apr  8 19:35:24.994: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on (
from FULL to DOWN, Neighbor Down: Interface down or detached
*Apr  8 19:35:24.994: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on (
from FULL to DOWN, Neighbor Down: Interface down or detached
*Apr  8 19:35:24.994: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on (
from FULL to DOWN, Neighbor Down: Interface down or detached
R1(config-router)# interface g0/0/0
R1(config-if)# ip ospf 1 area 0
R1(config-if)# interface g0/0/1
R1(config-if)# ip ospf 1 area 0
R1(config-if)#
*Apr  8 19:35:52.970: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on (
from LOADING to FULL, Loading Done
R1(config-if)# interface g0/0/2
R1(config-if)# ip ospf 1 area 0
R1(config-if)#
*Apr  8 19:36:13.362: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on (
from LOADING to FULL, Loading Done
R1(config-if)# interface g0/0/3
R1(config-if)# ip ospf 1 area 0
R1(config-if)#
*Apr  8 19:37:05.308: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on (
```

```
Apr  6 19:57:05.596: %OSPF-5-ADJCHG: PROCESS 1, NBR 4.4.4.4 OFF V
from LOADING to FULL, Loading Done
R1(config-if)#
```

When reading the example, read from top to bottom, and also consider the details about the failed and recovered neighbor relationships shown in the log messages. Removing the network command disabled OSPF on all interfaces on Router R1, causing all three neighbor relationships to fail. The example then shows the addition of the **ip ospf 1 area 0** command on the LAN interface, which enables OSPF but does not cause a neighbor relationship to form, because no other OSPF routers exist in that subnet. Then the example shows the same command added to each of the WAN links in succession, and in each case, the OSPF neighbor available over that WAN link comes up (as noted in the log messages).

Note

A router's configuration can include both a **network** router subcommand and an **ip ospf** interface subcommand that enable OSPF on the same interface. If those commands refer to different area numbers, IOS uses the area number from the **ip ospf** interface subcommand. Additionally, multiple **network** commands can match the same interface. In that case, IOS uses the order in which the commands appear in OSPF configuration mode.

Verifying OSPF Interface Configuration

OSPF operates the same way whether you use the new style or old style of configuration. The OSPF area design works

the same, neighbor relationships form the same way, routers negotiate to become the DR and BDR the same way, and so on. However, you can see a few small differences in **show** command output when using the newer OSPFv2 configuration if you look closely.

The **show ip protocols** command relists most of the routing protocol configuration, so it does list some different details if you use interface configuration versus the **network** command. With the **ip ospf** interface subcommands, the output lists the phrase “Interfaces Configured Explicitly,” as highlighted in [Example 22-13](#). The example first shows the relevant parts of the **show ip protocols** command when using interface configuration on Router R1, and then lists the same portions of the command from when R1 used **network** commands.

Example 22-13 *Differences in show ip protocols Output: Old- and New-Style OSPFv2 Configuration*

[Click here to view code image](#)

! First, with the new interface configuration

R1# **show ip protocols**

! ... beginning lines omitted for brevity

Routing for Networks:

Routing on Interfaces Configured Explicitly (Area 0):

GigabitEthernet0/0/0

GigabitEthernet0/0/1

GigabitEthernet0/0/2

GigabitEthernet0/0/3

Routing Information Sources:

Gateway	Distance	Last Update
---------	----------	-------------

4.4.4.4	110	00:09:30
---------	-----	----------

2.2.2.2	110	00:10:49
---------	-----	----------

3.3.3.3	110	05:20:07
---------	-----	----------

Distance: (default is 110)

```
! For comparison, the old results with the use of the OSPF network
R1# show ip protocols
! ... beginning lines omitted for brevity
  Routing for Networks:
    10.1.0.0 0.0.255.255 area 0
! ... ending line omitted for brevity
```

Another small piece of different output exists in the **show ip ospf interface** *[interface]* command. The command lists details about OSPF settings for the interface(s) on which OSPF is enabled. The output also makes a subtle reference to whether that interface was enabled for OSPF with the old or new configuration style. [Example 22-14](#) also begins with output based on interface configuration on Router R1, followed by the output that would exist if R1 still used the old-style **network** command.



Example 22-14 *Differences in show ip ospf interface Output with OSPFv2 Interface Configuration*

[Click here to view code image](#)

```
! First, with the new interface configuration
R1# show ip ospf interface g0/0/1
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.12.1/24, Area 0, Attached via Interface I
! Lines omitted for brevity
```

```
! For comparison, the old results with the use of the OSPF network
R1# show ip ospf interface g0/0/1
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.12.1/24, Area 0, Attached via Network St
! ... ending line omitted for brevity
```

Other than these small differences in a few **show** commands, the rest of the commands show nothing different depending on the style of configuration. For instance, the **show ip ospf interface brief** command does not change depending on the configuration style, nor do the **show ip ospf database**, **show ip ospf neighbor**, or **show ip route** commands.

Chapter Review

One key to doing well on the exams is to perform repetitive spaced review sessions. Review this chapter's material using either the tools in the book or interactive tools for the same material found on the book's companion website. Refer to the "[Your Study Plan](#)" element for more details. [Table 22-3](#) outlines the key review elements and where you can find them. To better track your study progress, record when you completed these activities in the second column.

Table 22-3 Chapter Review Tracking

Review Element	Review Date(s)	Resource Used
Review key topics		Book, website
Review key terms		Book, website
Answer DIKTA questions		Book, PTP

Review Element	Review Date(s)	Resource Used
Review Config Checklists		Book, website
Review command tables		Book
Do labs		Blog
Watch video		Website

Review All the Key Topics



Table 22-4 Key Topics for [Chapter 22](#)

Key Topic Element	Description	Page Number
Figure 22-1	Organization of OSPFv2 configuration with the network command	565
List	Example OSPF wildcard masks and their meaning	567
Figure 22-3	OSPF verification commands	569
Example 22-4	Example of the show ip ospf neighbor command	569
List	Neighbor states and their meanings	570

Key Topic Element	Description	Page Number
List	Rules for setting the router ID	574
Example 22-14	Differences in show ip ospf interface output with OSPF interface configuration	579

Key Terms You Should Know

OSPF neighbor
OSPF router-id

Command References

Tables 22-5 and 22-6 list configuration and verification commands used in this chapter. As an easy review exercise, cover the left column in a table, read the right column, and try to recall the command without looking. Then repeat the exercise, covering the right column, and try to recall what the command does.

Table 22-5 Chapter 22 Configuration Command Reference

Command	Description
router ospf <i>process-id</i>	Global command that enters OSPF configuration mode for the listed process
network <i>ip-address wildcard-mask area</i> <i>area-id</i>	Router subcommand that enables OSPF on interfaces matching the address/wildcard combination and sets the OSPF area

Command	Description
ip ospf <i>process-id</i> area <i>area-number</i>	Interface subcommand to enable OSPF on the interface and to assign the interface to a specific OSPF area
ip ospf cost <i>interface-cost</i>	Interface subcommand that sets the OSPF cost associated with the interface
bandwidth <i>bandwidth</i>	Interface subcommand that directly sets the interface bandwidth (Kbps)
auto-cost reference-bandwidth <i>number</i>	Router subcommand that tells OSPF the numerator in the $\text{Reference_bandwidth}/\text{Interface_bandwidth}$ formula used to calculate the OSPF cost based on the interface bandwidth
router-id <i>id</i>	OSPF command that statically sets the router ID
interface loopback <i>number</i>	Global command to create a loopback interface and to navigate to interface configuration mode for that interface

Table 22-6 [Chapter 22](#) EXEC Command Reference

Command	Description
show ip ospf	Lists information about the OSPF process running on the router, including the OSPF router ID, areas to which the router connects, and the number of interfaces in each area.

Command	Description
show ip ospf interface brief	Lists the interfaces on which the OSPF protocol is enabled (based on the network commands), including passive interfaces.
show ip ospf interface <i>[type number]</i>	Lists a long section of settings, status, and counters for OSPF operation on all interfaces, or on the listed interface, including the Hello and Dead Timers.
show ip protocols	Shows routing protocol parameters and current timer values.
show ip ospf neighbor <i>[type number]</i>	Lists brief output about neighbors, identified by neighbor router ID, including current state, with one line per neighbor; optionally, limits the output to neighbors on the listed interface.
show ip ospf neighbor <i>neighbor-ID</i>	Lists the same output as the show ip ospf neighbor detail command, but only for the listed neighbor (by neighbor RID).
show ip ospf database	Lists a summary of the LSAs in the database, with one line of output per LSA. It is organized by LSA type (first type 1, then type 2, and so on).
show ip route	Lists all IPv4 routes.

Command Description	
show ip route ospf	Lists routes in the routing table learned by OSPF.
clear ip ospf process	Resets the OSPF process, resetting all neighbor relationships and also causing the process to make a choice of OSPF RID.

Chapter 23

Implementing Optional OSPF Features

This chapter covers the following exam topics:

3.0 IP Connectivity

3.2 Determine how a router makes a forwarding decision by default

3.2.b Administrative distance

3.2.c Routing protocol metric

3.4 Configure and verify single area OSPFv2

3.4.a Neighbor adjacencies

3.4.b Point-to-point

3.4.c Broadcast (DR/BR selection)

3.4.d Router ID

The previous chapter showed how to configure the core OSPF settings to make OSPF work. This chapter examines a variety of optional OSPF settings, chosen for two reasons. First, the CCNA exam topics mention or imply coverage of many of the optional features mentioned in this chapter.

Second, the optional features listed here happen to be relatively popular in production networks.

The chapter begins with a section about OSPF network types. As a setting on each interface, the OSPF network type dictates whether the router attempts to dynamically discover neighbors, and once discovered, whether routers on the link use a designated router (DR) or not. The section also discusses how to influence which router wins the DR election using OSPF priority and router IDs (RIDs).

The final section then moves on to discuss a variety of smaller optional OSPF configuration topics. The features include topics such as how to use passive interfaces, how to change OSPF costs (which influences the routes OSPF chooses), and how to create a default route advertised by OSPF.

“Do I Know This Already?” Quiz

Take the quiz (either here or use the PTP software) if you want to use the score to help you decide how much time to spend on this chapter. The letter answers are listed at the bottom of the page following the quiz. [Appendix C](#), found both at the end of the book as well as on the companion website, includes both the answers and explanations. You can also find both answers and explanations in the PTP testing software.

Table 23-1 “Do I Know This Already?” Foundation Topics Section-to-Question Mapping

Foundation Topics Section	Questions
OSPF Network Types	1-4
Additional Optional OSPFv2 Features	5-6

1. Routers R1 and R2, with router IDs 1.1.1.1 and 2.2.2.2, connect over an Ethernet WAN link. If using all default OSPF settings, if the WAN link initializes for both routers at the same time, which of the following answers are true? (Choose two answers.)

- a. Router R1 will become the DR.
- b. Router R1 will dynamically discover the existence of Router R2.
- c. Router R2 will be neither the DR nor the BDR.
- d. Router R1's **show ip ospf neighbor** command will list R2 with a state of "FULL/DR."

2. Routers R1 and R2, with router IDs 1.1.1.1 and 2.2.2.2, connect over an Ethernet WAN link. The configuration uses all defaults, except giving R1 an interface priority of 11 and changing both routers to use OSPF network type point-to-point. If the WAN link initializes for both routers at the same time, which of the following answers are true? (Choose two answers.)

- a. Router R1 will become the DR.
- b. Router R1 will dynamically discover the existence of Router R2.
- c. Router R2 will be neither the DR nor the BDR.
- d. Router R2's **show ip ospf neighbor** command will list R1 with a state of "FULL/DR."

3. Per the command output, with how many routers is Router R9 fully adjacent over its Gi0/0 interface?

[Click here to view code image](#)

```
# show ip ospf interface brief
Interface  PID  Area  IP Address/Mask  Cost  State  Nbrs
Gi0/0      1    0     10.1.1.1/24      1     DROTH 2/5
```



a. 7

b. 0

c. 5

d. 2

4. Routers R1 and R2, which use default priority settings, become neighbors, with R1 as the DR and R2 as the BDR. The engineer then configures R2's interface to use OSPF priority 100. Which answers correctly predict any changes in the OSPF neighbor relationship?

a. Router R2 will immediately become the DR.

b. Router R2 will become the DR after the neighbor relationship fails.

c. Router R2 will immediately stop filling the BDR role.

d. Router R2 will become the DR after four OSPF Hello intervals.

5. Which of the following configuration settings on a router does not influence which IPv4 route a router chooses to add to its IPv4 routing table when using OSPFv2?

a. **auto-cost reference-bandwidth**

b. **delay**

c. **bandwidth**

d. **ip ospf cost**

6. A network engineer configures the **ip ospf hello-interval 15** subcommand under the interfaces that connect OSPF neighbors R1 and R2 but with no use of

the **ip ospf dead-interval** subcommand. Eventually, Router R1's OSPF process fails, but the link between R1 and R2 remains working. How long after Router R1's last Hello does R2 consider its neighbor relationship with R1 to fail?

- a. 10 seconds
- b. 15 seconds
- c. 40 seconds
- d. 60 seconds

Answers to the “Do I Know This Already?” quiz:

1 B, D

2 B, C

3 D

4 B

5 B

6 D

Foundation Topics

OSPF Network Types

Two CCNA 200-301 V1.1 exam topics might be completely misunderstood without taking a closer look at some default OSPF settings. In particular, the following exam topics refer to a specific per-interface OSPF setting called the *network type*—even listing the keywords used to configure the setting in the exam topics:

3.4.b: **point-to-point**

3.4.c: **broadcast** (DR/BDR selection)

OSPF includes a small number of network types as a setting on each OSPF-enabled interface. The setting tells the router whether or not to dynamically discover OSPF neighbors (versus requiring the static configuration of the neighboring router's IP address) and whether or not the router should attempt to use a designated router (DR) and backup designated router (BDR) in the subnet. Of the two OSPF network types included in the CCNA exam topics, both cause routers to dynamically discover neighbors, but one calls for the use of a DR, whereas the other does not. [Table 23-2](#) summarizes the features of the two OSPF network types mentioned in the exam topics.



Table 23-2 Two OSPF Network Types and Key Behaviors

Network Type Keyword	Dynamically Discovers Neighbors	Uses a DR/BDR
broadcast	Yes	Yes
point-to-point	Yes	No

The rest of this first major section of the chapter explores each type.

The OSPF Broadcast Network Type

OSPF defaults to use a **broadcast network type** on all types of Ethernet interfaces. Note that all the Ethernet interfaces in the examples in [Chapter 22, “Implementing Basic OSPF Features,”](#) relied on that default setting.

To see all the details of how the OSPF broadcast network type works, this chapter begins with a different design than the examples in [Chapter 22](#), instead using a single-area design that connects four routers to the same subnet, as shown in [Figure 23-1](#). All links reside in area 0, making the design a single-area design.

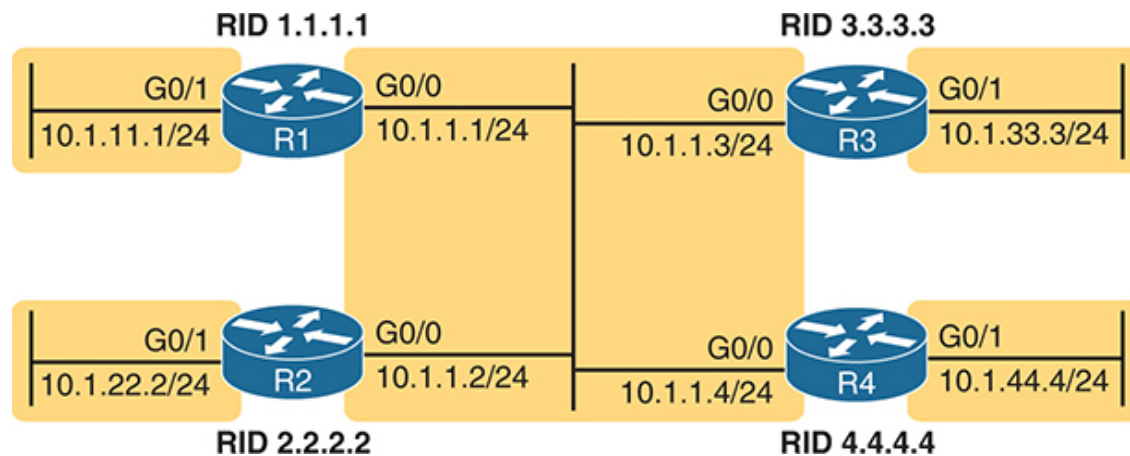


Figure 23-1 *The Single-Area Design Used in This Chapter*

To get a sense for how OSPF operates with the broadcast network type, imagine that all four routers use a straightforward OSPF interface configuration like the Router R1 configuration shown in [Example 23-1](#). Both GigabitEthernet interfaces on all four routers default to use network type broadcast. Note that the configuration on routers R2, R3, and R4 mirrors R1's configuration except that they use router IDs 2.2.2.2, 3.3.3.3, and 4.4.4.4, respectively, and they use the IP addresses shown in the figure.

Example 23-1 *R1 OSPF Configuration to Match [Figure 23-1](#)*
[Click here to view code image](#)

```
router ospf 1
  router-id 1.1.1.1
```

```
!  
interface gigabitEthernet0/0  
 ip ospf 1 area 0  
!  
interface gigabitEthernet0/1  
 ip ospf 1 area 0
```

This simple design gives us a great backdrop from which to observe the results of the broadcast network type on each router. Both interfaces (G0/0 and G0/1) on each router use the broadcast network type and perform the following actions:

- Attempt to discover neighbors by sending OSPF Hellos to the 224.0.0.5 multicast address (an address reserved for sending packets to all OSPF routers in the subnet)
- Attempt to elect a DR and BDR on each subnet
- On the interface with no other routers on the subnet (G0/1), become the DR
- On the interface with three other routers on the subnet (G0/0), be either DR, BDR, or a DROther router
- When sending OSPF messages to the DR or BDR, send the messages to the all-OSPF-DRs multicast address 224.0.0.6

[Example 23-2](#) shows some of the results using the **show ip ospf neighbor** command. Note that R1 lists R2, R3, and R4 as neighbors (based on their 2.2.2.2, 3.3.3.3, and 4.4.4.4 router IDs), confirming that R1 dynamically discovered the other routers. Also, note that the output lists 4.4.4.4 as the DR and 3.3.3.3 as the BDR.

Example 23-2 *R1's List of Neighbors*

[Click here to view code image](#)

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	
2.2.2.2	1	2WAY/DROTHER	00:00:35	10.1.1.2	(
3.3.3.3	1	FULL/BDR	00:00:33	10.1.1.3	(
4.4.4.4	1	FULL/DR	00:00:35	10.1.1.4	(

Verifying Operations with Network Type Broadcast

As discussed in the section “[Using Designated Routers on Ethernet Links](#)” in [Chapter 21](#), “[Understanding OSPF Concepts](#),” all discovered routers on the link should become neighbors and at least reach the 2-way state. For all neighbor relationships that include the DR and/or BDR, the neighbor relationship should further reach the *full* state. That section defined the term *fully adjacent* as a special term that refers to neighbors that reach this full state.

The design in [Figure 23-1](#), with four routers on the same LAN, provides just enough routers so that one neighbor relationship will remain in a 2-way state and not reach the full state, as a perfectly normal way for OSPF to operate. [Figure 23-2](#) shows the current conditions when the **show** commands in this chapter were gathered, with R4 as the DR, R3 as the BDR, and with R1 and R2 as DROther routers.

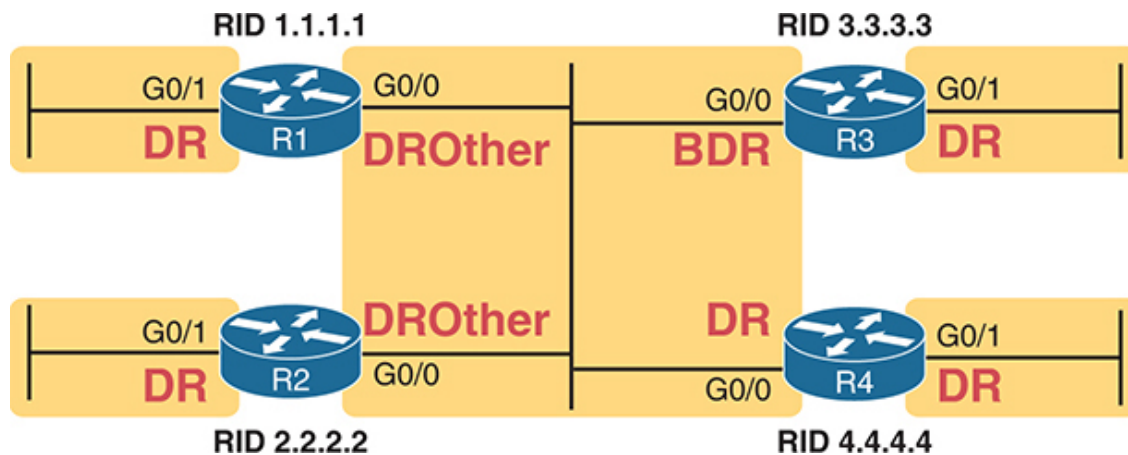


Figure 23-2 OSPF DR, BDR, and DROther Roles in the Network

Now consider Router R1's neighbors as listed in [Example 23-2](#). R1 has three neighbors, all reachable out its G0/0 interface. However, R1's **show ip ospf neighbor** command refers to the state of R1's relationship with the neighbor: 2-way with router 2.2.2.2. Because both R1 and R2 currently serve as DROther routers—that is, they wait ready to become the BDR if either the DR or BDR fails—their neighbor relationship remains in a 2-way state.

Examining [Example 23-2](#) one last time, R1, as a DROther router itself, has two neighbor relationships that reach a full state: R1's neighbor adjacency with DR R4 and R1's neighbor adjacency with BDR R3. But R1 has a total of three neighbors, all reachable off R1's G0/0 interface.

The next example emphasizes that R1 has three neighbors off its G0/0 interface, with only two as fully adjacent. The far right of the **show ip ospf interface brief** command output in [Example 23-3](#) shows "2/3," meaning two fully adjacent neighbors and three total neighbors on that interface. Also, note that this command's "State" column differs from the **show ip ospf neighbor** commands, because it lists the local router's role on the interface, with R1's G0/1 acting as DR and R1's G0/0 acting as a DROther router.



Example 23-3 Router R1 OSPF Interfaces: Local Role and Neighbor Counts

[Click here to view code image](#)

```
R1# show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State
Gi0/1	1	0	10.1.11.1/24	1	DR
Gi0/0	1	0	10.1.1.1/24	1	DROTH

So far, this topic has described the effect of the OSPF broadcast network type by taking advantage of the default setting on Ethernet interfaces. To see the setting, use the **show ip ospf interface** command, as shown in [Example 23-4](#). The first highlighted item identifies the network type. However, this command's output restates many of the facts seen in both the **show ip ospf neighbor** and **show ip ospf interface brief** commands in [Examples 23-2](#) and [23-3](#), so take the time to browse through all of [Example 23-4](#) and focus on the additional highlights to see those familiar items.

Example 23-4 Displaying OSPF Network Type Broadcast

[Click here to view code image](#)

```
R1# show ip ospf interface g0/0
```

GigabitEthernet0/0 is up, line protocol is up

Internet Address 10.1.1.1/24, Area 0, Attached via Interface Ethernet0/0, Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1

Topology-MTID	Cost	Disabled	Shutdown	Topology Name
0	1	no	no	Broadcast

```

v      1      110      110      base
Enabled by interface config, including secondary ip addresses
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 4.4.4.4, Interface address 10.1.1.4
Backup Designated router (ID) 3.3.3.3, Interface address 10.1.1.4
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit
  oob-resync timeout 40
  Hello due in 00:00:00
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1/1, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 3, Adjacent neighbor count is 2
  Adjacent with neighbor 3.3.3.3 (Backup Designated Router)
  Adjacent with neighbor 4.4.4.4 (Designated Router)
Suppress hello for 0 neighbor(s)

```

Although you would not need to configure an Ethernet interface to use the broadcast network type, for reference, IOS defaults to that setting on Ethernet interfaces per default command **ip ospf network broadcast**.

Using Priority and RID to Influence the DR/BDR Election

In some cases, you might want to influence the OSPF DR election. However, before deciding that makes sense in every case, note that OSPF DR/BDR election rules will not result in a specific router always being the DR, and another always being the BDR, assuming that each is up and working. In short, here are the rules once a DR and BDR have been elected:

- If the DR fails, the BDR becomes the DR, and a new BDR is elected.
- When a better router enters the subnet, no preemption of the existing DR or BDR occurs.

As a result of these rules, while you can configure a router to be the best (highest priority) router to become the DR in an election, doing so only increases that router's statistical chances of being the DR at a given point in time. If the router with the highest priority fails, other routers will become DR and BDR, and the best router will not be DR again until the current DR and BDR fail, causing new elections.

However, in some cases, you may want to influence the DR/BDR election. To do so, use these settings, listed here in order of precedence:

- **The highest OSPF interface priority:** The highest value wins during an election, with values ranging from 0 to 255. (A value of 0 prevents the router from ever becoming the DR.)



- **The highest OSPF Router ID:** If the priority ties, the election chooses the router with the highest OSPF RID.

For example, imagine all four routers in the design shown in [Figure 23-1](#) trying to elect the DR and BDR at the same time—for instance, after a power hit in which all four routers power off and back on again. No prior DR or BDR exists at this point. They all participate in the election. They all tie with default priority values of 1 (see [Example 23-4](#) for R1's priority in the **show ip ospf interface** command output). In

this case, R4 becomes the DR based on the numerically highest RID of 4.4.4.4, and R3 becomes the BDR based on the next highest RID of 3.3.3.3.

To influence the election, you could set the various RIDs with your preferred router with the highest RID value. However, many networks choose OSPF router IDs to help identify the router easily rather than choosing to make one value higher than its neighbor. Instead, using the **OSPF priority** setting makes better sense. For instance, if an engineer preferred that R1 be the DR, the engineer could add the configuration in [Example 23-5](#) to set R1's interface priority to 99.

Example 23-5 *Influencing DR/BDR Election Using OSPF Priority*

[Click here to view code image](#)

```
R1# configure terminal
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface g0/0
R1(config-if)# ip ospf priority 99
R1(config-if)# ^Z
R1#
R1# show ip ospf interface g0/0 | include Priority
    Transmit Delay is 1 sec, State DROTHER, Priority 99

R1# show ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address
2.2.2.2          1    2WAY/DROTHER    00:00:36   10.1.1.2
3.3.3.3          1    FULL/BDR        00:00:30   10.1.1.3
4.4.4.4          1    FULL/DR         00:00:37   10.1.1.4

R1# show ip ospf interface brief
Interface      PID   Area           IP Address/Mask    Cost   State
Gi0/0/1        1     0              10.1.1.1/24        1      DR
```

Gi0/1	1	0	10.1.1.1/24	1	DR
Gi0/0	1	0	10.1.1.1/24	1	DROTH

The top of the example shows R1's interface priority value now as 99, and the **show ip ospf interface G0/0** command that follows confirms the setting. However, the last two commands confirm that OSPF does not preempt the existing DR or BDR. Note that the **show ip ospf neighbor** command still lists R4's state as DR, meaning R4 still acts as the DR, so R1, with a higher priority, did not take over. The final command, **show ip ospf interface brief**, lists R1's State (role) as DROTH.

Just to complete the process and show R1 winning a DR election, [Example 23-6](#) shows the results after forcing a free election by failing the LAN switch that sits between the four routers. As expected, R1 wins and becomes DR due to its higher priority, with the other three routers tying based on priority. R4 wins between R2, R3, and R4 due to its higher RID to become the BDR.

Example 23-6 *Results of a Completely New DR/BDR Election*

[Click here to view code image](#)

```
! Not shown: LAN fails, and then recovers, causing a new OSPF Election
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address
2.2.2.2	1	FULL/DROTHER	00:00:37	10.1.1.2
3.3.3.3	1	FULL/DROTHER	00:00:38	10.1.1.3
4.4.4.4	1	FULL/BDR	00:00:38	10.1.1.4

```
R1# show ip ospf interface brief
```

Interface	RID	Area	DR Address/Mask	Cost	State
-----------	-----	------	-----------------	------	-------

Interface	PID	Area	IP Address/Mask	Cost	Status
Gi0/1	1	0	10.1.11.1/24	1	DR
Gi0/0	1	0	10.1.1.1/24	1	DR

Note

If you have begun to mentally compare OSPF DR elections to STP elections, keep some key differences in mind. First, STP uses lowest-is-best logic, whereas OSPF uses highest-is-best. STP elections allow preemption—for instance, if a new switch appears with a superior (lower) bridge ID (BID) than the current root, the new switch becomes the root switch. OSPF does not preempt, so a new router on a link, with the highest priority, does not take over as DR or BDR. Instead, it wins future elections, eventually becoming the DR.

The OSPF Point-to-Point Network Type

The other OSPF network type mentioned in the CCNA 200-301 V1.1 blueprint, point-to-point, works well for data links that by their nature have just two routers on the link. For example, consider the topology in [Figure 23-3](#), which shows Router R1 with three WAN links—two Ethernet WAN links and one serial link.

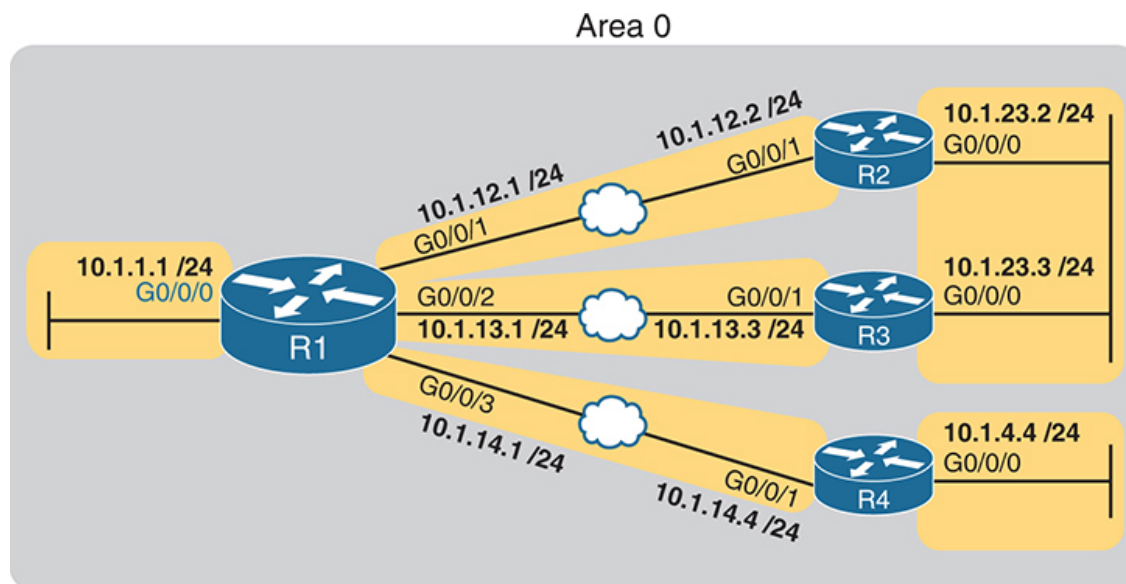


Figure 23-3 *Sample OSPF Design for Upcoming Examples*

First, consider the older-style serial links (not used in the figure). Serial links between routers do not support the capability to add a third router to the link. With only two devices on the link, using a DR/BDR provides no advantage; instead, it adds a little extra convergence time. Using a network type of point-to-point tells the router to not use a DR/BDR on the link, which makes sense on serial links. In fact, IOS defaults to a setting of **ip ospf network point-to-point** on serial interfaces.

While you might not see many serial links in networks today, some other point-to-point WAN topologies exist, including some Ethernet WAN links. All the Ethernet WAN links used in this book happen to use a point-to-point Ethernet WAN service called an Ethernet Private Wire Service or simply an Ethernet Line (E-Line). For that service, the service provider will send Ethernet frames between two devices (routers) connected to the service, but only those two devices. In other words, an E-Line is a point-to-point service in concept. So, like serial links, Ethernet WAN links with only two routers connected gain no advantage by using

a DR/BDR. As a result, many engineers prefer to instead use an OSPF **point-to-point network type** on Ethernet WAN links that in effect act as a point-to-point link.

[Example 23-7](#) shows the configuration of Router R1's G0/0/1 interface in [Figure 23-3](#) to use OSPF network type point-to-point. R2, on the other end of the WAN link, would need the same configuration command on its matching interface.

Example 23-7 *OSPF Network Type Point-to-Point on an Ethernet WAN Interface on R1*

[Click here to view code image](#)

```
R1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface g0/0/1
R1(config-if)# ip ospf network point-to-point
R1(config-if)#

R1# show ip ospf interface g0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  Internet Address 10.1.12.1/24, Area 0, Attached via Interface I
  Process ID 1, Router ID 1.1.1.1, Network Type POINT_TO_POINT, (
  Topology-MTID      Cost      Disabled      Shutdown      Topology Nar
           0           4         no           no           Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retran:
    oob-resync timeout 40
    Hello due in 00:00:01
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/3/3, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)

  Last flood scan length is 1, maximum is 2
```

```

Last flood scan length is 1, maximum is 3
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 2.2.2.2
Suppress hello for 0 neighbor(s)

```

Note the highlighted portions of the **show** command in [Example 23-7](#). The first two highlights note the network type. The final highlight with two lines notes that R1 has one neighbor on the interface, a neighbor with which it has become fully adjacent per the output.

[Example 23-8](#) closes this section with a confirmation of some of those facts with two more commands. Note that the **show ip ospf neighbor** command on R1 lists Router R2 (RID 2.2.2.2) with a full state, but with no DR or BDR designation, instead listing a -. The - acts as a reminder that the link does not use a DR/BDR. The second command, **show ip ospf interface brief**, shows the state (the local router's role) as P2P, which is short for point-to-point, with a counter of 1 for the number of fully adjacent neighbors and total number of neighbors.



Example 23-8 *OSPF Network Type Point-to-Point on an Ethernet WAN Interface on R1*

[Click here to view code image](#)

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	
2.2.2.2	0	Full /	00:00:30	10.1.12.2	/

```
2.2.2.2          0 FULL/ -          00:00:39          10.1.12.2          0
! lines omitted for brevity

R1# show ip ospf interface brief
Interface      PID    Area          IP Address/Mask    Cost    State
Gi0/0/1        1      0             10.1.12.1/24       4       P2P
! lines omitted for brevity
```

When using Ethernet WAN links that behave as a point-to-point link, consider using OSPF network type point-to-point rather than using the default broadcast type.

Additional Optional OSPFv2 Features

This final major section of the chapter discusses some popular but optional OSPFv2 configuration features, as listed here in their order of appearance:

- Passive interfaces
- Default routes
- Metrics
- Hello and Dead intervals

OSPF Passive Interfaces

Once OSPF has been enabled on an interface, the router tries to discover neighboring OSPF routers and form a neighbor relationship. To do so, the router sends OSPF Hello messages on a regular time interval (called the Hello interval). The router also listens for incoming Hello messages from potential neighbors.

Sometimes, a router does not need to form neighbor relationships with neighbors on an interface. Often, no other

routers exist on a particular link, so the router has no need to keep sending those repetitive OSPF Hello messages. In such cases, an engineer can make the interface passive, which means

- OSPF continues to advertise about the subnet that is connected to the interface.



- OSPF no longer sends OSPF Hellos on the interface.
- OSPF no longer processes any received Hellos on the interface.

The result of enabling OSPF on an interface but then making it passive is that OSPF still advertises about the connected subnet, but OSPF also does not form neighbor relationships over the interface.

To configure an interface as passive, two options exist. First, you can add the following command to the configuration of the OSPF process, in router configuration mode:

passive-interface *type number*

Alternately, the configuration can change the default setting so that all interfaces are passive by default and then add a **no passive-interface** command for all interfaces that need to not be passive:

passive-interface default

no passive-interface *type number*

For example, in the sample internetwork in [Figure 23-4](#), Router R1 connects to a LAN with its G0/0/0 interface. With no other OSPF routers on that LAN, R1 will never discover an OSPF neighbor on that interface. [Example 23-9](#) shows two

alternative configurations to make R1's G0/0/0 passive to OSPF.

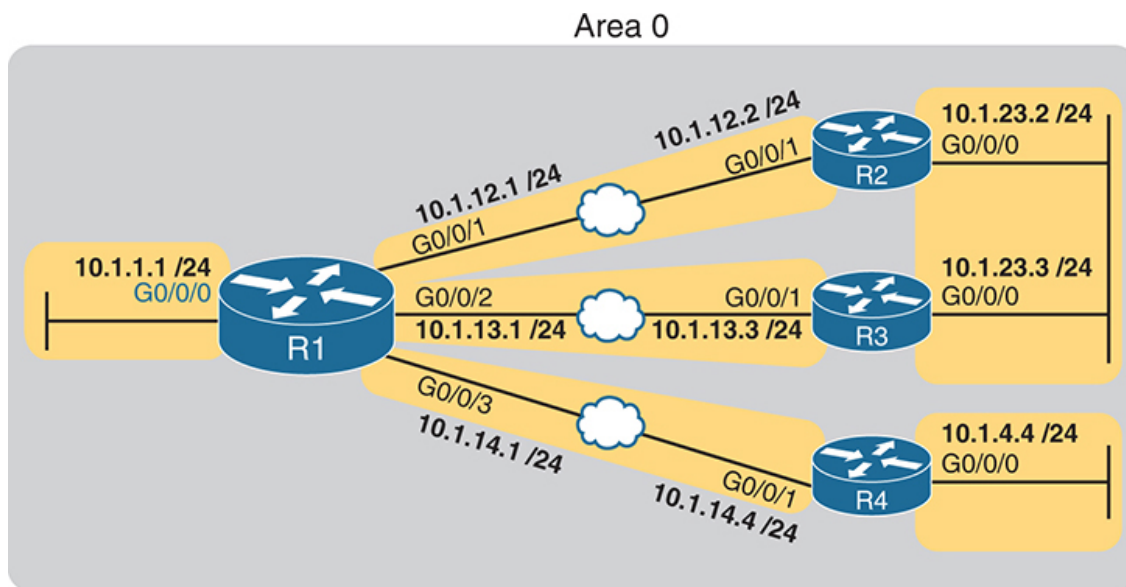


Figure 23-4 Sample OSPF Design for Upcoming Examples

Example 23-9 Configuring Passive Interfaces on R1 from Figure 23-4

[Click here to view code image](#)

```
! First, make each interface passive directly
```

```
router ospf 1  
  passive-interface GigabitEthernet0/0/0
```

```
! Or, change the default to passive, and make the other interface
```

```
router ospf 1  
  passive-interface default  
  no passive-interface GigabitEthernet0/0/1  
  no passive-interface GigabitEthernet0/0/2  
  no passive-interface GigabitEthernet0/0/3
```

In real internetworks, use the configuration style that requires the least number of commands. For example, a router with 20 interfaces, 18 of which are passive to OSPF, has far fewer configuration commands when using the **passive-interface default** command to change the default to passive. If only two of those 20 interfaces need to be passive, use the default setting, in which all interfaces are not passive, to keep the configuration shorter.

Interestingly, OSPF makes it a bit of a challenge to use **show** commands to find out whether an interface is passive. The **show running-config** command lists the configuration directly, but if you cannot get into enable mode to use this command, note these two facts:

The **show ip ospf interface brief** command lists all interfaces on which OSPF is enabled, including **passive interfaces**.

The **show ip ospf interface** command lists several output lines per interface with a single line that mentions that the interface is passive.

[Example 23-10](#) shows these commands on Router R1, based on the configuration shown in the top of [Example 23-9](#). Note that passive interface G0/0/0 appears in the output of **show ip ospf interface brief**.

Example 23-10 *Displaying Passive Interfaces*

[Click here to view code image](#)

R1# show ip ospf interface brief					
Interface	PID	Area	IP Address/Mask	Cost	State
Gi0/0/0	1	0	10.1.1.1/24	1	DR
Gi0/0/1	1	0	10.1.12.1/24	1	BDR
Gi0/0/2	1	0	10.1.13.1/24	1	BDR
Gi0/0/3	1	0	10.1.14.1/24	1	DR

```

R1# show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.1.1/24, Area 0, Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost:
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
            0          1          no           no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.1.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit
  oob-resync timeout 40
  No Hellos (Passive interface)
! Lines omitted for brevity

```

OSPF Default Routes

Chapter 17, “Configuring IPv4 Addresses and Static Routes,” showed some of the uses and benefits of default routes, with examples of static default routes. For those exact same reasons, networks can use OSPF to advertise default routes.

The most classic case for using a routing protocol to advertise a default route has to do with an enterprise’s connection to the Internet. As a strategy, the enterprise engineer uses these design goals:

- All routers learn specific (nondefault) routes for subnets inside the company; a default route is not needed when forwarding packets to these destinations.
- One router connects to the Internet, and it has a default route that points toward the Internet.
- All routers should dynamically learn a default route, used for all traffic going to the Internet, so that all

packets destined to locations in the Internet go to the one router connected to the Internet.

Figure 23-5 shows the idea of how OSPF advertises the default route, with the specific OSPF configuration. In this case, a company connects to an ISP with its Router R1. That router has a static default route (destination 0.0.0.0, mask 0.0.0.0) with a next-hop address of the ISP router. Then the use of the OSPF **default-information originate** command (Step 2) makes the router advertise a default route using OSPF to the remote routers (B1 and B2).

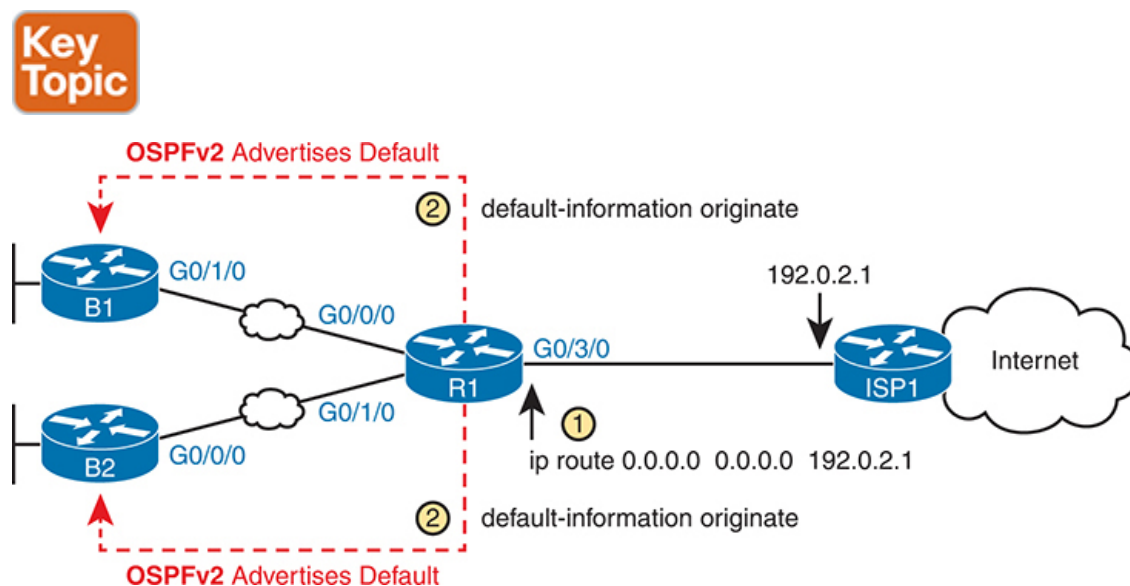


Figure 23-5 *Using OSPF to Create and Flood a Default Route*

Figure 23-6 shows the default routes that result from OSPF's advertisements in Figure 23-5. On the far left, the branch routers all have OSPF-learned default routes, pointing to R1. R1 itself also needs a default route, pointing to the ISP router, so that R1 can forward all Internet-bound traffic to the ISP.

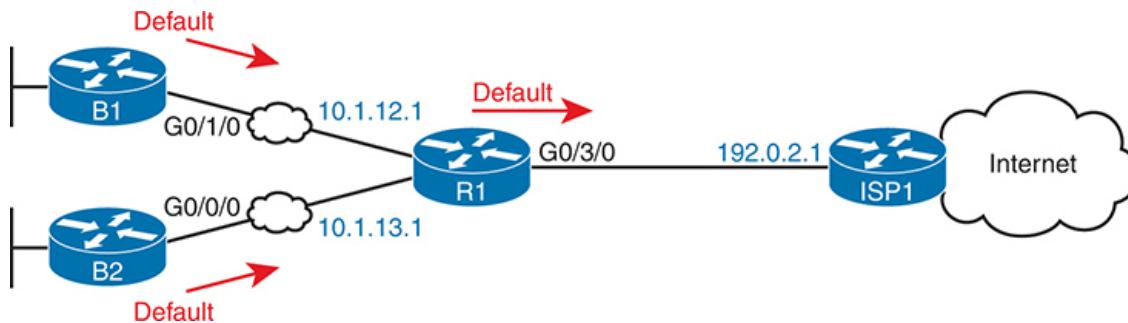


Figure 23-6 *Default Routes Resulting from the default-information originate Command*

[Example 23-11](#) first highlights the two commands relevant to the default route on Router R1 in [Figure 23-6](#). The router has a default static route referring to the ISP router as the next-hop router (192.0.2.1). It also has a **default-information originate** command configured under the OSPF process, telling the router to advertise a default route to other OSPF routers—but only if the router currently has a default route.

Example 23-11 *Relevant OSPF Configuration on Internet Edge Router R1*

[Click here to view code image](#)

```

! Excerpt from a show running-config command on router R1
ip route 0.0.0.0 0.0.0.0 192.0.2.1
!
router ospf 1
  network 10.0.0.0 0.255.255.255 area 0
  router-id 1.1.1.1
  default-information originate
  
```

[Example 23-12](#) shows the status based on the configuration shown in [Example 23-11](#). First, the top of the example confirms that R1 has a default route—that is, a route to 0.0.0.0/0. The “Gateway of last resort,” which refers to the

default route currently used by the router, points to next-hop IP address 192.0.2.1, which is the ISP router's IP address.

Example 23-12 *Default Routes on Routers R1 and B1*

[Click here to view code image](#)

```
! The next command is from Router R1. Note the static code for the
R1# show ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
! Rest of the legend omitted for brevity

Gateway of last resort is 192.0.2.1 to network 0.0.0.0

S*    0.0.0.0/0 [254/0] via 192.0.2.1
```

```
! The next command is from router B1; notice the External route (
B1# show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external ty
      E1 - OSPF external type 1, E2 - OSPF external type 2
! Rest of the legend omitted for brevity

Gateway of last resort is 10.1.12.1 to network 0.0.0.0

O*E2   0.0.0.0/0 [110/1] via 10.1.12.1, 00:20:51, GigabitEthernet
        10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.3.0/24 [110/3] via 10.1.12.1, 00:20:51, GigabitEthernet
O       10.1.13.0/24 [110/2] via 10.1.12.1, 00:20:51, GigabitEthernet
```

The bottom half of the example shows router B1's OSPF-learned default route. B1 lists a route for 0.0.0.0/0 as well.

The next-hop router in this case is 10.1.12.1, which is Router R1's IP address on the WAN link. The code on the far left is O*E2, meaning an OSPF-learned route, which is a default route, and is specifically an external OSPF route. B1's gateway of last resort setting uses that one OSPF-learned default route, with next-hop router 10.1.12.1.

Finally, the OSPF subcommand **default-information originate** makes the logic conditional: only advertise a default route into OSPF if you already have some other default route. Adding the **always** keyword to the command (**default-information originate always**) tells OSPF to always advertise a default route into OSPF regardless of whether a default route currently exists.

OSPF Metrics (Cost)

The section “[Calculating the Best Routes with SPF](#)” in [Chapter 21](#) discussed how SPF calculates the metric for each route, choosing the route with the best metric for each destination subnet. OSPF routers can influence that choice by changing the OSPF interface cost using three different configuration options:

- Directly, using the interface subcommand **ip ospf cost x**.
- Using the default calculation per interface, and changing the **interface bandwidth** setting, which changes the calculated value.
- Using the default calculation per interface, and changing the OSPF **reference bandwidth** setting, which changes the calculated value.

Setting the Cost Directly

Setting the cost directly requires a simple configuration command, as shown in [Example 23-13](#). The example sets

the cost of two interfaces on Router R1. This example uses the [Figure 23-4](#) topology, with a single area, with OSPF enabled on all interfaces shown in the figure. The **show ip ospf interface brief** command that follows details the cost of each interface. Note that the **show** command confirms the cost settings.

Example 23-13 Confirming OSPF Interface Costs

[Click here to view code image](#)

```
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface g0/0/1
R1(config-if)# ip ospf cost 4
R1(config-if)# interface g0/0/2
R1(config-if)# ip ospf cost 5
R1(config-if)# ^Z
R1#
R1# show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State
Gi0/0/0	1	0	10.1.1.1/24	1	DR
Gi0/0/1	1	0	10.1.12.1/24	4	DR
Gi0/0/2	1	0	10.1.13.1/24	5	BDR
Gi0/0/3	1	0	10.1.14.1/24	1	DR

The output also shows a cost value of 1 for the other Gigabit interfaces, which is the default OSPF cost for any interface faster than 100 Mbps. The next topic discusses how IOS determines the default cost values.

Setting the Cost Based on Interface and Reference Bandwidth

Routers use a per-interface *bandwidth* setting to describe the speed of the interface. Note that the interface bandwidth setting does not influence the actual transmission speed. Instead, the interface bandwidth acts as a configurable setting to represent the speed of the interface, with the option to configure the bandwidth to match the actual transmission speed...or not. To support this logic, IOS sets a default interface bandwidth value that matches the physical transmission speed when possible, but also allows the configuration of the interface bandwidth using the **bandwidth speed** interface subcommand.

OSPF (as well as other IOS features) uses the interface bandwidth to make decisions, with OSPF using the interface bandwidth in its calculation of the default OSPF cost for each interface. IOS uses the following formula to choose an interface's OSPF cost for interfaces that do not have an **ip ospf cost** command configured. IOS puts the interface's bandwidth in the denominator and an OSPF setting called the *reference bandwidth* in the numerator:

$$\text{Reference_bandwidth} / \text{Interface_bandwidth}$$

Note that while you can change both the interface bandwidth and reference bandwidth via configuration, because several IOS features make use of the interface bandwidth setting, you should avoid changing the interface bandwidth to influence the default OSPF cost.

Today, most companies override the default IOS reference bandwidth setting. Cisco chose the default setting (100 Mbps) decades ago in an era with much slower links. As a result, when using that default, any interface bandwidth of 100 Mbps or faster results in a calculated OSPF cost of 1. So, when you're relying on the OSPF cost calculation, it helps to configure the reference bandwidth to a speed faster than the fastest speed link in the network.

To see the issue, consider [Table 23-3](#), which lists several types of interfaces, the default interface bandwidth on those interfaces, and the OSPF cost calculated with the default OSPF reference bandwidth of 100 Mbps (that is, 100,000 Kbps). (OSPF rounds up for these calculations, resulting in a lowest possible OSPF interface cost of 1.)

Table 23-3 Faster Interfaces with Equal OSPF Costs

Interface	Interface Default Bandwidth (Kbps)	Formula (Kbps)	OSPF Cost
Serial	1544 Kbps	$100,000 / 1544$	64
Ethernet	10,000 Kbps	$100,000 / 10,000$	10
Fast Ethernet	100,000 Kbps	$100,000 / 100,000$	1
Gigabit Ethernet	1,000,000 Kbps	$100,000 / 1,000,000$	1
10 Gigabit Ethernet	10,000,000 Kbps	$100,000 / 10,000,000$	1
100 Gigabit Ethernet	100,000,000 Kbps	$100,000 / 100,000,000$	1

As you can see from the table, with a default reference bandwidth, all interfaces from Fast Ethernet's 100 Mbps and faster tie with their default OSPF cost. As a result, OSPF would treat a 100-Mbps link as having the same cost as a 10- or 100-Gbps link, which is probably not the right basis for choosing routes.

You can still use OSPF's default cost calculation (and many do) just by changing the reference bandwidth with the **auto-cost reference-bandwidth** *speed* OSPF mode subcommand. This command sets a value in a unit of megabits per second (Mbps). Set the reference bandwidth value to a value at least as much as the fastest link speed in the network, but preferably higher, in anticipation of adding even faster links in the future.

For instance, in an enterprise whose fastest links are 10 Gbps (10,000 Mbps), you could set all routers to use **auto-cost reference-bandwidth 10000**, meaning 10,000 Mbps or 10 Gbps. In that case, by default, a 10-Gbps link would have an OSPF cost of 1, while a 1-Gbps link would have a cost of 10, and a 100-Mbps link a cost of 100.

Better still, in that same enterprise, use a reference bandwidth of a faster speed than the fastest interface in the network, to allow room for higher speeds. For instance, in that same enterprise, whose fastest link is 10 Gbps, set the reference bandwidth to 40 Gbps or even 100 Gbps to be ready for future upgrades to use 40-Gbps links, or even 100-Gbps links. (For example, use the **auto-cost reference-bandwidth 100000** command, meaning 100,000 Mbps or 100 Gbps.) That causes 100-Gbps links to have an OSPF cost of 1, 40-Gbps links to have a cost of 2, 10-Gbps links to have a cost of 10, and 1-Gbps links to have a cost of 100.

Note

Cisco recommends making the OSPF reference bandwidth setting the same on all OSPF routers in an enterprise network.

For convenient study, the following list summarizes the rules for how a router sets its OSPF interface costs:

1. Set the cost explicitly, using the **ip ospf cost** x interface subcommand, to a value between 1 and 65,535, inclusive.



2. Although it should be avoided, change the interface bandwidth with the **bandwidth** *speed* command, with *speed* being a number in kilobits per second (Kbps).
3. Change the reference bandwidth, using router OSPF subcommand **auto-cost reference-bandwidth** *ref-bw*, with a unit of megabits per second (Mbps).

OSPF Hello and Dead Intervals

OSPF does a lot of interesting work with the OSPF Hello message when initializing neighbors. As explained in the section, “[Meeting Neighbors and Learning Their Router ID](#),” in [Chapter 21](#), an OSPF router uses Hello messages to announce the router’s presence on the link, with those messages sent to the 224.0.0.5 all-OSPF-routers multicast address. The router also listens for incoming Hello messages from potential neighbors. Upon hearing Hellos from each other, two routers check the settings revealed in the Hello, and if compatible, the two routers can proceed to become neighbors.

This section looks at the other end of the story: what happens when the neighbor relationship fails.

First, while the neighbor relationship continues to work, routers send Hello messages regularly per a per-interface **Hello interval** (also called the *Hello timer*). Cisco IOS defaults to a 10-second Hello interval on all Ethernet interfaces. Why? When a router no longer receives the

incoming Hello messages from a neighbor, the router believes the neighbor has failed, and takes the neighbor relationship down.

The process to decide when a neighbor fails uses the assumption on continual incoming Hello messages from the neighbor, combined with a second timer: the **Dead interval** (also called the *Dead timer*). The Dead timer tells the interface how long to wait. That is, how long since receiving a Hello message should a router wait before deciding that the neighbor failed.

Figure 23-7 shows an example of the interaction, focusing on what happens if Router R1 fails, and how Router R2 views the messages and timers. In this case, Router R1 loses power, but R2's Ethernet WAN link remains up because the link between R2 and the Ethernet WAN services does not have any problem. The sequence of events follows the circled step numbers as follows, using the default settings of a 10-second Hello timer and 40-second Dead timer:

1. Router R2 has a working neighbor relationship with R1 (full state).
2. R1 sends a Hello message, as usual; R2 resets its Dead timer to 40.
3. R1 sends another Hello message, 10 seconds later; R2 resets its Dead timer to 40.
4. R1 loses power; R2 receives no further Hellos from R1.
5. The Dead timer counts down to 0.
6. Router R2 changes the neighbor state for neighbor R1 to down and reconverges.

Under normal working conditions, with default settings, the Dead timer gets reset to 40, counts down to 30, and is reset to 40, over and over. However, the process also means that

you might want to configure to use a lower Dead interval to improve the speed with which routers react to failures.

Example 23-14 shows the straightforward interface subcommands to set the Hello and Dead intervals. The example is based on the network in [Figure 23-4](#), showing the configuration in Router R1, with the three WAN interfaces that connect to routers R2, R3, and R4. Although not shown, the neighboring routers must also be configured with the same Hello and Dead timer settings.

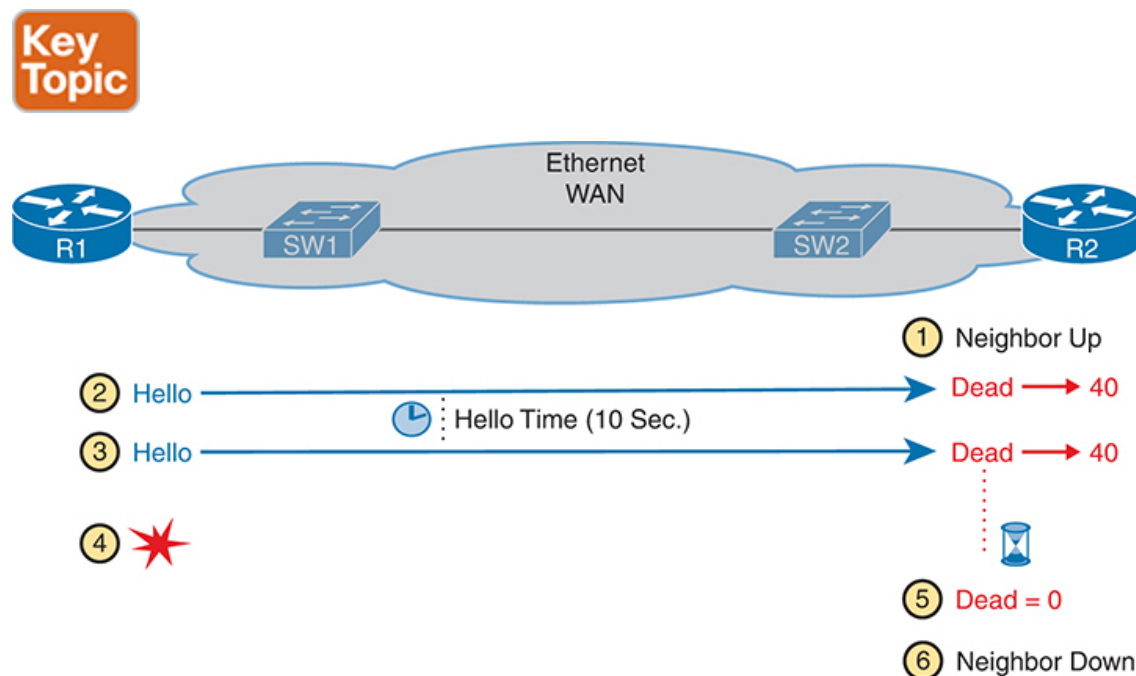


Figure 23-7 *An Example: R2's Use of the Hello and Dead Intervals*

Example 23-14 *Configuring OSPF Hello and Dead Intervals*

[Click here to view code image](#)

```
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
! Link connected to router R2
```

```
R1(config)# interface g0/0/1
R1(config-if)# ip ospf hello-interval 5
! Link connected to router R3
R1(config)# interface g0/0/2
R1(config-if)# ip ospf dead-interval 20
! Link connected to router R4
R1(config)# interface g0/0/3
R1(config-if)# ip ospf hello-interval 5
R1(config-if)# ip ospf dead-interval 15
R1(config-if)# Ctrl-z
R1#
```

[Example 23-15](#) confirms the settings configured in [Example 23-14](#), but with a twist on interface G0/0/1. IOS uses a Hello interval default of 10 seconds on Ethernet interfaces, but a default of four times the Hello interval for the Dead interval. For interface G0/0/1, because only the Hello interval was configured, IOS uses a default dead interval based on $4 \times$ Hello, or 20. However, the other two interfaces follow more predictable logic, summarized by this list, and then confirmed in the highlights in [Example 23-15](#):

- G0/0/1: Hello = 5 (configured) and Dead = 20 (calculated by IOS as $4 \times$ Hello)
- G0/0/2: Hello = 10 (IOS default) and Dead = 20 (configured)
- G0/0/3: Hello = 5 and Dead = 15 (both configured)

Example 23-15 *Confirming New OSPF Hello and Dead Intervals, Per Interface*

[Click here to view code image](#)

```
R1# show ip ospf interface g0/0/1 | include Hello
Timer intervals configured, Hello 5, Dead 20, Wait 20, Retransr
Hello due in 00:00:02

R1# show ip ospf interface g0/0/2 | include Hello
Timer intervals configured, Hello 10, Dead 20, Wait 20, Retran:
Hello due in 00:00:04

R1# show ip ospf interface g0/0/3 | include Hello
Timer intervals configured, Hello 5, Dead 15, Wait 15, Retransr
Hello due in 00:00:03
```

IOS enables us to make poor configuration choices with these two settings. As a fair warning, consider these points:

- The default settings create a 4:1 ratio between the dead:hello timers. That ratio requires four consecutive lost Hellos before the Dead timer would expire. No matter the numbers, consider using the same 4:1 ratio, or at least a 3:1 ratio for your chosen numbers.
- Two neighboring routers must use the same Dead and Hello timer settings. However, IOS allows you to configure different settings on potential neighbors, preventing them from becoming neighbors, so take care to ensure the settings match.
- IOS enables you to make a poor choice to configure a Dead interval smaller than the Hello interval. In that case, the Dead interval expires before a router receives the next Hello. The neighbor relationship fails and recovers repeatedly once per Hello interval. Instead, always set the Dead interval higher than the Hello interval.

Chapter Review

One key to doing well on the exams is to perform repetitive spaced review sessions. Review this chapter's material using either the tools in the book or interactive tools for the same material found on the book's companion website. Refer to the "[Your Study Plan](#)" element for more details. [Table 23-4](#) outlines the key review elements and where you can find them. To better track your study progress, record when you completed these activities in the second column.

Table 23-4 Chapter Review Tracking

Review Element	Review Date(s)	Resource Used
Review key topics		Book, website
Review key terms		Book, website
Answer DIKTA questions		Book, PTP
Review Config Checklists		Book, website
Review command tables		Book
Do labs		Blog
Watch videos		Website

Review All the Key Topics



Table 23-5 Key Topics for [Chapter 23](#)

Key Topic Element	Description	Page Number
Table 23-2	Two OSPF network types and key behaviors	586
Example 23-3	OSPF interfaces, local roles, and neighbor counts	589
List	Rules for electing OSPF DR/BDR	590
Example 23-8	Evidence of OSPF network type point-to-point	594
List	Actions taken by OSPF when the interface is passive	595
Figure 23-5	Using OSPF to create and flood a default route	597
List	Rules for how OSPF sets its interface cost settings	601
Figure 23-7	Conceptual view of OSPF Hello and Dead timers	603

Key Terms You Should Know

[broadcast network type](#)

[Dead interval](#)

[Hello interval](#)

[interface bandwidth](#)

[OSPF priority](#)

[passive interface](#)

[point-to-point network type](#)

[reference bandwidth](#)

Command References

Tables 23-6 and 23-7 list configuration and verification commands used in this chapter. As an easy review exercise, cover the left column in a table, read the right column, and try to recall the command without looking. Then repeat the exercise, covering the right column, and try to recall what the command does.

Table 23-6 Chapter 23 Configuration Command Reference

Command	Description
ip ospf network { broadcast point-to-point }	Interface subcommand used to set the OSPF network type on the interface
ip ospf priority <i>value</i>	Interface subcommand that sets the OSPF priority, used when electing a new DR or BDR
passive-interface <i>type number</i>	Router subcommand that makes the interface passive to OSPF, meaning that the OSPF process will not form neighbor relationships with neighbors reachable on that interface
passive-interface <i>default</i>	OSPF subcommand that changes the OSPF default for interfaces to be passive instead of active (not passive)
no passive-interface <i>type number</i>	OSPF subcommand that tells OSPF to be active (not passive) on that interface or subinterface

Command	Description
default-information originate [always]	OSPF subcommand to tell OSPF to create and advertise an OSPF default route, as long as the router has some default route (or to always advertise a default, if the always option is configured)
ip ospf cost <i>interface-cost</i>	Interface subcommand that sets the OSPF cost associated with the interface
bandwidth <i>bandwidth</i>	Interface subcommand that directly sets the interface bandwidth (Kbps)
auto-cost reference-bandwidth <i>number</i>	Router subcommand that tells OSPF the numerator in the Reference_bandwidth / Interface_bandwidth formula used to calculate the OSPF cost based on the interface bandwidth
ip ospf hello-interval <i>time</i>	Interface subcommand to set the OSPF Hello interval
ip ospf dead-interval <i>time</i>	Interface subcommand to set the OSPF Dead interval

Table 23-7 [Chapter 23](#) EXEC Command Reference

Command	Description
---------	-------------

Command	Description
show ip ospf interface brief	Lists the interfaces on which the OSPF protocol is enabled (based on the network commands), including passive interfaces
show ip ospf interface <i>[type number]</i>	Lists a long section of settings, status, and counters for OSPF operation on all interfaces, or on the listed interface, including the Hello and Dead timers
show ip ospf neighbor <i>[type number]</i>	Lists brief output about neighbors, identified by neighbor router ID, including current state, with one line per neighbor; optionally, limits the output to neighbors on the listed interface
show ip ospf neighbor <i>neighbor-ID</i>	Lists the same output as the show ip ospf neighbor detail command, but only for the listed neighbor (by neighbor RID)
show ip route	Lists all IPv4 routes
show ip route ospf	Lists routes in the routing table learned by OSPF

Chapter 24

OSPF Neighbors and Route Selection

This chapter covers the following exam topics:

3.0 IP Connectivity

3.1 Interpret the components of routing table

3.1.e Administrative distance

3.2 Determine how a router makes a forwarding decision by default

3.2.a Longest prefix match

3.2.b Administrative distance

3.2.c Routing protocol metric

3.4 Configure and verify single area OSPFv2

3.4.a Neighbor adjacencies

3.4.b Point-to-point

3.4.c Broadcast (DR/BDR selection)

3.4.d Router ID

[Chapter 21, “Understanding OSPF Concepts,”](#) and [Chapter 22, “Implementing Basic OSPF Features,”](#) discuss the required and most common optional OSPF configuration settings, along with the many verification commands to show how OSPF works with those settings. This chapter continues with more OSPF implementation topics, both to round out the discussion of OSPF and to focus even more on the specific CCNA 200-301 exam topics.

The first major section focuses on neighbors and neighbor adjacencies, as mentioned in yet another of the OSPF exam topics. OSPF routers cannot exchange LSAs with another router unless they first become neighbors. This section discusses the various OSPF features that can prevent OSPF routers from becoming neighbors and how you can go about discovering if those bad conditions exist—even if you do not have access to the running configuration.

The chapter closes with a section called “[Route Selection,](#)” which discusses OSPF logic plus IP routing logic. This section tackles the question of what route the router should choose, focusing on the cases in which multiple routes exist. The text discusses how OSPF chooses between competing routes and how routers match packet destination addresses to the routes that already exist in the IP routing table.

“Do I Know This Already?” Quiz

Take the quiz (either here or use the PTP software) if you want to use the score to help you decide how much time to spend on this chapter. The letter answers are listed at the bottom of the page following the quiz. [Appendix C](#), found both at the end of the book as well as on the companion website, includes both the answers and explanations. You can also find both answers and explanations in the PTP testing software.

Table 24-1 “Do I Know This Already?” Foundation Topics
Section-to-Question Mapping

Foundation Topics Section	Questions
OSPF Neighbor Relationships	1-3
Route Selection	4-7

- 1.** An engineer connects Routers R11 and R12 to the same Ethernet LAN and configures them to use OSPFv2. Which answers describe a combination of settings that would prevent the two routers from becoming OSPF neighbors? (Choose two answers.)
 - a.** R11’s interface uses area 11 while R12’s interface uses area 12.
 - b.** R11’s OSPF process uses process ID 11 while R12’s uses process ID 12.
 - c.** R11’s interface uses OSPF priority 11 while R12’s uses OSPF priority 12.
 - d.** R11’s interface uses an OSPF Hello timer value of 11 while R12’s uses 12.
- 2.** An engineer connects Routers R13 and R14 to the same Ethernet LAN and configures them to use OSPFv2. Which answers describe a combination of settings that would prevent the two routers from becoming OSPF neighbors?
 - a.** Both routers’ interface IP addresses reside in the same subnet.
 - b.** Both routers’ OSPF process uses process ID 13.
 - c.** Both routers’ OSPF process uses router ID 13.13.13.13.

- d.** Both routers' interfaces use an OSPF Dead interval of 40.
- 3.** Router R15 has been a working part of a network that uses OSPFv2. An engineer then issues the **shutdown** command in OSPF configuration mode on R15. Which of the following occurs?
 - a.** R15 empties its IP routing table of all OSPF routes but keeps its LSDB intact.
 - b.** R15 empties its OSPF routes and LSDB but keeps OSPF neighbor relationships active.
 - c.** R15 keeps OSPF neighbors open but does not accept new OSPF neighbors.
 - d.** R15 keeps all OSPF configuration but ceases all OSPF activities (routes, LSDB, neighbors).
- 4.** Using OSPF, Router R1 learns three routes to subnet 10.1.1.0/24. It first calculates a route through Router R2 with metric 15000, then a route through Router R3 with metric 15001, and then a route through Router R4 with metric 15000. Which routes does the router place into its routing table, assuming all default configuration settings for any features that would impact the answer?
 - a.** Only the route with R2 as the next-hop router
 - b.** Both routes with metric 15000
 - c.** All three routes to subnet 10.1.1.0/24
 - d.** Only the router with R3 as the next-hop router
- 5.** Router R2 runs both EIGRP and OSPF. It learns two OSPF routes to subnet 172.16.1.0/24, one with metric 1000 and one with metric 2000. It learns two EIGRP

routes with metrics 1,000,000 and 2,000,000. If using default settings for any settings that might impact the answer, which route(s) will the router place into the IP routing table?

- a. The metric 1000 OSPF route and the metric 1,0000,000 EIGRP route
 - b. The metric 2000 OSPF route and the metric 2,0000,000 EIGRP route
 - c. The metric 1000 OSPF route only
 - d. The metric 1,0000,000 EIGRP route only
6. Router R3 receives a packet with destination IP address 172.20.89.100. How many of the address ranges defined in the routes per the **show ip route** command match the packet's destination address?

[Click here to view code image](#)

```
3# show ip route
Gateway of last resort is 172.20.15.5 to network 0.0.0.0

0*E2  0.0.0.0/0 [110/1] via 172.20.15.5, 00:04:56, GigabitEthernet0/24
      172.20.0.0/16 is variably subnetted, 12 subnets, 4 masks
S      172.20.90.9/32 [1/0] via 172.20.11.1
0 IA   172.20.88.0/23 [110/3] via 172.20.12.2, 00:03:44, GigabitEthernet0/24
0 IA   172.20.80.0/20 [110/3] via 172.20.13.3, 00:04:55, GigabitEthernet0/24
0 IA   172.20.0.0/16 [110/6] via 172.20.14.4, 00:02:14, GigabitEthernet0/24
```

- a. 1
- b. 2
- c. 3
- d. 4
- e. 5

- 7.** Router R3 receives a packet with destination IP address 172.20.90.1. Which next-hop IP address does Router R3 use when forwarding the packet?

[Click here to view code image](#)

```
3# show ip route
Gateway of last resort is 172.20.15.5 to network 0.0.0.0

0*E2  0.0.0.0/0 [110/1] via 172.20.15.5, 00:04:56, GigabitEthernet0/24
      172.20.0.0/16 is variably subnetted, 12 subnets, 4 masks
S      172.20.90.9/32 [1/0] via 172.20.11.1
0 IA   172.20.88.0/23 [110/3] via 172.20.12.2, 00:03:44, GigabitEthernet0/24
0 IA   172.20.80.0/20 [110/3] via 172.20.13.3, 00:04:55, GigabitEthernet0/24
0 IA   172.20.0.0/16 [110/6] via 172.20.14.4, 00:02:14, GigabitEthernet0/24
```

- a.** 172.20.11.1
- b.** 172.20.12.2
- c.** 172.20.13.3
- d.** 172.20.14.4
- e.** 172.20.15.5

Answers to the “Do I Know This Already?” quiz:

- 1** A, D
- 2** C
- 3** D
- 4** B
- 5** D
- 6** D
- 7** C

Foundation Topics

OSPF Neighbor Relationships

A router's OSPF configuration enables OSPF on a set of interfaces. IOS then attempts to discover other neighbors on those interfaces by sending and listening for OSPF Hello messages. However, once discovered, two routers may not become neighbors. They must have compatible values for several settings as listed in the Hellos exchanged between the two routers. This second major section of the chapter examines those reasons.

OSPF Neighbor Requirements

After an OSPF router hears a Hello from a new neighbor, the routing protocol examines the information in the Hello and compares that information with the local router's own settings. If the settings match, great. If not, the routers do not become neighbors. Because there is no formal term for all these items that a routing protocol considers, this book just calls them *neighbor requirements*. [Table 24-2](#) lists the neighbor requirements for OSPF, with some comments about the various issues following the table.



Table 24-2 Neighbor Requirements for OSPF

Requirement	Required for OSPF	Neighbor Missing If Incorrect
Interfaces must be in an up/up state.	Yes	Yes

Requirement	Required for OSPF	Neighbor Missing If Incorrect
Access control lists (ACL) must not filter routing protocol messages.	Yes	Yes
Interfaces must be in the same subnet.	Yes	Yes
Neighbors must pass routing protocol neighbor authentication (if configured).	Yes	Yes
Hello and dead timers must match.	Yes	Yes
Router IDs (RID) must be unique.	Yes	Yes
Neighbors must be in the same area.	Yes	Yes
OSPF process must not be shut down.	Yes	Yes
OSPF must not be shut down on the interface.	Yes	Yes
Neighboring interfaces must use same MTU setting.	Yes	No
Neighboring interfaces must use same OSPF network type.	Yes	No
Neighboring interfaces cannot both use priority 0.	Yes	No

First, consider the meaning of the two rightmost columns. The column labeled “Required for OSPF” means that the

item must be working correctly for the neighbor relationship to work correctly. The last column heading notes whether the neighbor will be missing (“yes”) in the list of OSPF neighbors in commands like the **show ip ospf neighbor** command.

The table breaks into three sections. The first section lists non-OSPF configuration while the second lists OSPF configuration—all of which prevents OSPF neighbor relationships from forming. The third section lists settings that allow OSPF neighbor relationships, but with other related problems that prevent the addition of correct OSPF routes to the IP routing table.

For reference, [Table 24-3](#) relists some of the requirements from [Table 24-2](#), along with the most useful commands to find the related settings.



Table 24-3 OSPF Neighbor Requirements and the Best **show/debug** Commands

Requirement	Best show Command
Hello and dead timers must match.	show ip ospf interface
Neighbors must be in the same area.	show ip ospf interface brief
RIDs must be unique.	show ip ospf
Neighbors must pass any neighbor authentication.	show ip ospf interface

Requirement	Best show Command
OSPF process must not be shut down.	show ip ospf, show ip ospf interface

The rest of this section looks at some of the items from [Table 24-3](#) in a little more detail.

Note

One configuration choice that people sometimes think is an issue, but is not, is the process ID as defined by the **router ospf process-id** command. Neighboring routers can use the same process ID values, or different process ID values, with no impact on whether two routers become OSPF neighbors.

Issues That Prevent Neighbor Adjacencies

The next few pages look at three neighbor issues from [Table 24-2](#), using [Figure 24-1](#)'s topology in the examples. R1 begins with all correct configuration as listed in [Example 24-1](#). However, later examples introduce configuration mistakes on Routers R2, R3, and R4 as follows:

- R2 has been reconfigured to place both LAN interfaces in area 1, whereas the other three routers' G0/0 interfaces remain in area 0.
- R3 has been reconfigured to use the same RID (1.1.1.1) as R1.
- R4 has been reconfigured with Hello/Dead timers of 5/20 on its G0/0 interface, instead of the default settings of 10/40 used by R1, R2, and R3.

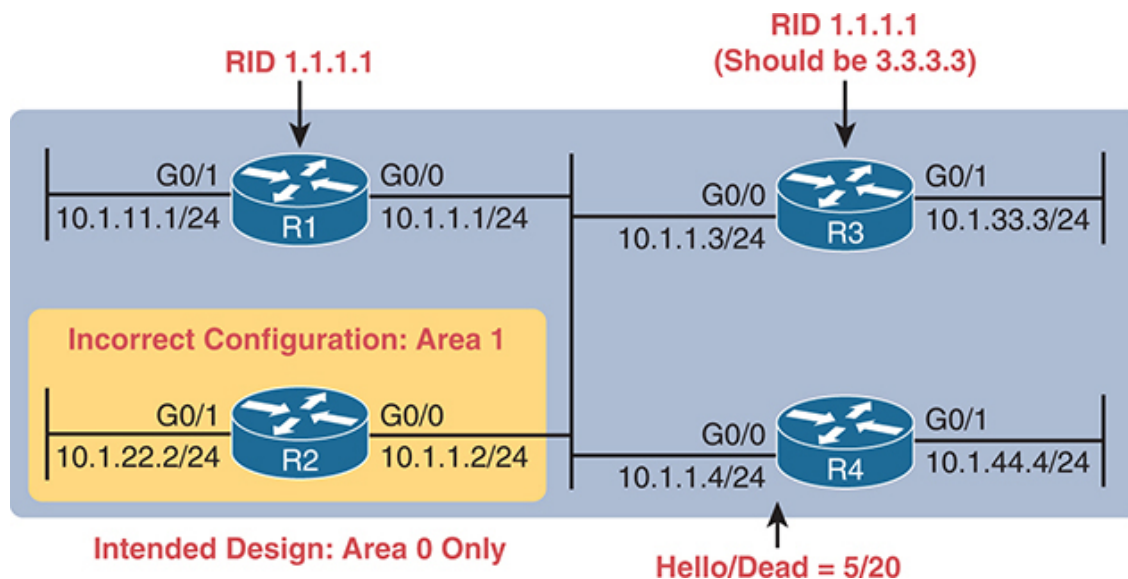


Figure 24-1 Summary of Problems That Prevent OSPF Neighbors on the Central LAN

Example 24-1 Router R1 Configuration with No Configuration Issues

[Click here to view code image](#)

```
router ospf 1
  router-id 1.1.1.1
!
  interface gigabitEthernet0/0
    ip address 10.1.1.1 255.255.255.0
    ip ospf 1 area 0
!
  interface gigabitEthernet0/1
    ip address 10.1.11.1 255.255.255.0
    ip ospf 1 area 0
```

Finding Area Mismatches

To create an area mismatch, the configuration on some router must place the interface into the wrong area per the

design. [Figure 24-1](#) shows the intent to make that mistake on Router R2, placing both its interfaces into area 1 instead of area 0. [Example 24-2](#) shows the configuration, which uses the correct syntax (and is therefore accepted by the router) but sets the wrong area number.

Example 24-2 *Setting Area 1 on R2's Interfaces, When They Should Be in Area 0*

[Click here to view code image](#)

```
router ospf 1
  router-id 2.2.2.2
!
interface gigabitEthernet0/0
  ip address 10.1.1.2 255.255.255.0
  ip ospf 1 area 1
!
interface gigabitEthernet0/1
  ip address 10.1.22.2 255.255.255.0
  ip ospf 1 area 1
```

With an area mismatch error, the **show ip ospf neighbor** command will not list the neighbor. Because you see nothing in the OSPF neighbor table, to troubleshoot this problem, you need to find the area configuration on each interface on potentially neighboring routers. To do so:

- Check the output of **show running-config** to look for:
 - **ip ospf process-id area area-number** interface subcommands
 - **network** commands in OSPF configuration mode
- Use the **show ip ospf interface [brief]** command to list the area number

Finding Duplicate OSPF Router IDs

Next, [Example 24-3](#) shows R1 and R3 both trying to use RID 1.1.1.1. Due to the duplicate RIDs, neither router will list the other in the output of the **show ip ospf neighbor** command. Interestingly, both routers automatically generate a log message for the duplicate OSPF RID problem between R1 and R3; the end of [Example 24-3](#) shows one such message. For the exams, just use the **show ip ospf** commands on both R3 and R1 to easily list the RID on each router, noting that they both use the same value.

Example 24-3 Comparing OSPF Router IDs on R1 and R3

[Click here to view code image](#)

```
! Next, on R3: R3 lists the RID of 1.1.1.1
```

```
!
```

```
R3# show ip ospf
```

```
Routing Process "ospf 1" with ID 1.1.1.1
```

```
Start time: 00:00:37.136, Time elapsed: 02:20:37.200
```

```
! lines omitted for brevity
```

```
! Back to R1: R1 also uses RID 1.1.1.1
```

```
R1# show ip ospf
```

```
Routing Process "ospf 1" with ID 1.1.1.1
```

```
Start time: 00:01:51.864, Time elapsed: 12:13:50.904
```

```
! lines omitted for brevity
```

```
*May 29 00:01:25.679: %OSPF-4-DUP_RTRID_NBR: OSPF detected duplicate  
1.1.1.1 from 10.1.1.3 on interface GigabitEthernet0/0
```

First, focus on the problem: the duplicate RIDs. The first line of the **show ip ospf** command on the two routers quickly shows the duplicate use of 1.1.1.1. To solve the problem,

assuming R1 should use 1.1.1.1 and R3 should use another RID (maybe 3.3.3.3), change the RID on R3 and restart the OSPF process. To do so, use the **router-id 3.3.3.3** OSPF subcommand and the EXEC mode command **clear ip ospf process**. (OSPF will not begin using a new RID value until the process restarts, either via command or reload.) At that point, the routers should become neighbors again and be displayed in the output of the **show ip ospf neighbor** command.

Note

There are cases in which routers in different areas can use the same RID and cause no problems in OSPF. However, to be safe, use unique OSPF RIDs throughout the entire OSPF domain (that is, among all routers in your enterprise that use OSPF).

Finding OSPF Hello and Dead Timer Mismatches

First, as a reminder from chapters past:

- **Hello interval/timer:** The per-interface timer that tells a router how often to send OSPF Hello messages on an interface.
- **Dead interval/timer:** The per-interface timer that tells the router how long to wait without having received a Hello from a neighbor before believing that neighbor has failed. (Defaults to four times the Hello timer.)

Next, consider the problem created on R4, with the configuration of a different Hello timer and dead timer (5 and 20, respectively) as compared with the default settings on R1, R2, and R3 (10 and 40, respectively). A Hello or Dead interval mismatch prevents R4 from becoming neighbors

with any of the other three OSPF routers. Routers list their Hello and Dead interval settings in their Hello messages and choose not to become neighbors if the values do not match. As a result, none of the routers become neighbors with Router R4 in this case.

[Example 24-4](#) shows the easiest way to find the mismatch using the **show ip ospf interface** command on both R1 and R4. This command lists the Hello and Dead timers for each interface, as highlighted in the example. Note that R1 uses 10 and 40 (Hello and Dead), whereas R4 uses 5 and 20.

Example 24-4 *Finding Mismatched Hello/Dead Timers*

[Click here to view code image](#)

```
R1# show ip ospf interface G0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.1.1.1/24, Area 0, Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID Cost Disabled Shutdown Topology Name
             0      1      no      no      Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.1.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  ! lines omitted for brevity
```

```
! Moving on to R4 next
```

```
!
```

```
R4# show ip ospf interface Gi0/0
```

```
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.1.1.4/24, Area 0, Attached via Network Statement
  Process ID 4, Router ID 10.1.44.4, Network Type BROADCAST, Cost: 1
  Topology-MTID Cost Disabled Shutdown Topology Name
```

```
0          1          no          no          Base
Transmit Delay is 1 sec, State DR, Priority 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 10.1.44.4, Interface address 10.1.1.4
No backup designated router on this network
Timer intervals configured, Hello 5, Dead 20, Wait 20, Retransm
! lines omitted for brevity
```

Shutting Down the OSPF Process

Like administratively disabling and enabling an interface, IOS also allows the OSPFv2 routing protocol process to be disabled and enabled with the **shutdown** and **no shutdown** router mode subcommands, respectively. When a routing protocol process is shut down, IOS does the following:

- Brings down all neighbor relationships and clears the OSPF neighbor table
- Clears the LSDB
- Clears the IP routing table of any OSPF-learned routes

At the same time, shutting down OSPF does retain some important details about OSPF, in particular:

- IOS retains all OSPF configuration.
- IOS still lists all OSPF-enabled interfaces in the OSPF interface list (**show ip ospf interface**) but in a DOWN state.

Shutting down the OSPF routing protocol process allows the network engineer to stop using the routing protocol on that router without having to remove all the configuration. Once the process is shut down, the **show ip ospf interface [brief]** command should still list some output, as will the

show ip ospf command, but the rest of the commands will list nothing.

Example 24-5 shows an example on Router R5, as shown in Figure 24-2. R5 is a different router than the one used in earlier examples, but it begins the example with two OSPF neighbors, R2 and R3, with Router IDs 2.2.2.2 and 3.3.3.3. The example shows the OSPF process being shut down, the neighbors failing, and those two key OSPF **show** commands: **show ip ospf neighbor** and **show ip ospf interface brief**.

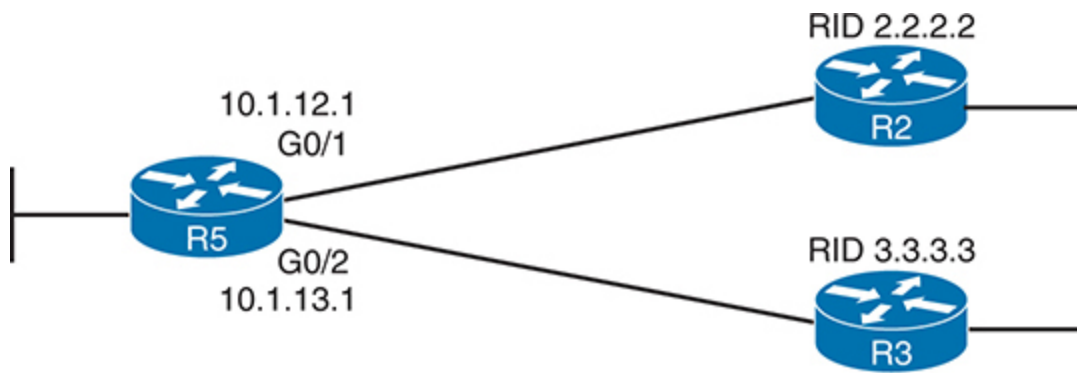


Figure 24-2 Example Network to Demonstrate OSPF Process Shutdown

Example 24-5 Shutting Down an OSPF Process, and the Resulting Neighbor States

[Click here to view code image](#)

```
R5# show ip ospf neighbor
Neighbor ID    Pri   State           Dead Time   Address      Interface
2.2.2.2        1     FULL/DR         00:00:35   10.1.12.2    G0/1
3.3.3.3        1     FULL/DR         00:00:33   10.1.13.3    G0/2
R5# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)# router ospf 1
R5(config-router)# shutdown
R5(config-router)# ^Z
```

```

*Mar 23 12:43:30.634: %0SPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on (
from FULL to DOWN, Neighbor Down: Interface down or detached
*Mar 23 12:43:30.635: %0SPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on (
from FULL to DOWN, Neighbor Down: Interface down or detached
R5# show ip ospf interface brief
Interface      PID      Area      IP Address/Mask  Cost  State Nb
Gi0/1          1        0         10.1.12.1/24    1     DOWN  0/(
Gi0/2          1        0         10.1.13.1/24    1     DOWN  0/(
R5# show ip ospf
Routing Process "ospf 1" with ID 5.5.5.5
Start time: 5d23h, Time elapsed: 1d04h
Routing Process is shutdown

! lines omitted for brevity
R5# show ip ospf neighbor
R5#
R5# show ip ospf database
                OSPF Router with ID (3.3.3.3) (Process ID 1)
R5#

```

First, before the **shutdown**, the **show ip ospf neighbor** command lists two neighbors. After the **shutdown**, the same command lists no neighbors at all. Second, the **show ip ospf interface brief** command lists the interfaces on which OSPF is enabled, on the local router's interfaces. However, it lists a state of DOWN, which is a reference to the local router's state. Also, note that the **show ip ospf** command positively states that the OSPF process is in a shutdown state, while the **show ip ospf database** command output lists only a heading line, with no LSAs.

Shutting Down OSPF on an Interface

IOS also supports a feature to disable OSPF on an interface without having to remove any OSPF configuration to do so. The feature has the same motivations as the **shutdown** command in router configuration mode: to allow OSPF configuration to remain while stopping OSPF. However, shutting down OSPF on an interface ceases all OSPF operations on that interface rather than all OSPF operations on the router.

To shut down OSPF on an interface, use the **ip ospf shutdown** interface subcommand, and to enable it again, use the **no ip ospf shutdown** command. After you use the **ip ospf shutdown** command, the router changes as follows:

- The router stops sending Hellos on the interface, allowing existing OSPF neighbor relationships to time out.
- The neighbor failure(s) triggers OSPF reconvergence, resulting in the removal of any routes that use the interface as an outgoing interface.
- The router also stops advertising about the subnet on the link that is shut down.

If those ideas sound familiar, the feature works much like OSPF passive interfaces, except that when shut down, OSPF also stops advertising about the connected subnet.

Interestingly, using the **ip ospf shutdown** interface subcommand also does not change a few commands about interfaces. For instance, after you configure interface G0/0/0 with the **ip ospf shutdown** command, the **show ip ospf interface** and **show ip ospf interface brief** commands still show G0/0/0 in the list of interfaces.

Issues That Allow Neighbors but Prevent IP Routes

Some configuration mismatches prevent learning OSPF routes, but they do allow routers to become neighbors, as noted in the final section of [Table 24-2](#). The issues are

- A mismatched MTU setting



- A mismatched OSPF network type
- Both neighbors using OSPF Priority 0

The next few pages explain the issues, with an example that shows all three misconfigurations along with the resulting **show** commands.

Mismatched MTU Settings

The maximum transmission unit (MTU) size defines a per-interface setting used by the router for its Layer 3 forwarding logic, defining the largest network layer packet that the router will forward out each interface. The IPv4 MTU size of an interface defines the maximum size IPv4 packet that the router can forward out an interface, and similarly, the IPv6 MTU size defines the largest IPv6 packet.

Routers often use a default IP MTU size of 1500 bytes, with the ability to set the value as well. The **ip mtu** size interface subcommand defines the IPv4 MTU setting, and the **ipv6 mtu** size command sets the equivalent for IPv6 packets.

Alternatively, you can set the MTU size for IPv4 and IPv6 using the **mtu** size interface subcommand. That command sets the MTU for IPv4 on the interface if the **ip mtu** size

command does not appear and for IPv6 if the **ipv6 mtu size** command does not appear.

With different IPv4 MTU settings, two OSPFv2 routers become OSPF neighbors; however, they fail to complete regular OSPF database exchange, reaching other interim OSPF neighbor states and then failing to a down state. Over time, they repeat the process to become neighbors, try to exchange their LSDBs, fail, and fall back to a down state.

Mismatched OSPF Network Types

In the section, “[OSPF Network Types](#),” in [Chapter 23](#), “[Implementing Optional OSPF Features](#),” you read about the OSPF broadcast network type, which uses a DR/BDR, and the OSPF point-to-point network type, which does not. Interestingly, if you misconfigure network type settings such that one router uses broadcast, and the other uses point-to-point, the two routers become neighbors and reach a full state. They remain stable in a full state, which means they exchanged their LSDBs; however, neither router can use routes based on LSAs learned from the neighbor.

The reason for not adding the routes has to do with the details of LSAs and how the use of a DR (or not) changes those LSAs. Basically, the two routers expect different details in the LSAs, and the SPF algorithm notices those differences and cannot resolve those differences when calculating routes.

Both Neighbors Using OSPF Priority 0

OSPF interface priority allows us to influence which router wins a DR or BDR election when using the broadcast network type. The highest priority router wins the election, with 1 as the default setting and allowed values ranging from 0 to 255 decimal.

Priority 0 acts as a special case meaning that the router will not serve as DR or BDR. For instance, in a topology with many routers sharing the same subnet, you could choose to make some routers use priority 0, effectively refusing to become DR or BDR so that you have more predictability when operating the network.

A problem occurs when you make the poor configuration choice to make all routers on a subnet use priority 0. If making that mistake, you have a subnet that must have a DR but for which all routers refuse the role. The routers proceed as normal to use Hellos to discover each other. They list each other as neighbors and reach a 2WAY state. However, because no router serves as the DR, the database exchange process stops at that point. The typical symptom is a stable neighbor in a 2WAY/DROTHER state.

Examples That Show OSPF Neighbors but No Routes

[Figure 24-3](#) shows a router topology used in an upcoming example. In this case, R1 uses poor configuration choices that cause the problems discussed in the previous few pages. In particular:

R1 G0/0/1: Uses network type point-to-point; R2 uses default setting broadcast.

R1 G0/0/2: Uses an IPv4 MTU of 1600; R3 uses the default setting 1500.

R1 G0/0/3: Both R1 and R4 use an OSPF priority of 0.

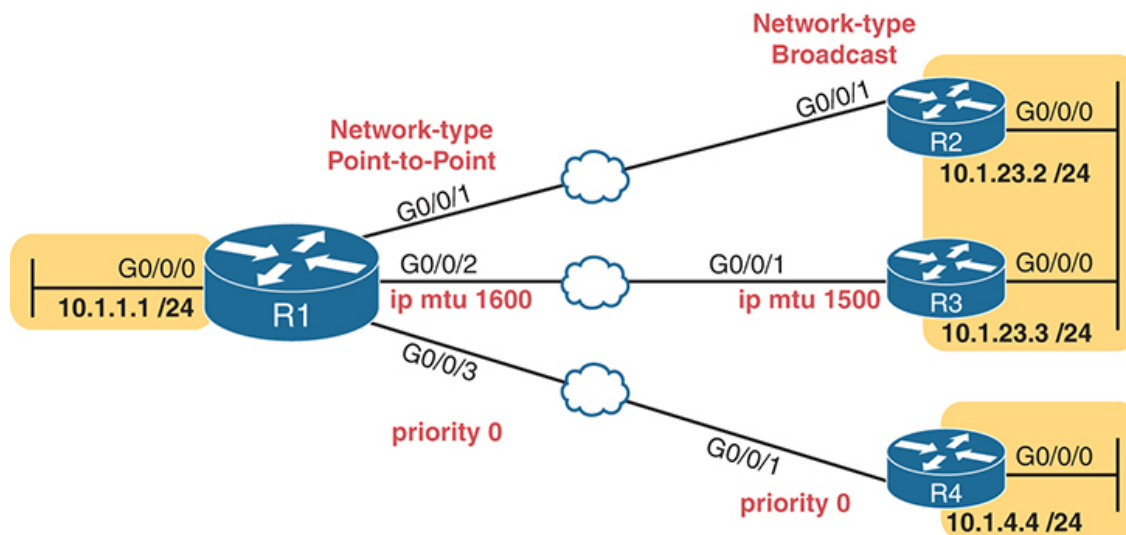


Figure 24-3 *OSPF Mistakes: Neighbors Formed, but Routes Not Learned*

To begin, [Example 24-6](#) shows the Router R1 configuration to cause the problems shown in the figure. Routers R2, R3, and R4 use all default configuration related to these parameters other than R4, which also requires the **ip ospf priority 0** interface subcommand. The example shows only the relevant configuration.

[Example 24-7](#) shows the results. First, note that Routers R2, R3, and R4 use OSPF RIDs 2.2.2.2, 3.3.3.3, and 4.4.4.4, respectively. The example begins with a **show ip ospf neighbor** command at a time when the neighbor relationships reach their best states.

Example 24-6 *OSPF Configuration Settings Matching Figure 24-3*

[Click here to view code image](#)

```
R1# show running-config interface g0/0/1
! Neighboring router R2 defaults to network type broadcast
interface GigabitEthernet0/0/1
ip address 10.1.12.1 255.255.255.0
```

```
ip ospf network point-to-point
ip ospf 1 area 0
```

```
R1# show running-config interface g0/0/2
```

```
! Neighboring router R3 defaults to IP MTU 1500
interface GigabitEthernet0/0/2
 ip address 10.1.13.1 255.255.255.0
 ip mtu 1600
 ip ospf 1 area 0
```

```
R1# show running-config interface g0/0/3
```

```
! Neighboring router R4 is also configured for OSPF priority 0
interface GigabitEthernet0/0/3
 ip address 10.1.14.1 255.255.255.0
 ip ospf priority 0
 ip ospf 1 area 0
```

Example 24-7 Resulting OSPF Neighbor Relationships on Router R1

[Click here to view code image](#)

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
4.4.4.4	0	2WAY/DROTHER	00:00:38	10.1.14.4	G0/0/3
3.3.3.3	0	EXCHANGE/DROTHER	00:00:38	10.1.13.3	G0/0/2
2.2.2.2	0	FULL/-	00:00:39	10.1.12.2	G0/0/1

```
R1#
```

```
*Nov  2 21:38:34.046: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on G0/0/2 from EXCHANGE to DOWN, Neighbor Down: Too many retransmissions
```

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
4.4.4.4	0	2WAY/DROTHER	00:00:31	10.1.14.4	G:
3.3.3.3	0	DOWN/DROTHER	-	10.1.13.3	G:
2.2.2.2	0	FULL/ -	00:00:34	10.1.12.2	G:

In this case, R1's neighbor relationship with Router R3 (3.3.3.3) cycles through different states. The middle of the example shows messages about how this neighbor relationship fails. Then the end of the example shows neighbor 3.3.3.3 now in a DOWN state. Over time, the neighbor relationship cycles through repeated attempts.

Note that none of the three neighbor relationships result in any IP routes. In this case, the **show ip route ospf** command on R1 lists zero routes.

Route Selection

When OSPF calculates routes to a specific subnet, the competition between routes might be simple and obvious. A router might calculate a single route for a subnet and use that route. A router might calculate multiple routes to reach a subnet, so it finds the route with the lowest metric (cost), placing that route into the IP routing table.

However, other scenarios occur both inside and outside of OSPF, requiring more understanding of the logic used by a router. This final major section of the chapter discusses three topics that make us think about the following:

1. How one router chooses between multiple equal-cost OSPF routes for one subnet
2. How one router chooses between multiple routes for one subnet that were learned by different routing protocols

3. When forwarding packets, how routers match routes in the routing table, particularly when a packet's destination address matches more than one route

Equal-Cost Multipath OSPF Routes

Consider the routes that one OSPF router calculates for one destination subnet. When more than one route to that one subnet exists, one route may have the lowest metric, making OSPF's logic simple: Add that route to the IP routing table. However, when the metrics tie for multiple routes to the same subnet, the router applies different logic: It places multiple **equal-cost routes** in the routing table.

IOS limits the concurrent equal-cost OSPF routes for each destination subnet based on the **maximum-paths** *number* subcommand under **router ospf**. The default varies depending on the router platform, with a common default setting of four concurrent routes.

For example, consider the most recent [Figure 24-3](#), but with all configuration errors fixed, all routers learn routes to all subnets. Router R1 has only one possible route to reach subnet 10.1.4.0/24, located off Router R4. However, R1 has two possible routes to reach subnet 10.1.23.0/24: one route through neighbor R2 and one through neighbor R3. If using OSPF costs 10 for all GigabitEthernet interfaces, both R1's routes to subnet 10.1.23.0/24 have a cost of 20. [Example 24-8](#) shows that result with a list of OSPF-learned routes on Router R1.

Take an extra moment to look closely at the last two lines of output. Note that the output does not repeat the subnet ID and mask, listing it once, but leaving spaces in that position in the last line. That format should help you identify cases like this of multiple routes for the same subnet. Also, note the second number within brackets lists the metric, 20 for both routes, confirming they have an equal metric (cost).

Example 24-8 OSPF Routes on Router R1 from [Figure 24-3](#)

[Click here to view code image](#)

```
R1# show ip route ospf
! Legend omitted for brevity

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
0       10.1.4.0/24 [110/20] via 10.1.14.4, 00:00:17, GigabitEtl
0       10.1.23.0/24 [110/20] via 10.1.13.3, 00:00:20, GigabitEi
        [110/20] via 10.1.12.2, 00:00:23, GigabitEi
```

Now, think about packet forwarding instead of thinking about the logic of placing routes into the routing table. Which of the two routes to subnet 10.1.23.0/24 should R1 use? How should it take advantage of the two equal-cost routes? A router could load balance the packets on a per-packet basis, but that is a poor choice for a few reasons. Instead, by default, routers balance based on the individual destination address. The router sends packets for one destination IP address using one route, another destination address using the other route, and so on. Using destination-based load balancing allows for much less router overhead and avoids some of the problems that occur with per-packet load balancing.

Note

The logic in this section often goes by the name **equal-cost multipath (ECMP)**.

Multiple Routes Learned from Competing Sources

A typical enterprise router first learns connected routes, based on interface IP addresses. Those routes happen to be the best routes to reach those subnets because those subnets connect directly to the router. Additionally, each enterprise router uses one routing protocol (for example, OSPF) to dynamically learn all other routes.

However, in several legitimate cases, one router learns routes using more than connected routes plus one routing protocol. For instance, some routers also have static routes, as discussed in [Chapter 17, “Configuring IPv4 Addresses and Static Routes.”](#) A router configuration could define a static route to a subnet, while the IGP also learns a route to that same subnet. Which route is better? In other cases, one router might run both OSPF and BGP. Again, if both learn a route to the same subnet, which is better? Or a router might use multiple IGPs, like both OSPF and EIGRP. Again, both might learn a route to the same subnet, begging the same question.

Note

To consider routes to be to the same subnet, they must refer to the same subnet ID and subnet mask.

Routing protocol metrics do not help a router choose between competing routes in these cases. For instance, EIGRP commonly assigns metrics with values in the millions and billions, with OSPF using hundreds or thousands. Additionally, connected and static routes have no metrics. So metrics cannot help the router choose the best route in

these cases, so IOS needs another method to choose between routes from different sources.

When IOS must choose between routes learned using different routing protocols, IOS uses a concept called **administrative distance**. Administrative distance (AD) is a number that denotes how believable an entire routing protocol is on a single router. The lower the number, the better, or more believable, the routing protocol. For example, RIP has a default administrative distance of 120, OSPF uses a default of 110, and EIGRP defaults to 90. When using OSPF and EIGRP, the router will believe the EIGRP route instead of the OSPF route (at least by default). The administrative distance values are configured on a single router and are not exchanged with other routers. [Table 24-4](#) lists the various sources of routing information, along with the default administrative distances.



Table 24-4 Default Administrative Distances

Route Type	Administrative Distance
Connected	0
Static	1
BGP (external routes [eBGP])	20
EIGRP (internal routes)	90
OSPF	110
IS-IS	115
RIP	120

Route Type	Administrative Distance
EIGRP (external routes)	170
BGP (internal routes [iBGP])	200
DHCP default route	254
Unusable	255

Note

The **show ip route** command lists each route's administrative distance as the first of the two numbers inside the brackets. The second number in brackets is the metric.

Figure 24-4 shows what might happen inside each routing process and the choice to prefer the EIGRP route instead of the OSPF route. The left side shows how OSPF learns three routes to subnet 10.1.1.0/24, while EIGRP learns two routes to the same subnet. Each routing protocol chooses the lowest metric route to offer as the best route. However, an additional logic step then considers the administrative distance (AD) of the route. With default settings, the router chooses the EIGRP (AD 90) route instead of the OSPF (AD 110) route for subnet 10.1.1.0/24.

You might wonder at the choice to use more than one routing protocol. One classic case occurs when two companies merge into one company, as shown in Figure 24-5. Company A on the left uses EIGRP with addresses from private network 10.0.0.0. Company B on the right uses OSPF with addresses from private network 172.16.0.0. As a first step to connect the networks, the network engineers install

new WAN links between the A1 and B1 router plus the A2 and B2 router as shown in the figure.

**Key
Topic**

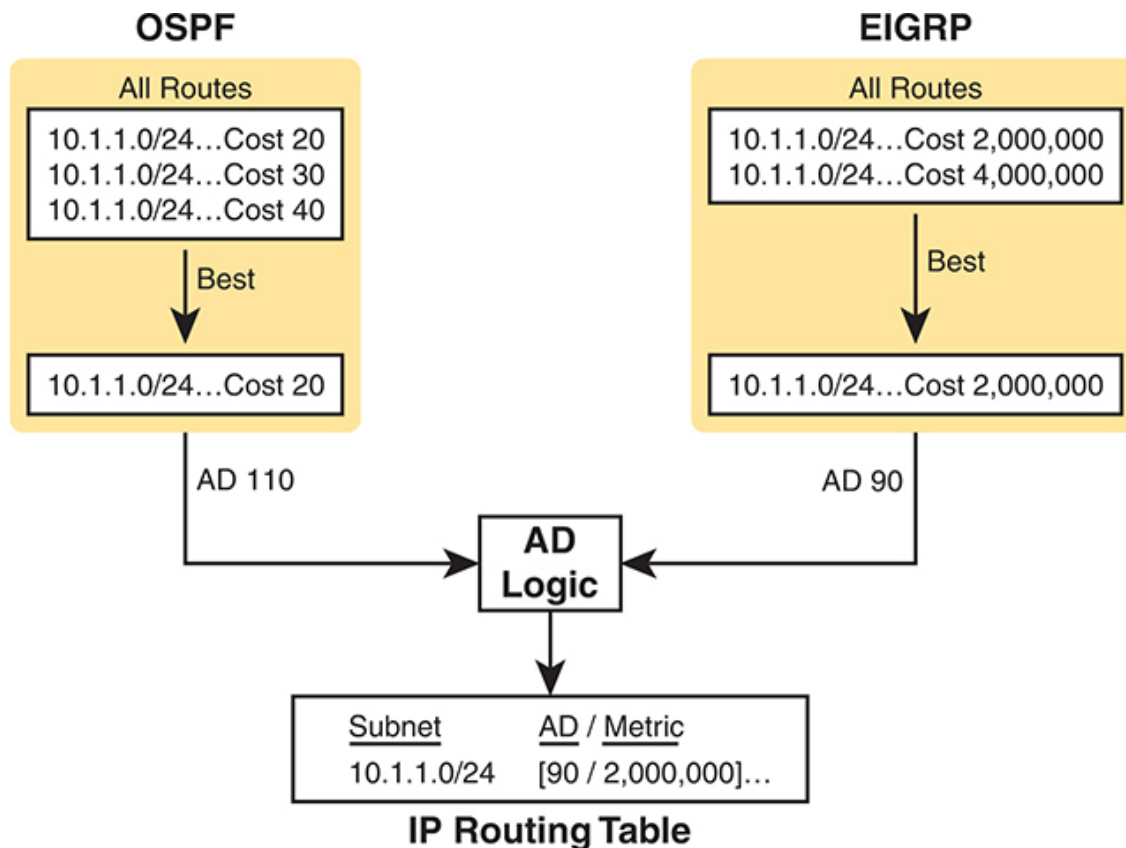


Figure 24-4 Logic: Choosing the Lowest AD Route Between OSPF and EIGRP

Next, the routers need to learn routes from both companies. The EIGRP routers on the left need to learn the company B routes known to OSPF on the right, and vice versa. To do that, the networking staff uses a feature called *route redistribution*, in which a *small set of routers run both routing protocols*. Internal to those routers, the router redistributes (takes routes from one protocol and advertises into the other), taking OSPF routes and advertising those subnets using EIGRP, and vice versa.

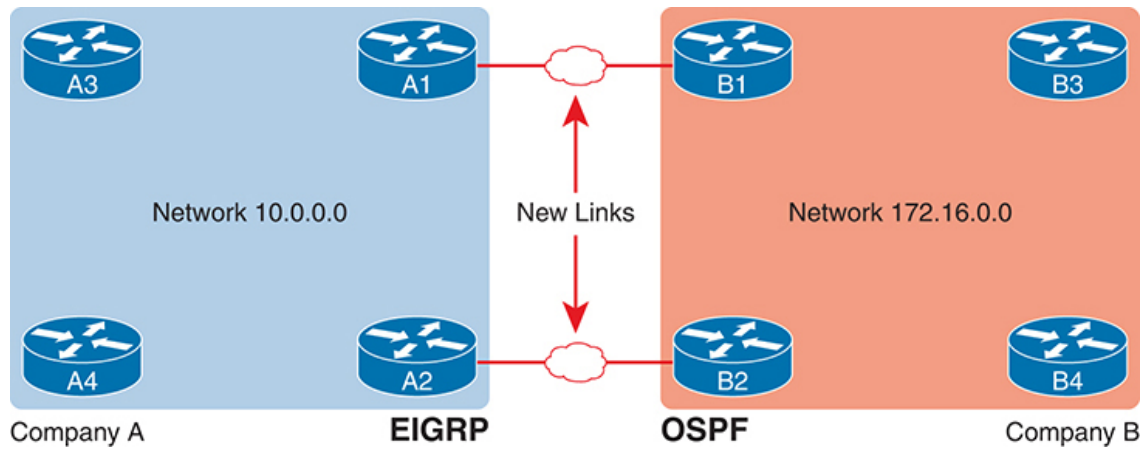


Figure 24-5 *First Step: Adding WAN Links Between Existing Company A and B Routers*

Figure 24-6 shows the updated scenario, with Routers A1 and A2 performing route redistribution. Most routers continue to run only OSPF or only EIGRP; in this case, Routers B1, B2, B3, and B4 continue to use only OSPF, whereas A3 and A4 use only EIGRP. However, the redistribution process on Routers A1 and A2 advertises routes so that all learn routes to all subnets in network 10.0.0.0 and network 172.16.0.0.

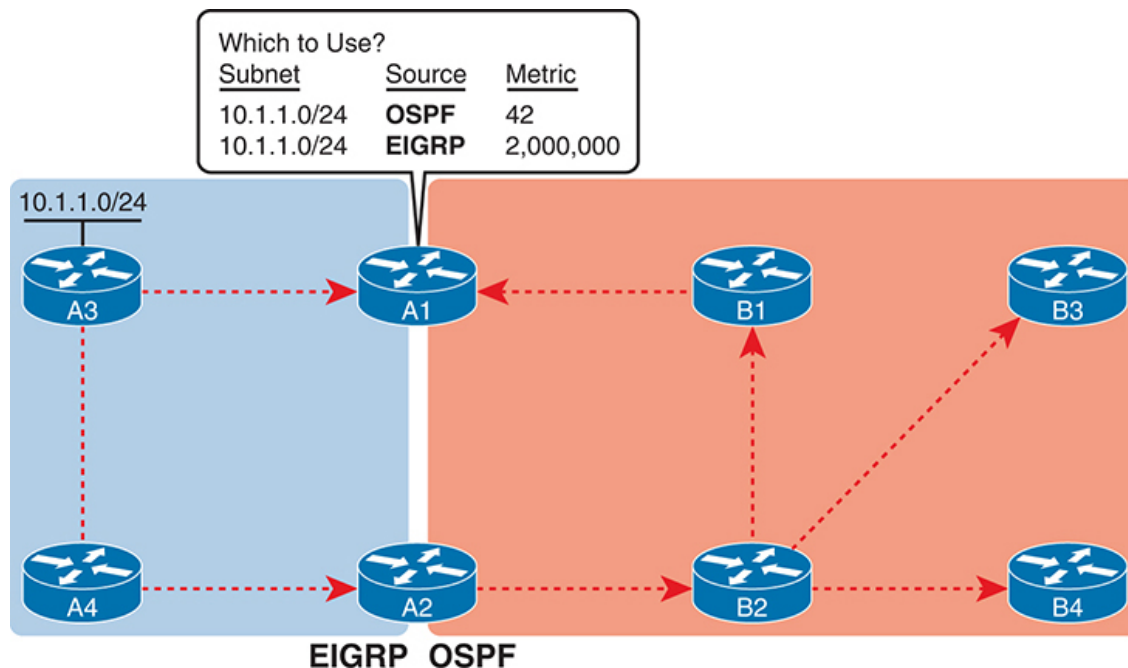


Figure 24-6 *Example: Router A1 Learns an EIGRP and OSPF Route to Subnet 10.1.1.0/24*

The figure uses dashed lines to show the propagation of routing information about subnet 10.1.1.0/24. First, focusing on the left, all four EIGRP routers in former Company A learn a route for subnet 10.1.1.0/24 using EIGRP. Next, follow the dashed lines along the bottom of the drawing, with Router A2 redistributing a route for 10.1.1.0/24 into OSPF. The OSPF routers all learn routes for 10.1.1.0/24 using OSPF. Now Router A1 has a route for subnet 10.1.1.0/24 learned with EIGRP and another learned with OSPF, creating a real-life scenario in which it must use administrative distance.

Note

The CCNP Enterprise certification discusses route redistribution, both concept and configuration. However, the CCNA V1.1 blueprint does not. It is mentioned here only to show a realistic case when a router needs to use administrative distance.

Note

The section “[Floating Static Routes](#)” in [Chapter 17](#) discusses another example of how routers use the AD to choose routes. Take a moment to review that section if you do not recall the details.

IP Forwarding with the Longest Prefix Match

For the final few pages of this chapter, focus on how a router matches a packet’s destination IP address to one of the routes already placed in the routing table. That process

goes by a few terms, like *IP routing*, *forwarding*, or the *data plane*. Regardless, focus on the logic.

A router's IP routing process requires that the router compare the destination IP address of each packet with the existing contents of that router's IP routing table. Often, only one route matches a particular destination address. When only one route matches the packet's destination, the action is obvious: forward the packet based on the details listed in that route.

In some cases, multiple routes exist for the exact same subnet, that is, for the exact same subnet and mask. The earlier section, "[Equal-Cost Multipath OSPF Routes](#)," discussed how OSPF would choose to add these multiple routes to the routing table, and how a router will load balance packets that match those routes.

This section discusses a different case in which a set of routes lists different subnets whose address ranges overlap. In that case, some packets' destination addresses match multiple routes in the IP routing table. For instance, one route might list subnet 10.1.0.0/16, another 10.1.1.0/25, and another 10.1.1.1/32. The range of addresses in each of those subnets includes 10.1.1.1, so a packet sent to IP address 10.1.1.1 would match all those routes.

Many legitimate router features can cause these multiple overlapping routes to appear in a router's routing table, including

- Static routes (including host routes)
- Route autosummarization
- Manual route summarization
- Default routes

In this case, a router chooses the best route as follows:

When a particular destination IP address matches more than one route in a router's IPv4 routing table, and those routes list different rather than identical subnets (different subnet IDs and masks), the router uses **longest prefix match** logic to match the most specific route—the route with the longest prefix length mask.



Using Your Subnetting Math Skills to Predict the Choice of Best Route

One way to predict which route a router uses requires you to use your subnetting skills plus the output from the **show ip route** command. To see how it works, an upcoming example uses several overlapping routes learned by Router R1 in [Figure 24-7](#). [Example 24-9](#) focuses on routes that match PC D's IP address (172.16.10.4), matching four routes on Router R1.

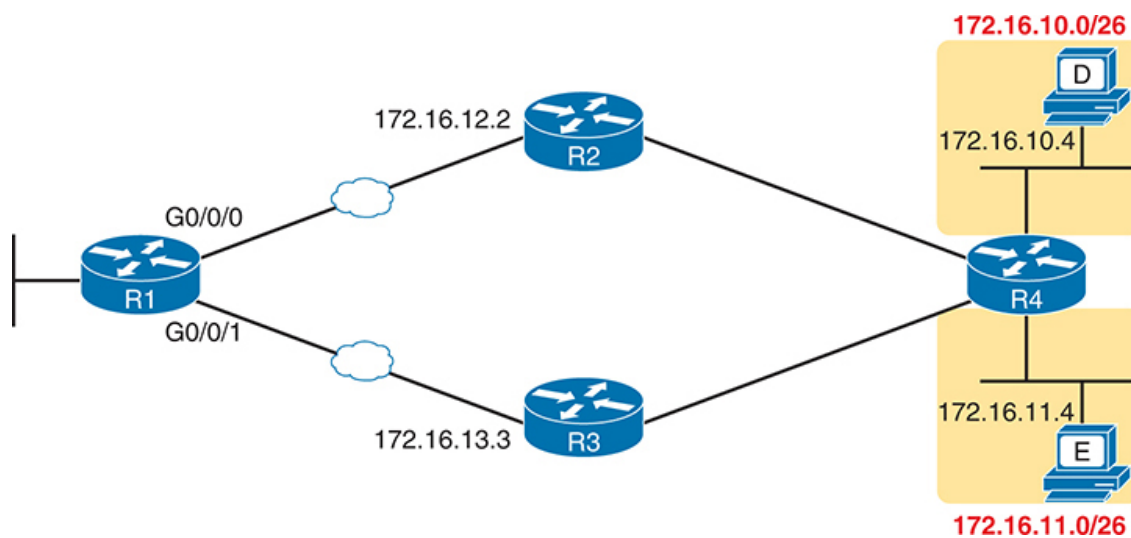


Figure 24-7 *Topology, Interfaces, and Next-Hop Addresses Used with [Example 24-9](#)*



Example 24-9 show ip route Command with Overlapping Routes

[Click here to view code image](#)

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external ty
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS
       ia - IS-IS inter area, * - candidate default, U - per-user
       o - ODR, P - periodic downloaded static route, H - NHRP,
       + - replicated route, % - next hop override

Gateway of last resort is 172.16.13.3 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 172.16.13.3, 00:04:56, GigabitEthernet0/24
       172.16.0.0/16 is variably subnetted, 12 subnets, 4 masks
S      172.16.10.4/32 [1/0] via 172.16.12.2
O IA   172.16.10.0/26 [110/3] via 172.16.13.3, 00:04:56, GigabitEthernet0/24
O IA   172.16.10.0/23 [110/6] via 172.16.12.2, 00:04:56, GigabitEthernet0/24
O IA   172.16.11.0/26 [110/3] via 172.16.13.3, 00:04:56, GigabitEthernet0/24
! Non-overlapping routes omitted for brevity
```

On the exam, or in real life, you would look at each route's subnet ID and mask and do the math to list the range of addresses in each subnet. [Table 24-5](#) shows the result to ensure the math does not get in the way of your understanding the concepts. All the subnets listed in the table match the destination address 172.16.10.4 (PC D).

Table 24-5 Analysis of Address Ranges for the Subnets in [Example 24-9](#)

Subnet/Prefix of Address Range a Route		Next-Hop
172.16.10.4/32	172.16.10.4 (just this one address)	172.16.12.2 (R2)
172.16.10.0/26	172.16.10.0-172.16.10.63	172.16.13.3 (R3)
172.16.10.0/23	172.16.10.0-172.16.11.255	172.16.12.2 (R2)
0.0.0.0/0	0.0.0.0-255.255.255.255 (all addresses)	172.16.13.3 (R3)

Note

The route listed as 0.0.0.0/0 is the default route.

Working through the logic, note that packet destination 172.16.10.4 matches all routes highlighted in [Example 24-9](#) (the routes also listed in [Table 24-5](#)). The various prefix lengths (masks) range from /0 to /32. The longest prefix (largest /P value, meaning the best and most specific route) is /32. So, a packet sent to 172.16.10.4 uses the route to 172.16.10.4/32, not the other routes.

It helps to think through a few more examples. The list identifies other destination addresses and explains why the router matches a specific route in [Example 24-9](#).

172.16.10.4: PC D. Matches all four highlighted routes; the longest prefix is /32, the route to router 172.16.10.4/32.

172.16.10.1: A different host in PC D's subnet, this destination matches the default route (0 prefix length) and the routes with the /23 and /26 prefix lengths. R1 uses the /26 route for subnet 172.16.10.0/26.

172.16.10.100: Per [Table 24-5](#), this address resides in the range of addresses for the /23 route, and it matches the default route as always. R1 uses the route with /23 prefix length for subnet 172.16.10.0/23.

172.16.12.1: Matches only the default route with /0 prefix length.

Using show ip route *address* to Find the Best Route

A second method to identify the route used by a router does not require you to use any subnetting math skills. Instead, use the **show ip route** *address* command, with the packet's destination IP address as the final parameter in the command. The router replies by listing its route when forwarding packets to that address.

For instance, [Example 24-10](#) lists the output of the **show ip route 172.16.10.1** command on the same router used in [Example 24-9](#). The first line of (highlighted) output lists the matched route: the route to 172.16.10.0/26. The rest of the output lists the details of that particular route, including the outgoing interface of GigabitEthernet0/0/1 and the next-hop router of 172.16.13.3.

Example 24-10 show ip route Command with Overlapping Routes

[Click here to view code image](#)

```
R1# show ip route 172.16.10.1
Routing entry for 172.16.10.0/26
  Known via "ospf 1", distance 110, metric 2, type inter area
```

```
known via ospf 1, distance 110, metric 3, type inter area
Last update from 172.16.13.3 on GigabitEthernet0/0/1, 00:44:09
Routing Descriptor Blocks:
* 172.16.13.3, from 3.3.3.3, 00:44:09 ago, via GigabitEthernet0/0/1
  Route metric is 3, traffic share count is 1
```

Certainly, if answering a lab question on the exam, use this command because it tells you what the router will choose without you doing the subnetting math.

Interpreting the IP Routing Table

Here at the end of three consecutive book parts about IP and IP routing, before moving on to [Part VII, “IP Version 6,”](#) this final topic reviews the most critical router command at the center of the discussion: the **show ip route** command. You have learned the various components of the command output through many examples. This final topic of the chapter pulls the concepts together in one place for easier reference and study.

[Figure 24-8](#) shows the output of a sample **show ip route** command. The figure numbers various parts of the command output for easier reference, with [Table 24-6](#) describing the output noted by each number.

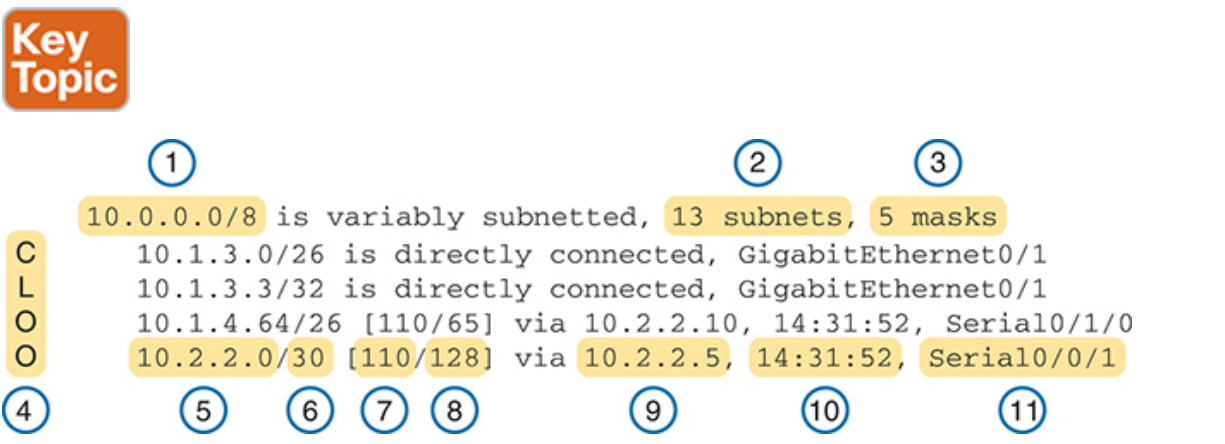


Figure 24-8 show ip route Command Output
Reference



Table 24-6 Descriptions of the **show ip route** Command Output (refer to [Figure 24-8](#))

Item	Idea	Value in the Figure	Description
1	Classful network	10.0.0.0/8	The routing table is organized by classful network. This line is the heading line for classful network 10.0.0.0; it lists the default mask for Class A networks (/8).
2	Number of subnets	13 subnets	The number of routes for subnets of the classful network known to this router, from all sources, including local routes—the /32 routes that match each router interface IP address.
3	Number of masks	5 masks	The number of different masks used in all routes known to this router inside this classful network.

Item	Idea	Value in the Figure	Description
4	Legend code	C, L, O	A short code that identifies the source of the routing information. <i>O</i> is for OSPF, <i>D</i> for EIGRP, <i>C</i> for Connected, <i>S</i> for static, and <i>L</i> for local. (See Example 24-11 for a sample of the legend.)
5	Prefix (Subnet ID)	10.2.2.0	The subnet number of this particular route.
6	Prefix length (Mask)	/30	The prefix mask used with this subnet.
7	Administrative distance	110	If a router learns routes for the listed subnet from more than one source of routing information, the router uses the source with the lowest administrative distance (AD).
8	Metric	128	The metric for this route.
9	Next-hop router	10.2.2.5	For packets matching this route, the IP address of the next router to which the packet should be forwarded.

Item	Idea	Value in the Figure	Description
10	Timer	14:31:52	For OSPF and EIGRP routes, this is the time since the route was first learned.
11	Outgoing interface	Serial0/0/1	For packets matching this route, the interface out which the packet should be forwarded.

You should also have a good mastery of the most common codes found in the legend at the beginning of the output from the **show ip route** command. [Example 24-11](#) closes the chapter with one final example, showing the legend but without any routes. Make an effort to commit the highlighted codes to memory.

Example 24-11 show ip route—Most Common Legend Codes

[Click here to view code image](#)

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external ty
       E1 - OSPF external type 1, E2 - OSPF external type 2, m -
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS
       ia - IS-IS inter area, * - candidate default, U - per-user
       H - NHRP, G - NHRP registered, g - NHRP registration summa
       o - ODR, P - periodic downloaded static route, l - LISP
       ~ - application route
```

```
a - application route
+ - replicated route, % - next hop override, p - override
& - replicated local route overrides by connected
! Lines omitted for brevity
```

Chapter Review

One key to doing well on the exams is to perform repetitive spaced review sessions. Review this chapter's material using either the tools in the book or interactive tools for the same material found on the book's companion website. Refer to the "[Your Study Plan](#)" element for more details. [Table 24-7](#) outlines the key review elements and where you can find them. To better track your study progress, record when you completed these activities in the second column.

Table 24-7 Chapter Review Tracking

Review Element	Review Date(s)	Resource Used
Review key topics		Book, website
Review command tables		Book
Review memory tables		Website
Watch video		Website

Review All the Key Topics



Table 24-8 Key Topics for [Chapter 24](#)

Key Topic Element	Description	Page Number
Table 24-2	OSPF neighbor requirements	611
Table 24-3	show commands to display facts for OSPF neighbor requirements	612
List	Reasons why routers can become OSPF neighbors but fail to exchange routing information	618
Table 24-4	Default Cisco router administrative distance settings	623
Figure 24-4	Concept of how a router uses administrative distance to choose an EIGRP route over an OSPF route	624
Paragraph	Router logic when a packet's destination address matches multiple IP routes	626
Example 24-9	Example of an IP routing table with overlapping IP routes	627
Figure 24-8	The Cisco IP routing table field reference	629
Table 24-6	Explanations for Figure 24-8 's IP routing table	629

Key Terms You Should Know

[administrative distance](#)

equal-cost multipath (ECMP)
equal-cost route
longest prefix match

Command References

Tables 24-9 and 24-10 list configuration and verification commands used in this chapter. As an easy review exercise, cover the left column in a table, read the right column, and try to recall the command without looking. Then repeat the exercise, covering the right column, and try to recall what the command does.

Table 24-9 Chapter 24 Configuration Command Reference

Command	Description
router ospf <i>process-id</i>	Global command that enters OSPF configuration mode for the listed process
ip ospf <i>process-id</i> area <i>area-number</i>	Interface subcommand to enable OSPF on the interface and to assign the interface to a specific OSPF area
router-id <i>id</i>	OSPF command that statically sets the router ID
ip ospf hello-interval <i>seconds</i>	Interface subcommand that sets the interval for periodic Hellos
ip ospf dead-interval <i>number</i>	Interface subcommand that sets the OSPF dead timer

Command	Description
[no] shutdown	An OSPF configuration mode command to disable (shutdown) or enable (no shutdown) the OSPF process
[no] ip ospf shutdown	An interface subcommand to disable or enable OSPF functions on the selected interface
mtu size	An interface subcommand to set the largest packet size (MTU) for all Layer 3 protocols enabled on the interface
ip mtu size	An interface subcommand to set the largest packet size (MTU) for IPv4 packets on the interface, overriding the setting of the mtu size subcommand
ip ospf priority value	Interface subcommand that sets the OSPF priority, used when electing a new DR or BDR
ip ospf network {broadcast point-to-point}	Interface subcommand used to set the OSPF network type on the interface
maximum-paths number	OSPF router subcommand that defines the maximum number of equal-cost multipath (ECMP) routes, learned by OSPF, to be added to the routing table at one time

Table 24-10 [Chapter 24 show](#) Command Reference

Command	Description

Command	Description
show ip protocols	Shows routing protocol parameters and current timer values, including an effective copy of the routing protocols' network commands and a list of passive interfaces
show ip ospf interface brief	Lists the interfaces on which the OSPF protocol is enabled (based on the network commands), including passive interfaces
show ip ospf interface <i>[type number]</i>	Lists detailed OSPF settings for all interfaces, or the listed interface, including Hello and Dead timers and OSPF area
show ip ospf neighbor	Lists neighbors and current status with neighbors, per interface
show ip ospf	Lists a group of messages about the OSPF process itself, listing the OSPF Router ID in the first line
show interfaces	Lists a long set of messages, per interface, that lists configuration, state, and counter information
show ip ospf database	Displays the contents of the router's OSPF LSDB
show ip route	Lists all IPv4 routes

Command Description	
show ip route ospf	Lists the OSPF-learned IPv4 routes in the routing table
show ip route <i>address</i>	Lists details about the one route this router would match for packets destined to the listed address
clear ip ospf process	Resets the OSPF process, resetting all neighbor relationships and also causing the process to make a choice of OSPF RID

Part VI Review

Keep track of your part review progress with the checklist in [Table P6-1](#). Details about each task follow the table.

Table P6-1 [Part VI Part Review](#) Checklist

Activity	1st Date Completed	2nd Date Completed
Repeat All DIKTA Questions		
Answer Part Review Questions		
Review Key Topics		
Do Labs		
Watch Video		
Use Per-Chapter Interactive Review		

Repeat All DIKTA Questions

For this task, answer the “Do I Know This Already?” questions again for the chapters in this part of the book using the PTP software. See the section “How to View Only DIKTA Questions by Chapter or Part” in the Introduction to

this book to learn how to make the PTP software show you DIKTA questions for this part only.

Answer Part Review Questions

For this task, answer the Part Review questions for this part of the book using the PTP software. See the section “How to View Part Review Questions” in the Introduction to this book to learn how to make the PTP software show you DIKTA questions for this part only.

Review Key Topics

Review all Key Topics in all chapters in this part, either by browsing the chapters or by using the Key Topics application on the companion website.

Do Labs

Depending on your chosen lab tool, here are some suggestions for what to do in lab:

Pearson Network Simulator: If you use the full Pearson CCNA simulator, focus more on the configuration scenario and troubleshooting scenario labs associated with the topics in this part of the book. These types of labs include a larger set of topics and work well as Part Review activities. (See the Introduction for some details about how to find which labs are about topics in this part of the book.)

Blog: Config Labs: The author’s blog includes a series of configuration-focused labs that you can do on paper or with Cisco Packet Tracer in about 15 minutes. To find them, open <https://www.certskills.com> and look under the Labs menu item.

Other: If using other lab tools, here are a few suggestions: make sure to experiment heavily with OSPF configuration and all the optional settings that impact OSPF neighbor compatibility.

Watch Video

The companion website includes a variety of common mistake and Q&A videos organized by part and chapter. Use these videos to challenge your thinking, dig deeper, review topics, and better prepare for the exam. Make sure to bookmark a link to the companion website and use the videos for review whenever you have a few extra minutes.